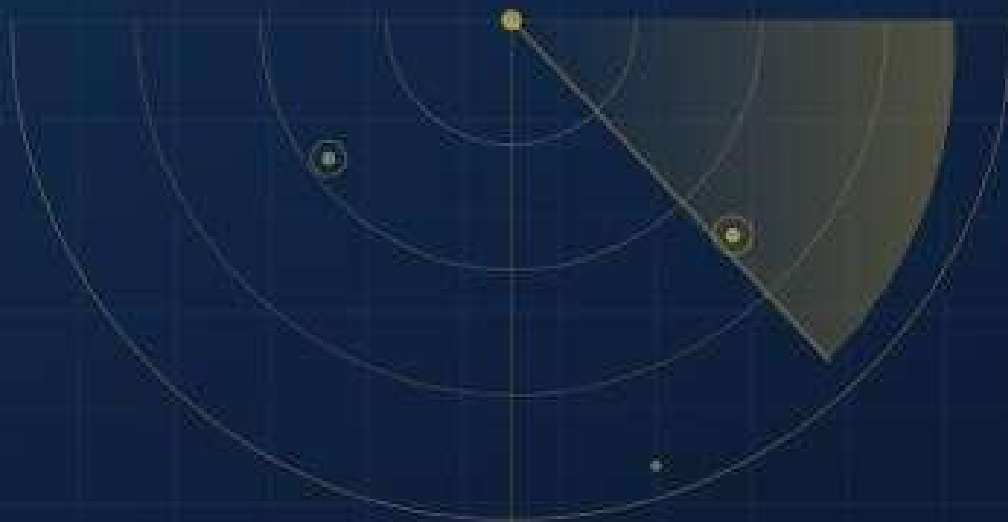


GRANT MANAGEMENT ASSOCIATES
DEFENSE & TECHNOLOGY PRACTICE

FROM DEFENSE STARTUP TO

Defense Industrial Powerhouse

A PLAYBOOK FOR FOUNDERS



Prepared by

Grant Management Associates

Defense & Technology Practice - in conjunction with BlackMountain Strategy Group

Authored by Mark Sue

Table of Contents

Executive Overview & Strategic Orientation	3
Rules of Engagement.....	—
Part 1: Year One — Building the Defense Capture Engine.....	14
1.1 What the PM Actually Evaluates	14
1.2 The Five Levels of Government Advocates	17
1.3 Case Studies	19
1.4 Tactical Outreach & Capture Engine Structure.....	23
1.5 Technology Readiness Levels — What TRL Actually Measures.....	27
Part 2: Year Two — Converting Relationships Into Programs	31
2.1 Crossing the Valley of Death.....	31
2.2 Capitol Hill — The Congressional Capture Strategy	33
2.3 Acquisition Pathways & Contract Vehicles	49
2.4 Designing High-Impact Field Demonstrations.....	51
2.5 Case Studies	53
Part 3: Year Three — Becoming a Major Defense Supplier.....	56
3.1 Scaling for Industrial Advantage.....	56
3.2 Manufacturing as a Competitive Advantage	58
3.3 The Back-Office Compliance Regimes That Blindside Scaling Companies.....	59
3.4 Congressional Engagement & Policy Alignment.....	63
3.5 Case Studies	64
Part 4: Domain-Specific Execution Matrix & Tactical Reference Manual	66
4.1 Cross-Domain Operational Matrix	66
4.2 The Strategic Founder Checklist	73
4.3 Common Failure Patterns	78
Part 5: Year Four and Beyond — Becoming an Enduring Powerhouse	79
5.1 Recompete Risk — Your Program Is Never Permanently Won	79
5.2 Tech Refresh and Spiral Development.....	81
5.3 International Expansion.....	82
5.4 M&A — Buyer or Target.....	82
Conclusion: The Winning Framework.....	86
Private Company Profiles	87
Space / Orbital.....	88
Commercial / Civil Focus	95
Aviation	97
Defense Hardware, Autonomy & Sensors.....	101
Robotics / Humanoids (Commercial & Industrial)	114
Appendix A: Founder Self-Assessment Scorecard.....	127
Appendix B: Glossary	129
Appendix C: Sources	132
Appendix D: Defense Agencies in Detail.....	134
Appendix E: Additional Resource Tools	151
Appendix F: Where This Leaves You.....	153

Executive Overview & Strategic Orientation

Who this is for. Written primarily for founders and capture leads at a post-Series B/C/D defense tech company, but most of it applies equally to a board member or investor evaluating GTM readiness in diligence — the PM Risk Matrix (1.1), the Five Levels of advocates (1.2), and the PPBE timeline (2.1) are as useful for assessing a company from the outside as for running one from the inside.

If you're reading this as an investor, skip the outreach templates in 1.5 and focus on the checklists in 4.2 and the scorecard in Appendix A — those are the fastest way to benchmark where a portfolio company actually stands against what its pitch deck claims.

Which Track Are You On?

This playbook is written primarily for the post-Series B/C/D track below, but the Year One-to-Four structure applies unevenly across company stages. Use the table below to weight which parts matter most for where you actually are.

Client segment	What you need most	Objective
Early entrant	Registration, market mapping, SBIR/STTR, customer discovery, readiness baseline	“Establish a credible defense entry point”
Transitioning dual-use company	Sponsor mapping, demonstrations, budget and contract pathway, teaming	“Convert validation into institutional adoption”
Scaling supplier	Manufacturing, compliance, sustainment, program capture, supply chain	“Build a durable defense-industrial platform”
Established contractor or investor	New-domain expansion, legislative landscape, recompetete strategy, diligence	“Defend and expand strategic position”

Congrats — you've raised your Series B or Series C. The bad news: this is exactly when most defense tech companies face their toughest transition. Up to this point, survival has depended almost entirely on technical validation — proving your stuff works: the algorithm runs on the edge, the airframe survives high-G maneuvers, and the autonomous surface vessel navigates GPS-denied environments. With \$100M–\$300M in the bank, history suggests this is when things get harder, not easier.

Most companies respond by doing more of what got them here: refining and polishing the technology. But beyond this stage, the primary cause of failure isn't technical — it's institutional adoption failure. The mindset has to change, and so do the resources.

Exhibit 1: The Inflection Point

Stage	Core Question
Technical Validation (Seed/A)	Can we build it and prove that it works?
Institutional Adoption (Series B-D)	Can we build an ecosystem inside government to buy it?
Program of Record	Can we transition to a funded program with a budget line?
Scaled Production	Can we deliver at scale and become a lasting defense industrial company?

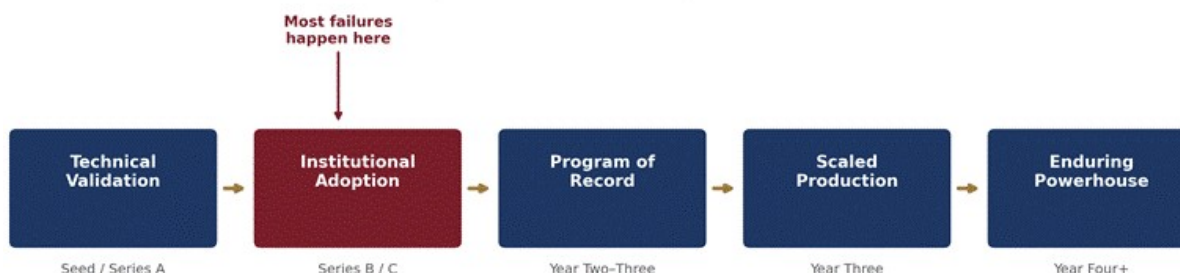
Source: GMA Defense & Technology Practice analysis

Exhibit 1 outlines the four stages at altitude. Exhibit 2 below breaks the middle two — Institutional Adoption and Program of Record — into the five sequential moves a company must execute to move from one to the next. That five-stage sequence is what the rest of this playbook is built around.

The primary cause of failure past Series B/C is institutional adoption failure, not technical failure.

Exhibit 2: The Five-Stage Capture Sequence

The Inflection Point — Five Stages, One Recurring Failure Gate



Source: GMA Defense & Technology Practice

Extending Exhibit 1 through Part 5: the same failure gate that sinks most companies at Series B/C recurs, in smaller form, at every later transition too.

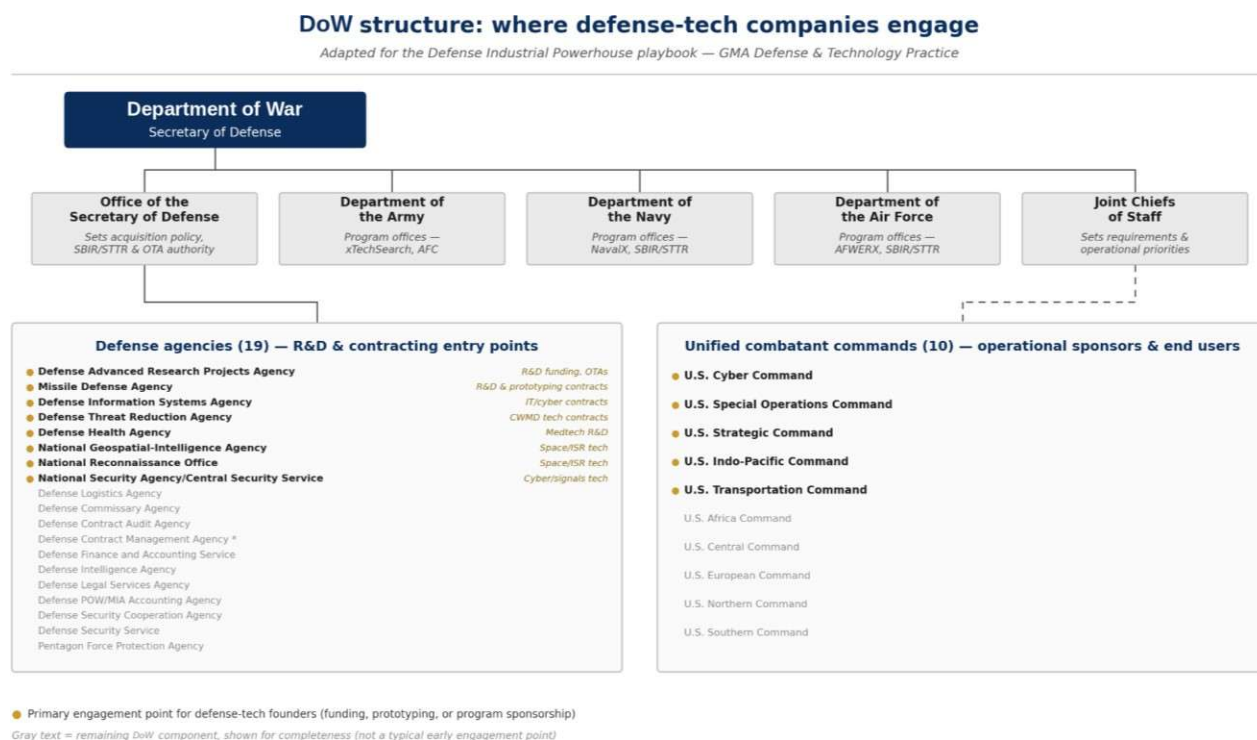
The Dept. Of War: It Doesn't Behave Like a Customer

A note on terminology: Executive Order 14347 established "Department of War" as a secondary designation for the Department of Defense. This playbook uses "DoW" (Department of War) throughout for consistency with that designation, and readers should confirm current usage in any solicitation, clause, or official correspondence they are relying on. DoW operates on structural, statutory, and cultural incentives that have no equivalent in commercial B2B:

- **Decoupled value chains.** The tactical end user who loves your product rarely controls the formal requirement, never controls the procurement line-item budget, and cannot sign the contract. These are three different people in three separate offices, often in three different buildings.
- **Asymmetric risk incentives.** Government decision makers are accountable for mission results, stewardship, oversight, schedule, and risk — not for a startup's commercial upside. A PM or PEO who backs an unproven technology bears career and program risk if it fails, misses a milestone, or triggers a congressional inquiry, with far less personal upside if it succeeds. That asymmetry is a common reason PMs lean toward incumbents, and it means companies should focus on making adoption low-risk and defensible, not just compelling.
- **Budget timelines exceed venture cycles.** The Planning, Programming, Budgeting, and Execution (PPBE) process runs on a rolling two-year forward horizon. Engage a PM in 2026 without a pre-existing budget line, and the earliest unallocated major procurement dollars typically sit in FY2028 — a full funding cycle later than a founder used to SaaS sales motions expects.

To scale into an enduring defense powerhouse, you have to stop operating as a software or hardware vendor and start operating as an institutional player within a system that runs on its own clock, its own risk logic, and its own chain of custody for money.

Exhibit 3: DoW Organizational Chart



Source: DoW

Government Traction as an Investment Signal

Institutional adoption isn't just a survival requirement anymore. It's become a leading indicator that investors price directly.

Take Shield AI. Its valuation rose roughly 140% over the twelve months following its March 2025 raise. That round closed about a month after the U.S. Air Force selected Hivemind for the Collaborative Combat Aircraft program in February 2026.

Anduril shows the same pattern over a longer timeline. Its first defense contract — a roughly \$12.5M Marine Corps award — came about a year after the company's 2017 founding. That was well before the multibillion-dollar valuations that followed.

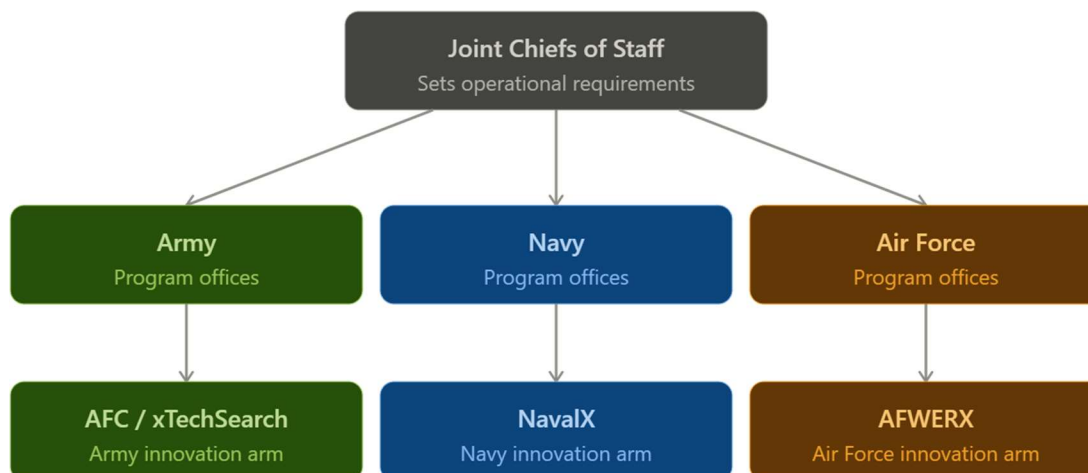
In both cases, a validated government relationship came first. The jump in private valuation came after, not the other way around.

Understanding the DoW you're selling into

The Department of War is not a single customer. It's a federation of components, each with its own budget, acquisition authority, and priorities.

Before a founder can identify a funding target or a program sponsor, it helps to understand the organization's basic structure. The Secretary of Defense sets department-wide policy and controls acquisition authorities such as SBIR/STTR and Other Transaction Agreements (OTAs). The three military departments — Army, Navy, and Air Force — operate their own program offices and innovation arms: AFC/xTechSearch, NavalX, and AFWERX, respectively. The Joint Chiefs of Staff set the operational requirements that ultimately drive what gets funded.

Exhibit 4: DoW Organizational Structure and Innovation Pathways



Source: GMA Defense & Technology Practice analysis; DoW

Defense Industrial Readiness Score: Where Does Your Company Stand?

A defense startup can have exceptional technology and still be years away from becoming a viable defense supplier.

The difference is **industrial readiness**.

Winning a defense contract requires more than demonstrating that a technology works. Companies must establish operational demand, secure program sponsorship, align with government budgets, demonstrate a credible transition pathway, build production capacity, address cybersecurity and compliance requirements, develop sustainment capabilities, establish prime-contractor relationships, and—where appropriate—prepare for international sales through Foreign Military Sales (FMS) or Direct Commercial Sales (DCS).

Exhibit 5: Industrial Readiness Assessment

Use the following score to determine where your company currently stands.

Dimension	0	1	2	3	Score
Operational Pull	None	Informal interest	Unit validation	Requirement owner	___
Technical Validation	Lab only	Limited field testing	Accredited evaluation	Independent validation	___
Program Management Sponsorship	None	Initial contact	Active PM engagement	Budget owner engaged	___
Budget Alignment	None	Potential PE	POM discussion	Funded line	___
Transition Path	None	Prototype	OTA / SBIR Phase III	Production pathway	___
Manufacturing Readiness	Prototype	Pilot production	Production capable	Surge-capable	___
Cybersecurity / Compliance	Early stage	In process	Certified / compliant	Mature	___
Sustainment	None	Conceptual	Costed / planned	Operational capability	___
Prime Strategy	None	Initial discussion	Teaming relationship	Protected workshare	___
International	None	Customer interest	Active FMS/DCS pathway	Established FMS/DCS capability	___
TOTAL					___/30

Source: GMA Defense & Technology Practice analysis

Your Readiness Level

Score: 0–10 — Technology Company

Be honest, you may have a compelling technology, but the company has not yet demonstrated sufficient evidence of defense demand, program sponsorship, transition potential, or industrial execution.

Primary objective: Create operational pull and establish a credible path into a defense program.

Focus on:

- Operator engagement
- Mission validation
- Relevant demonstrations
- Initial program-office relationships
- Acquisition pathway identification
- Early budget and requirement alignment

Score: 11–20 — Defense Entrant

The company has begun establishing a foothold in the defense market. There is evidence of customer interest, technical validation, or program engagement, but key elements of the transition and production strategy remain unresolved.

Primary objective: Convert interest into institutional commitment.

Focus on:

- Program-manager sponsorship
 - Requirement alignment
 - Budget/POM positioning
 - Formal evaluations
 - OTA, SBIR Phase III, or other transition mechanisms
 - Prime-contractor strategy
 - Production and supply-chain planning
-

Score: 21–25 — Institutionalizing

The company is moving beyond individual champions and beginning to establish the institutional infrastructure required to become a durable defense supplier.

Primary objective: Turn successful capability into a repeatable defense business.

Focus on:

- Program and budget alignment
- Production scale
- Cybersecurity and compliance
- Sustainment economics
- Contracting strategy
- Prime relationships
- Recompete positioning
- International market preparation

Score: 26–30 — Program-Ready

The company has demonstrated many of the characteristics of a mature defense supplier: validated operational demand, program sponsorship, budget alignment, a credible transition pathway, production capability, compliance readiness, sustainment planning, and a strategy for scaling through government and/or prime-contractor channels.

Primary objective: Capture, scale, and defend the position.

At this stage, the question is no longer simply: **Can the technology work?**

The question becomes:

Can the company become an indispensable part of the defense program?

Why This Score Matters

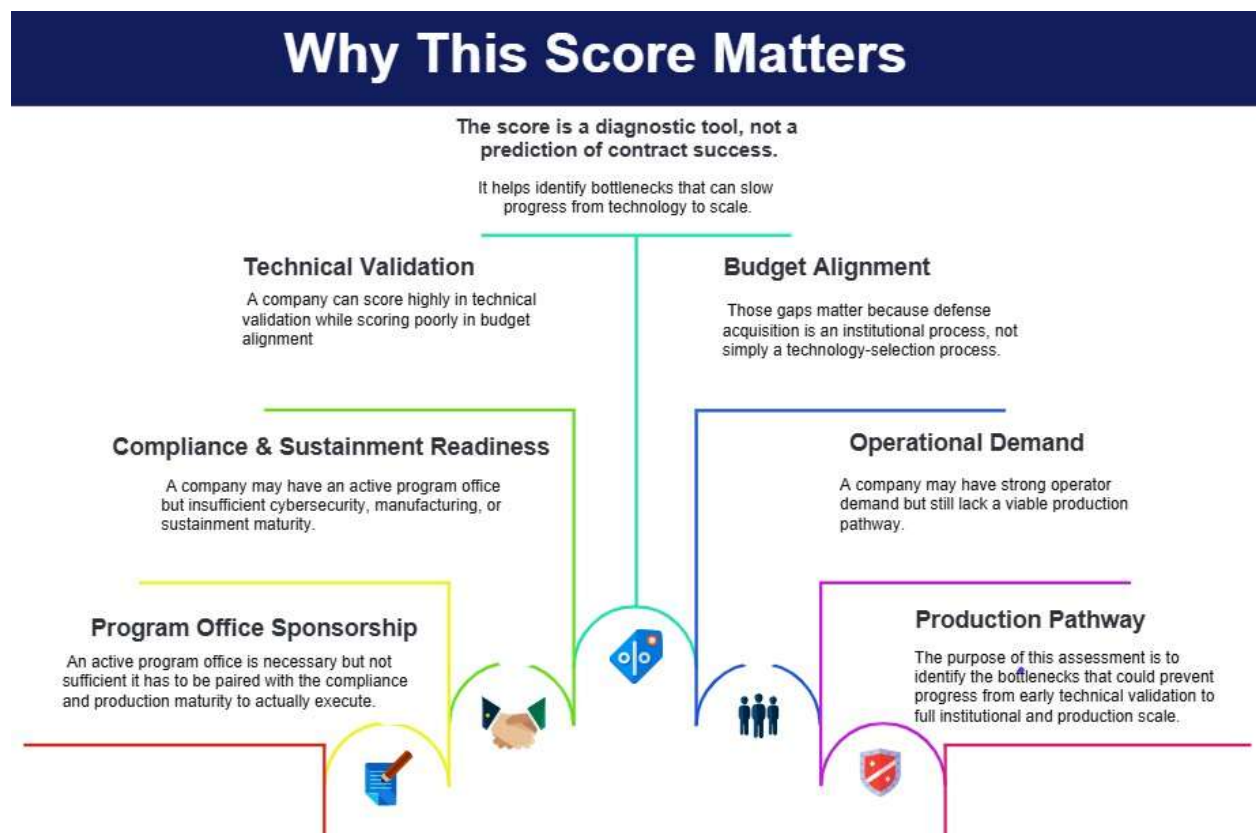
The score is not intended to predict whether a company will win a particular contract. It is a diagnostic tool.

A company can score highly in technical validation while scoring poorly in budget alignment. Another may have strong operator demand but lack a production pathway. A third may have an active program office but insufficient cybersecurity, manufacturing, or sustainment maturity.

Those gaps matter because **defense acquisition is an institutional process, not simply a technology-selection process.**

The purpose of this assessment is therefore to identify the bottlenecks that could prevent a promising company from early technical validation to full institutional and production scale,.

Exhibit 6: Why Scoring Matters



Source: GMA Defense & Technology Practice

The detailed Founder Self-Assessment Scorecard in **Appendix A** expands on these dimensions and provides a more comprehensive assessment of the company's defense readiness.

Defense Industrial Readiness CEO Dashboard

The following is a management tool for transitioning from technical validation to institutional adoption and for identifying areas of strength and weakness.

How to Use the Dashboard

Green = the company has credible evidence and an active institutional pathway.

Yellow = the pathway exists but is incomplete, informal, or dependent on a next decision.

Red = the pathway is absent, undefined, or unsupported by evidence.

Management rule: do not average the colors. A single RED in a critical adoption dimension can block progress even when the overall technology is strong.

CEO objective: move RED items to YELLOW and YELLOW items to GREEN. Each RED should have an owner, next action, and target date.

Recommended CEO Review Format

Exhibit 7: Executive Dashboard

RED / YELLOW ISSUE	OWNER	NEXT DECISION / ACTION	TARGET DATE

Scaled Production	Can we deliver at scale and become a lasting defense industrial company?
-------------------	--

Source: GMA Defense & Technology Practice analysis

Finding the Right Engagement Points

Beyond the military services and the Office of the Secretary of Defense, DoW includes 19 defense agencies — many of which run their own R&D and contracting programs, but only a handful are realistic entry points for an early-stage defense company.

The eight highlighted here (DARPA, MDA, DISA, DTRA, DHA, NGA, NRO, and NSA/CSS) actively fund prototyping, R&D, or technology contracts in areas such as OTAs, ISR, cyber, and CWMD — making them the agencies to prioritize for outreach.

The remaining eleven are largely administrative, logistical, or oversight functions (finance, audit, legal, POW/MIA accounting, force protection, etc.) with little to no direct R&D or contracting pathway for a startup. They are shown grayed out for completeness rather than as active targets.

Exhibit 8: DoW Engagement Points

Defense agencies (19) — R&D & contracting entry points	
● Defense Advanced Research Projects Agency	<i>R&D funding, OTAs</i>
● Missile Defense Agency	<i>R&D & prototyping contracts</i>
● Defense Information Systems Agency	<i>IT/cyber contracts</i>
● Defense Threat Reduction Agency	<i>CWMD tech contracts</i>
● Defense Health Agency	<i>Medtech R&D</i>
● National Geospatial-Intelligence Agency	<i>Space/ISR tech</i>
● National Reconnaissance Office	<i>Space/ISR tech</i>
● National Security Agency/Central Security Service	<i>Cyber/signals tech</i>
Defense Logistics Agency	
Defense Commissary Agency	
Defense Contract Audit Agency	
Defense Contract Management Agency *	
Defense Finance and Accounting Service	
Defense Intelligence Agency	
Defense Legal Services Agency	
Defense POW/MIA Accounting Agency	
Defense Security Cooperation Agency	
Defense Security Service	
Pentagon Force Protection Agency	

Source: Department of War

Exhibit 8b: DoW Engagement Pathways for Defense-Tech Founders

DoW Engagement Pathways for Defense-Tech Founders

Three parallel channels: direct R&D contracting, no-cost procurement assistance, and non-dilutive SBIR/STTR funding

Major R&D and Contracting

- **Defense Advanced Research Projects Agency**
R&D funding, OTAs
- **Missile Defense Agency**
R&D & prototyping contracts
- **Defense Information Systems Agency**
IT/cyber contracts
- **Defense Threat Reduction Agency**
CWMD tech contracts
- **Defense Health Agency**
Medtech R&D
- **National Geospatial-Intelligence Agency**
Space/ISR tech
- **National Reconnaissance Office**
Space/ISR tech
- **NSA/Central Security Service**
Cyber/signals tech

Primary engagement point

Direct R&D funding, prototyping contracts, and Other Transaction Agreements (OTAs) — the fastest path to a named program sponsor for early-stage defense tech.

APEX Accelerators

- **90+ APEX Accelerators nationwide**
formerly PTACs, est. 1986
- **Free procurement technical assistance**
SAM.gov registration, bid matching
- **Capability statements & teaming support**
one-on-one advisors
- **Administered under DoW's Office of**
Small Business Programs

Start here

apexaccelerators.us — find your regional Accelerator. A login.gov account lets you begin SAM.gov registration before you have a company profile.

SBIR/STTR

- **DoW components**
Army, Navy, Air Force, Space Force
- **DoW components (cont.)**
DARPA, MDA, DHA, DLA, DTRA, OSD, SOCOM
- **NASA**
a full participating agency — not just defense-adjacent
- **11 federal agencies government-wide**
incl. NSF, DOE, HHS, DHS, USDA, DOT

Check open topics

sbir.gov and each agency's own portal. DoW runs three BAA cycles/year (26.1/26.2/26.3); program reauthorized through FY2031.

Source: GMA analysis; apexaccelerators.us; sbir.gov; DoW component SBIR/STTR data (as of Sept. 2026)

These three channels aren't mutually exclusive — a well-run capture strategy runs them in parallel: APEX Accelerators for registration and paperwork, SBIR/STTR for non-dilutive early funding, and direct engagement with the agencies above once a capability is ready to demonstrate.

Where founders get in the door

DoW's 44 components, or the 19 specific agencies, aren't equally relevant to an early-stage company. Most defense agencies handle logistics, finance, legal, or administrative functions with no direct role in funding or contracting with tech startups. A much smaller set — DARPA, the Missile Defense Agency, DISA, DTRA, DHA, NSA, NRO, and NSA/CSS — are where non-dilutive R&D dollars, prototyping contracts, and OTAs actually originate.

On the operational side, the unified combatant commands are the end users who sponsor programs and pull technology through the pipeline once it's proven; Cyber Command, SOCOM, STRATCOM, INDOPACOM, and TRANSCOM see the most defense-tech relevance given their cyber, rapid-prototyping, space, ISR, and logistics missions. The figure below highlights these entry points against the full DoW structure, so founders can prioritize outreach rather than treating all 44 components as equally reachable.

Exhibit 9: Priority Combatant Commands for Defense Tech

Unified combatant commands (10) — operational sponsors & end users

- **U.S. Cyber Command**
Cyber/software
 - **U.S. Special Operations Command**
Rapid prototyping
 - **U.S. Strategic Command**
Space/nuclear deterrence
 - **U.S. Indo-Pacific Command**
Maritime/ISR
 - **U.S. Transportation Command**
Logistics tech
- U.S. Africa Command
U.S. Central Command
U.S. European Command
U.S. Northern Command
U.S. Southern Command

Source: Department of War

The practical implication for a founder: the capture discipline in Parts 1–3 isn't just about revenue. A lab evaluation, a directed-requirement win, or a Program of Record line item serves as the same evidence that a later funding round is priced against — which is exactly why the PM Risk Matrix and the Five Levels model below should be treated as investor-facing artifacts, not just internal BD tools.

Rules of Engagement

This playbook supports legitimate market research, capability education, requirements understanding, and lawful capture planning. It does not endorse, and should never be read as endorsing, improper influence, preferential access to non-public information, or attempts to bypass full and open competition.

Boundaries every client should observe

- Procurement Integrity Act — do not seek or use contractor bid or proposal information or source-selection information outside authorized channels, before or during an active procurement.
- Organizational conflicts of interest — disclose and manage relationships (including with former government personnel) that could bias, or appear to bias, a competition.
- Post-government-employment restrictions — confirm any former official's cooling-off period, recusal obligations, and lifetime/2-year/1-year restrictions before involving them in a specific pursuit.
- Lobbying disclosure and ethics rules — register and report as required under the Lobbying Disclosure Act and comply with agency-specific gift, gratuity, and ethics rules for any Hill or agency engagement.
- Nonpublic information — never seek or use nonpublic source-selection, budget, proprietary, classified, or controlled technical information obtained outside proper channels.
- Truthfulness in marketing and proposals — every claim about capability, past performance, funding, credentials, or a relationship with GMA, a partner, or a government official must be accurate and substantiated.
- Engagement during active procurements — follow the specific communication rules in effect once a solicitation is issued; industry days, RFIs, and technical interchange meetings each have their own boundaries.

What GMA does and does not do

GMA supports opportunity intelligence, capture planning, funding-pathway analysis, proposal development, strategic messaging, and coordination with qualified legal, government-affairs, cybersecurity, accounting, export-control, and security-cleared specialists where a client's needs require specialized, licensed, or credentialed advice. GMA does not provide legal advice on FAR/DFARS, data rights, or export controls; does not conduct lobbying on a client's behalf; and does not negotiate contracts. Where this playbook describes GMA "helping" with any of those activities, that help consists of preparation, coordination, and introductions to qualified specialists — not the specialized service itself.

A note on titles and authorities: named roles and authorities in this playbook (PM, PEO, technical evaluator, resource sponsor, and others) vary by military department, defense agency,

command, program, appropriation, and acquisition pathway. Companies should map the actual requirement owner, technical authority, program office, contracting activity, resource sponsor, and end user for each specific pursuit rather than assuming a fixed org chart applies everywhere.

Government Engagement Matrix

A quick reference for which channel fits which situation. This is a starting point — the specific rules in effect for any active procurement always govern.

Channel	When it fits	Practical note
RFI responses	Market research phase, before a formal solicitation	Standard, low-risk channel
Industry days	Pre-solicitation, agency-hosted	Good for visibility; not a substitute for a named sponsor
Technical interchange meetings	Once a technical dialogue is warranted	Confirm scope/ground rules with the government POC first
White papers	Unsolicited or in response to a BAA/CSO	Useful for shaping understanding; not a commitment vehicle
BAAs (Broad Agency Announcements)	Standing solicitations, rolling submission	Read the specific BAA's rules — they vary by agency
Unsolicited proposals	No solicitation fits; genuinely novel capability	Follow FAR 15.6 procedures; agencies scrutinize these closely
SBIR/STTR	Early-stage, non-dilutive funding pathway	See Exhibit 8b for the current agency landscape
CSOs	Commercial-item and dual-use technology	Typically faster than traditional BAAs
OTA consortia	Established or attainable consortium membership	Diligence membership fees/rules before joining
Subcontractor discussions	Teaming with a prime already on a program	Confirm no exclusivity conflicts with parallel pursuits
Demonstration invitations	Capability mature enough to show, not just describe	Treat as evidence-generating, not a sales pitch
Congressional briefings	Educating staff on capability/industrial-base relevance	Governed by the Rules of Engagement above — lobbying/ethics rules apply



Part 1: Year One — Building the Defense Capture Engine

“The primary cause of failure past Series B/C is institutional adoption failure, not technical failure.”

1.1 What the PM Actually Evaluates

Founders often pitch Defense PMs the way they'd pitch an investor or VC — revolutionary technical performance, disruptive cost structure, and novel architecture. It's an understandable habit: that's the pitch that raised the last funding round, and leading with leap-ahead capability is what founders are trained to do. But a PM isn't scoring a fundraising narrative. A PM is asking whether adopting this capability will blow up their budget, slip their schedule, or fail an audit — none of which has anything to do with how impressive the technology sounds in a deck.

This often fails because it answers questions the PM isn't asking. The founder walks in ready to defend the technology, while the PM silently runs through a checklist of program-management risks the pitch never addresses. Recognizing which list you're actually being scored against — and building the pitch around it, not the specs — is usually the difference between a second meeting and a polite pass.

Exhibit 10: What Founders Pitch vs. What PMs Actually Evaluate

What the Startup Thinks It's Selling	What the PM Is Actually Evaluating
Breakthrough AI performance	Cyber accreditation & ATO status
Higher sensor resolution	Sustainment & logistics burden
Novel platform design	Integration burden with existing systems
Lower unit cost	Budget line-item stability
Flashy operational demo	Repeatability & reliability

Source: GMA Defense & Technology Practice

The PM Risk Matrix. The five items on the right side of Exhibit 9 aren't a random list — they cluster into four formal risk categories that most program offices already score against internally, whether or not the PM ever says the word "risk" out loud in the meeting.

Knowing which category a question falls into matters because it tells you what kind of evidence actually answers it: a technical risk question wants test data, a company risk question wants a cap table and a runway number, and no amount of the other kind will substitute. In other words, be specific and answer the questions. Every new technology gets assessed across four risk categories:

Exhibit 11: Understanding Risk

Risk Category	Core Question
Technical Risk	Does it work under field stress — EW jamming, extreme thermal conditions, saltwater corrosion?
Integration Risk	How much work is required to connect this to existing C2 systems, power architectures, legacy radios, and logistics software?

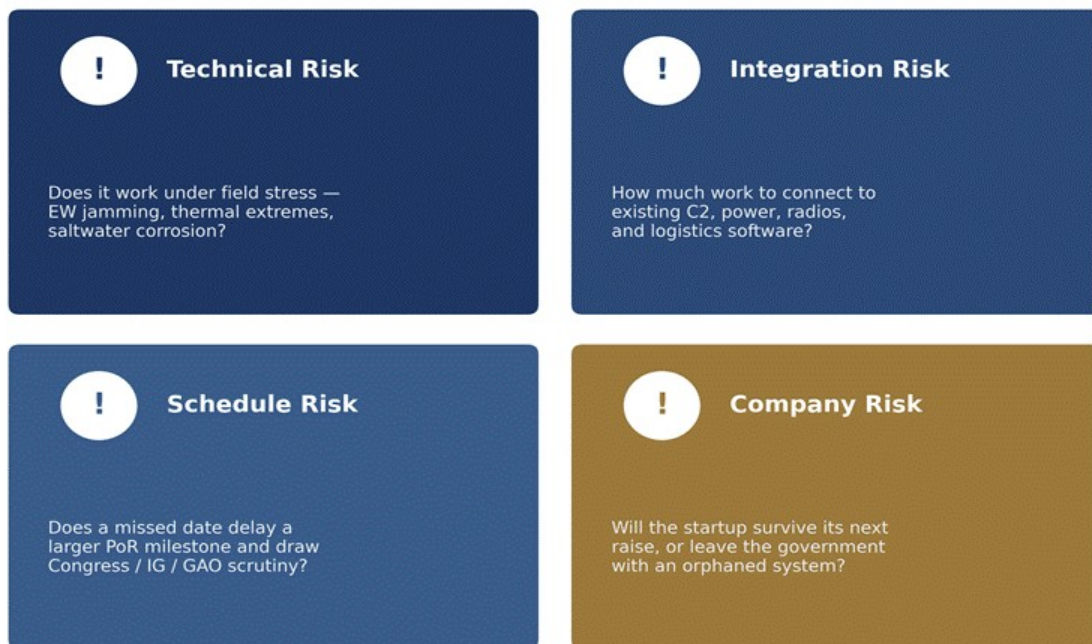
Risk Category	Core Question
Schedule Risk	If this vendor misses a delivery date, does it delay a larger Program of Record milestone and draw scrutiny from Congress, the IG, or GAO?
Company Risk	Will this startup survive its next venture round, or will it run out of runway, pivot, or get acquired — leaving the government with an orphaned system?

Source: GMA Defense & Technology Practice analysis

A pitch built around "breakthrough performance" answers none of these. A pitch built around the ATO pathway, integration burden, and company runway answers all four, and that's the pitch that gets a second meeting.

Exhibit 12: The Four Risk Questions

The PM Risk Matrix



These are the four questions behind every PM go/no-go decision, regardless of domain.

Source: (NIST) Cybersecurity Framework (CSF)

A Fuller Picture: Expanded PM/Acquisition Evaluation Framework

The four questions above are the core of it, but a program office's actual evaluation runs wider — across technical, schedule, cost, manufacturing, integration, cybersecurity, test, safety, sustainment, supply-chain, and data-rights posture. The table below is the fuller version worth preparing for once a pursuit moves past an initial conversation.

PM/acquisition question	Evidence an industry team should bring
What operational problem does this solve?	Mission thread, use case, user feedback, quantified mission outcome
Is the capability mature enough?	TRL evidence, test reports, prototype performance, configuration baseline
Can it integrate?	Interface-control info, architecture alignment, interoperability plan, cyber/data-flow diagrams
Can it be tested and certified?	Test strategy, dev/operational test plan, safety approach, spectrum plan where relevant
Can it be secured?	CUI handling plan, NIST SP 800-171 posture, DFARS compliance, ATO/RMF strategy if applicable
Can it be manufactured and delivered?	MRL assessment, bill of materials, capacity plan, quality controls, supplier qualifications
Can it be sustained?	Reliability data, repair concept, training, technical data, spares, lifecycle cost
Can the government afford it?	Cost model, unit economics, production curve, affordability analysis, funding profile
Can the program execute it?	Acquisition strategy, contracting vehicle, schedule realism, staffing, subcontractor roles
Who owns and can use the data?	IP/data-rights strategy, license assumptions, software documentation, escrow/continuity
Will the company remain viable?	Corporate stability plan, supply-chain stability, financing/runway, key-person continuity

1.2 The Five Levels of Government Advocates

Build advocates from the bottom up, not the top down. Passing the risk-matrix test earns a second meeting — it doesn't win a program. Most founders assume that meeting is the hard part and that a strong technical answer to the four questions in Exhibit 11 is what finally gets things moving. It isn't, or at least it isn't sufficient on its own.

Getting from a good meeting to a funded line item requires a chain of internal champions who carry the capability upward through five distinct levels of government decision-making — each with its own incentives, its own definition of proof, and its own ability to say no. Skip a level or try to compress the sequence, and the campaign stalls, no matter how strong the case was at the level you did reach.

Exhibit 13: The Five Levels of Government Advocates

The Five Levels of Government Advocates



Operational pull and technical validation are often essential; senior engagement can accelerate an evidence-based effort but rarely substitutes for requirements, funding, technical authority, and acquisition ownership.

Source: GMA Defense & Technology Practice analysis

The pyramid shows the direction advocacy has to move up. The table below breaks out who sits at each level and what each one needs from you before they'll carry the capability up to the next level.

Exhibit 14: L1–L5 Advocate Roles and Responsibilities

Level	Who	Role
L1 — Operational Users	Warrant officers, NCOs, squad leaders, pilots, intelligence analysts	Establish the operational gap; create tactical pull

Level	Who	Role
L2 — Technical Evaluators	Engineers at Warfare Centers (NSWC Crane, NAVAIR Pax River) and labs (AFRL, DEVCOM)	Verify environmental durability, software architecture, cybersecurity
L3 — Program Managers	O-5/O-6 officers or GS-15 civilians running specific Programs of Record	Own schedule, integration budget, contract execution
L4 — Program Executive Officers	General officers or SES leaders overseeing portfolios of related programs	Reallocate resources across program lines
L5 — Senior Defense Leadership	Service Secretaries, COCOM Commanders, Undersecretaries	Set strategic direction, signal priorities to Congress

Source: GMA Defense & Technology Practice analysis; DoW

Knowing the five roles isn't the same as using them in the right order. Most companies that stall here can name every level and still start the campaign at L5 — the comparison below shows what that mistake looks like next to the sequence that actually converts.

Exhibit 15: Failure Pattern vs. Successful Pattern

✗ Failure Pattern	✓ Successful Pattern
Start at L5 with "top-down direction"	Start at L1 to build tactical pull
Blocked by mid-level PMs — no owner, budget, or ATO	Validated by L2 technical evaluators
Verbal interest never converts to funded action	L3 PMs adopt the bottom-up demand; L4/L5 reallocate resources to match

Source: GMA Defense & Technology Practice analysis; industry case studies

A four-star's enthusiasm is real and worth having — but it is not a substitute for a PM who owns a budget line. Treat senior-leader interest as an accelerant for a bottom-up campaign already in motion, not as the campaign itself.

Reading the Strategy Documents for Acquisition Clues

The five levels aren't the only source of signal. Published service strategies — along with the National Defense Strategy and National Security Strategy — telegraph what each service is actually required to buy or fund in the coming years. These documents are written well before the budget lines that fund them appear, which makes them a leading indicator rather than a lagging one: a founder who reads the current service strategy alongside the NDS/NSS can often identify which capability gaps are about to become funded priorities before a formal RFI ever appears.

The five levels also don't have to be worked strictly one at a time. In practice, L1–L5 engagement can happen concurrently rather than as a strict handoff — a company can be building L1 operator pull and briefing an L4 PEO in the same quarter. What has to stay sequenced is the validation logic: an L4 or L5 champion still needs to be able to point to real L1/L2 evidence when asked, even if the conversations themselves overlapped in time.

1.3 Case Studies

Anduril Industries — "Build First, Demonstrate, Scale." Rather than waiting for a formal JCIDS requirement, Anduril used private venture capital to build fully functional systems (Lattice OS, Sentry Towers, Anvil) before seeking government funding, then demonstrated working hardware to L1 operators and L2 evaluators in field trials — forcing L3 PMs to respond to bottom-up operational demand instead of the reverse. The company later invested in dedicated production capacity (its Arsenal-1 facility, announced in 2024, well after the initial build-first phase) to prove it could deliver at volume.

Lesson: private capital lets you move down the risk curve faster than a traditional contractor. Showing up with working hardware compresses the evaluation cycle — but the manufacturing investment came after operational pull was established, not before.

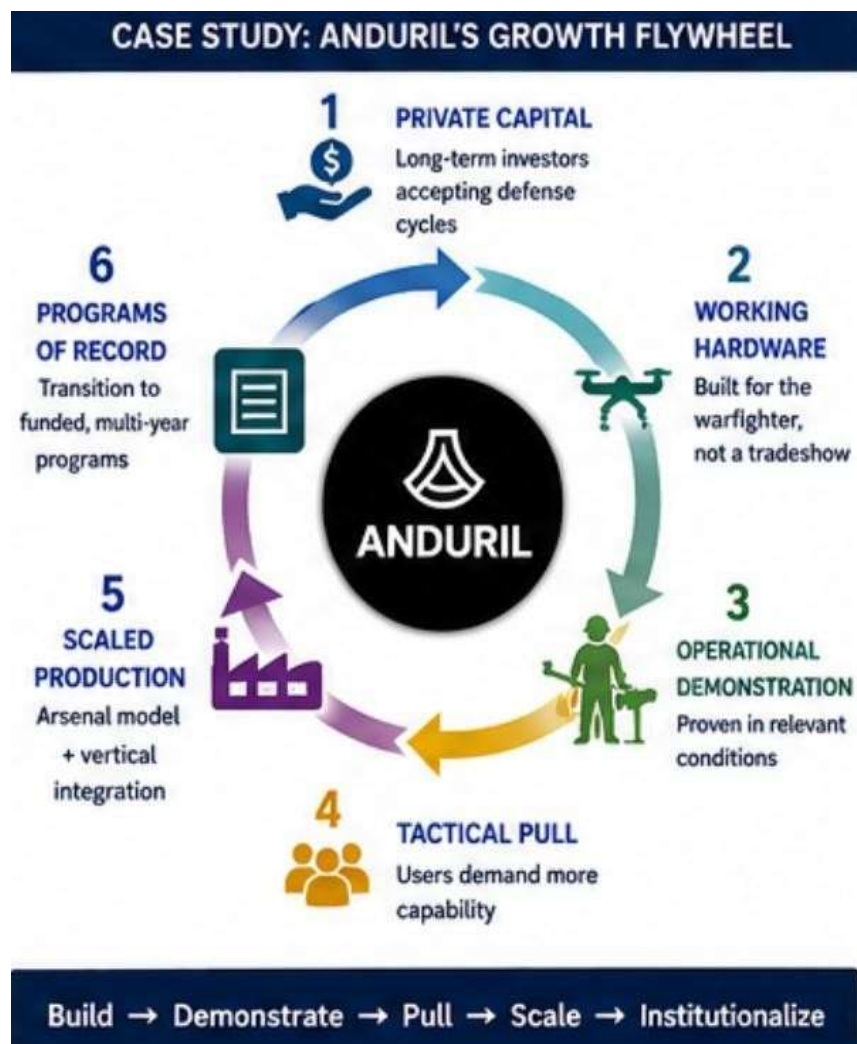
Exhibit 16: Anduril Ghost — Field-Deployed



Source: Anduril Industries, Ghost UAS product materials

The photo above is the outcome. The flywheel below is the mechanism that produced it — six repeatable stages Anduril cycles through with every new platform.

Exhibit 17: Anduril's Growth Flywheel



Source: GMA analysis of Anduril's publicly reported growth strategy and press coverage

Shield AI — "Platform-Agnostic Autonomy." Shield AI built its Hivemind autonomy stack and proved it first on its own aircraft (V-BAT, now a Shield AI product following the 2021 acquisition of V-BAT's original maker), then deliberately abstracted the software layer so it could run on platforms it doesn't own — including flying an X-62A VISTA (a modified F-16) through AI-driven dogfighting maneuvers under DARPA's ACE program, integrating onto General Atomics' MQ-20 Avenger, and being selected for the Air Force's Collaborative Combat Aircraft mission autonomy production contract.



Exhibit 18: Shield AI Autonomous Aircraft

Source: Shield AI public disclosures and press materials

Lesson: owning one flagship hardware platform to prove the software, then deliberately decoupling the software from that hardware, gives you two paths to revenue — prime vendor and essential subsystem provider to other primes' airframes.



Illustrative pattern — the "Perpetual Pilot" trap. A recurring, non-company-specific pattern across the sUAS space: a startup raises significant capital and secures a string of Phase I/II SBIRs, DIU prototype contracts, and tactical unit trials — real wins, each on its own. But leadership treats each as a sales win rather than a stepping stone, never engaging L3 PMs to align a pilot with a specific Program Element for multi-year procurement. When prototype funding expires, there's no transition budget in a Program of Record, and the company burns capital during the resulting procurement gap.

Lesson: prototype wins and pilot projects can mask the absence of a long-term strategy. Unless a pilot is explicitly tied to an acquisition pathway and a budget line item, it's a dead end, regardless of how many logos are on the win list.

Exhibit 19: Year One Do's and Don'ts

DO'S AND DON'TS FOR YEAR ONE	
DO	DON'T
● Target your outreach at Level 1 operators and Level 2 laboratory evaluators to build technical credibility before requesting major procurement meetings. ● Ask for technical exchanges rather than marketing meetings: "We want 30 minutes with your lead engineer to review our environmental testing data." ● Map out integration burden for software and hardware before your first meeting with a PM. ● Use SBIR contracts strategically as mechanisms to unlock sole-source Phase III follow-on contracts.	■ Don't rely on executive intros (e.g., retired Four-Stars) to force adoption without building support at Levels 1, 2 and 3 first. ■ Don't send generic pitches focusing on high-level concepts like "revolutionary AI" or "battlefield transformation." ■ Don't treat prototypes as business models; OTAs and SBIRs are short-term bridges to Programs of Record. ■ Don't give unrealistic demos under overly controlled conditions that do not reflect operational reality.

Source: GMA Defense & Technology Practice

1.4 Tactical Outreach & Capture Engine Structure

Anatomy of a high-conversion outreach:

1. Specific mission context — acknowledge the PM's exact operational gap, ongoing modernization effort, or public RFI focus area.
2. Empirical evidence & test data — flight hours, field trials, environmental stress tests, hardware-in-the-loop results.

3. Architectural integration & ATO pathway — compatibility with MOSA/CMOSS/OMS and current cyber accreditation status.
4. Risk reduction & transition readiness — production capacity, supply chain maturity, existing contract vehicles.
5. Low -friction, discovery-oriented ask — a targeted technical exchange with engineering leads, not a marketing pitch.

Exhibit 20: Cold Outreach Email Template

SUBJECT: Technical Exchange: Integration Data for [Specific Operational Gap]

PM [Last Name],

We've been tracking your office's work on [specific operational challenge — e.g., persistent tactical sensing in contested C2 environments].

Over the past [N] months, our [Capability Name] platform completed field evaluations with [Unit/Laboratory], demonstrating:

- Operational performance: [specific metric — e.g., 14 hours continuous untethered operation in GPS-denied, jammed environments]*
- Integration maturity: native compliance with [MOSA/SOSA/OMS], requiring zero proprietary interface modifications to existing C2 software*
- Cybersecurity & compliance: currently executing our ATO pathway under [NIST SP 800-171 / CMMC Level 2]*

We're not seeking an immediate procurement decision. We'd like 30 minutes with your engineering team to review our environmental test data, discuss integration lessons learned, and evaluate whether our performance envelope aligns with your upcoming modernization milestones.

Respectfully,

[Founder/Capture Lead Name] · [Title, Company] · [Phone] | [Website]

Source: GMA Defense & Technology Practice

A single well-crafted email won't sustain a capture campaign on its own — someone has to own the follow-through, translate the response into next steps, and navigate the contract mechanics that follow a PM's yes. The roles below show what that ownership actually looks like.

Exhibit 21: Structuring the Year One capture engine

Role	Description
Capture Director (Opportunity Owner)	Owns opportunities from discovery through award. Tracks budget line items, RFIs/RFPs, competitor positioning.
Former Operators (Mission Translators)	Tactical veterans (O-4/O-5 or senior NCO) with direct L1 access. Translate engineering capability into mission outcomes.
Acquisition Specialists (Pathway Navigators)	Former Contracting Officers or PEO staff. Own contract vehicles, SOWs, FAR compliance, transition funding.

Source: GMA Defense & Technology Practice analysis

A capture team structured this way still needs a budget that matches its job — a Capture Director tracking competitor positioning and an Acquisition Specialist navigating contract vehicles can't do much if the spend behind them is still allocated like a commercial launch. The comparison below shows where that mismatch typically shows up.

Exhibit 22: Year One Budget Allocation: Typical vs. Recommended

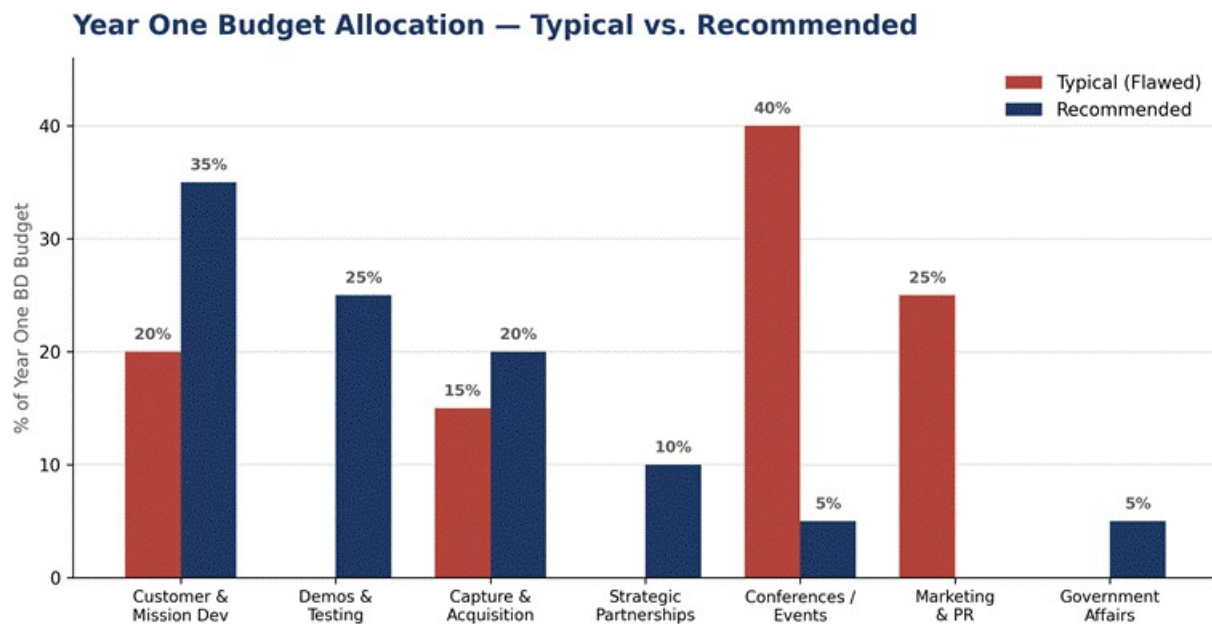
Category	Typical (Flawed)	Recommended
Customer & Mission Development	20%	35%
Demonstrations & Testing	0%	25%
Capture & Acquisition Strategy	15%	20%
Strategic Partnerships	0%	10%

Category	Typical (Flawed)	Recommended
Conferences / Events	40%	5%
Marketing & PR	25%	0%
Government Affairs	0%	5%

Source: GMA Defense & Technology Practice analysis

The “typical” column isn't hypothetical — it's the default a commercial-tech playbook produces when applied unmodified to defense: two-thirds of spend goes to visibility (conferences, marketing) and almost nothing to the demonstrations and testing that actually retire a PM's risk.

Exhibit 23: Budget Allocation Comparison Chart



The typical allocation over-indexes on visibility; the recommended allocation over-indexes on risk retirement.

Source: GMA Defense & Technology Practice analysis

Tool: The 90-Day Capture Plan

A one-page working document per active pursuit — meant to be revisited and updated as the pursuit develops, not written once and filed. A pursuit with unfilled fields at the 90-day mark is a candidate to drop.

Field	What goes here
Target organization	The specific office/program — not just “the Army” or “DARPA.”
Mission problem	The operational gap this pursuit addresses, in the sponsor’s own language.
Named stakeholders	Requirement owner, technical evaluator, program office contact, resource sponsor.
Required evidence	What has to be demonstrated or documented before the next decision point.
Validation activity	The specific test, demo, or lab evaluation planned to produce that evidence.
Budget hypothesis	Best current guess at funding source and rough size — flagged as hypothesis.
Contract pathway	SBIR/STTR, OTA, direct award, subcontract — and why it fits this pursuit.
Partnership strategy	Teaming, prime relationships, or subcontracting plan, if any.
Next decision point	The specific date or milestone at which this pursuit gets re-evaluated.
Client owner	Who is accountable for the pursuit moving forward.
Go/no-go criteria	The explicit conditions under which this pursuit continues, pauses, or is dropped.

1.5 Technology Readiness Levels — What TRL Actually Measures

TRL is the government's shorthand for "how proven is this technology," on a 1–9 scale originally developed by NASA and formally adopted DoW-wide under DoDI 5000.85. Nearly every solicitation a founder encounters — SBIR topics, OTA prototype announcements, BAAs — specifies a minimum TRL for eligibility, and PMs use it as a quick filter before agreeing to a meeting. Get the framing wrong, and you're either screened out for being "too early" or caught overstating maturity you can't defend in a technical exchange.

Exhibit 24: The Nine Technology Readiness Levels

TRL	Definition	What Exists
1	Basic principles observed	Scientific research findings, no application defined
2	Technology concept formulated	Applied research begins; theoretical, no proof
3	Analytical/experimental proof of concept	Lab-scale validation of critical function
4	Component validation in lab environment	Breadboard-level integration, controlled conditions
5	Component validation in relevant environment	Higher-fidelity integration, still not operational conditions
6	System/subsystem model demonstrated in relevant environment	Prototype tested near operational conditions
7	System prototype demonstrated in operational environment	Field or operational demo, not yet production-representative
8	Actual system completed and qualified through test and demonstration	Fully integrated, tested under real conditions
9	Actual system proven through successful mission operations	Deployed and operating in the field

Source: NASA, Technology Readiness Level Definitions, NPR 7123.1C (NASA Systems Engineering Processes and Requirements)

Why TRL Alone Doesn't Get You Funded

TRL indicates whether the technology works. It says nothing about whether the government can buy, integrate, sustain, or afford it — which is exactly the institutional-adoption gap this playbook is built around (Exhibit 1). A TRL 6 system with no accreditation pathway, no manufacturing plan, and no PM sponsor is no more fundable than a TRL 4 system with all three. PMs increasingly triangulate TRL against two companion scales:

- MRL (Manufacturing Readiness Level) — can this be produced at the volume and cost the program needs, not just built once in a lab?
- IRL (Integration Readiness Level) — does this actually interoperate with the C2 systems, radios, and power architecture it has to plug into?

A startup that can only speak to TRL, and goes blank when a PM asks about MRL or IRL, reads as unready regardless of the TRL number on the slide.

What's Required to Advance a Level, Not Just Claim One

Self-declared TRL is treated skeptically — PMs and lab evaluators have seen too many pitch decks round up. What actually moves a technology from one level to the next, and what a founder needs to have on hand to defend the claim in a technical exchange:

Exhibit 25: What It Takes to Advance a TRL Level

To Credibly Claim...	You Need
TRL 4–5	Documented test data from an accredited Service Laboratory or Warfare Center not internal-only testing (this is the same bar Appendix A's Stage 1 checklist sets)
TRL 6	A demonstration run under stressed, relevant-environment conditions (EW jamming, thermal extremes, contested C2 see 2.3's Risk-Reduction Demo Design), with failure-mode data captured, not just best-case results
TRL 7	Active-duty operators not your own engineers executing the system integrated into an operational network, per the "Operator control" standard in 2.3
TRL 8–9	Demonstrated production capacity (not modeled), sustainment priced into the contract structure, and NDAA-compliant sub-tier components mapped at the part level the same criteria as Appendix A's Stage 4 checklist

Source: NASA, [Technology Readiness Assessment Best Practices Guide \(NASA SP-20205003605\)](#); industry sources

Exhibit 26: Do's and Don'ts related to TRLs

DO'S AND DON'TS	
DO	DON'T
● Anchor every TRL claim to a named lab, unit, or evaluator who can independently verify it if a PM asks ● Pair every TRL statement with the corresponding MRL and IRL status PMs will ask for both once they trust the TRL number. ● Treat TRL advancement as both an engineering and fundraising/capture milestone, evidence of PM risk reduction and investor confidence.	■ Round up. Claiming TRL 6 for a demo that ran in a controlled lab environment is the single fastest way to lose credibility with an L2 technical evaluator. ■ Treat TRL as a solo metric. A high TRL with no MRL or IRL story invites the same "orphaned system" concern the PM Risk Matrix's Company Risk category is built to catch. ■ Wait until a solicitation's minimum-TRL requirement is the first time you're forced to articulate where you actually stand , use Appendix A's stage checklist to self-assess continuously.

Source: GMA Defense & Technology Practice

Resource Spotlight: APEX Accelerators

Founders navigating their first SBIR/STTR awards or DoW contracts don't have to do it alone — and shouldn't pay for help they can get for free. APEX Accelerators (apexaccelerators.us) is a federally funded network of 90+ regional offices established by Congress in 1985 (originally as Procurement Technical Assistance Centers) to help small businesses compete for federal, state, and local government contracts.

Funded through cooperative agreements with the Department of War and local host institutions (universities, economic development offices), APEX Accelerators provide free one-on-one counseling, bid-match services, proposal review, and help securing certifications (SDVOSB, 8(a), HUBZone). In 2023 alone, the network helped over 56,000 clients win more than \$24 billion in government contracts and subcontracts.

How GMA Helps — Part 1: Year One

Building L1–L3 advocacy and running a disciplined SBIR strategy only pays off if the paperwork behind it is airtight. GMA's Defense Technology team includes a retired Major General who has overseen portfolios up to \$17B, a national DoW grant strategist who has led \$230M+ in initiatives, and numerous veteran officers with direct operational leadership experience and the right connections.

That bench does three things for founders: decodes agency-specific eligibility rules and review criteria before the first outreach email goes out; translates an L1 operator's-eye view of a capability gap into language a PM will actually credit; and manages the SAM.gov, Grants.gov, and DIBBS registration and compliance work — so a strong L1/L2 relationship doesn't stall on an administrative disqualification.

We also help convert Phase I/II SBIR activity into a deliberate foot-in-the-door, not an accidental one. Our bench includes Sonia Vohnout, a nationally recognized SBIR/STTR and commercialization strategist with 30+ years of experience and a systems-engineering background, who has helped clients secure funding across DoW, Navy, DARPA, NASA, and NIH. We pair that expertise with Technical and Business Assistance (TABAs)-style commercialization coaching, so a Phase I award comes with an actual transition plan — not just a check.



Part 2: Year Two — Converting Relationships Into Programs

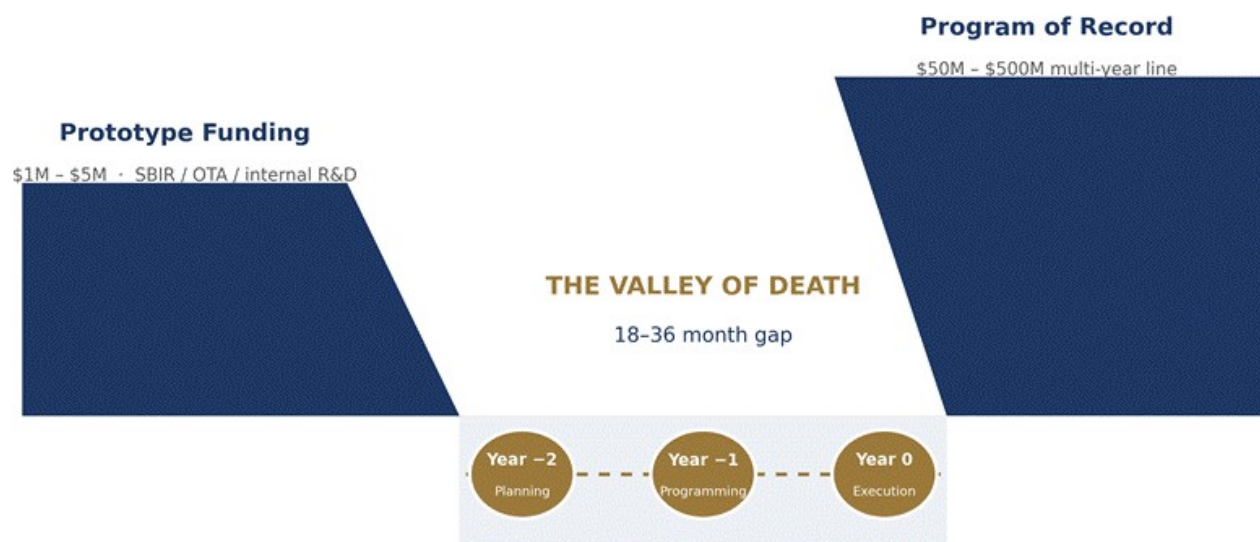
“For enduring program funding, PPBE/POM alignment is usually central — but bridge funding, reprogramming, production OTs, and Phase III awards can also help.”

2.1 Crossing the Valley of Death

The "Valley of Death" is the 18–36 month lag between a successful prototype demonstration (funded via SBIR, OTA, or internal R&D) and a dedicated, multi-year budget line item in a Program of Record.

Exhibit 27: The Valley of Death

Crossing the Valley of Death



For enduring program funding, PPBE/POM alignment is usually central; bridge funding, reprogramming, production OTs, Phase III awards, and other authorities may sometimes help but do not eliminate the need for sustained resourcing.

Source: GMA Defense & Technology Practice analysis

The graphic shows the shape of the gap. The table below puts dollar ranges on both sides of it, because a PM asking "how big a check are we talking about" wants figures, not a diagram.

Exhibit 28: Early-Stage R&D vs. Program of Record Funding

Early-Stage R&D / SBIR	Program of Record
Small pilot funding (\$1M–\$5M)	Multi-year procurement (\$50M–\$500M line)
The Valley of Death: 18–36 month gap between the two	

Source: GMA Defense & Technology Practice analysis

Surviving this gap means aligning capture activity with the government's own budget calendar PPBE (Planning, Programming, Budgeting, and Execution: DoW's resource allocation and budget cycle).

Exhibit 29: The Two-Year PPBE Timeline

Year -2 (Planning)	Year -1 (Programming)	Year 0 (Execution)
Service & Joint Staff define long-term strategic capability gaps	PEOs build the POM submission; specific line items set	Congress passes NDAA/Appropriations; contract award occurs
Action: shape problem definition via RFIs/white papers with requirements teams	Action: lock capability into POM line items with PMs and requirements offices	Action: execute transition bridge contracts (OTAs, Phase III SBIRs)

Source: GMA Defense & Technology Practice analysis; DoW

The practical implication: a PM conversation in Year 0 about Year 0 money is already too late. The conversation that funds you must happen roughly two years before the fiscal year you're trying to be paid in.

Is the Valley of Death Narrowing?

The 18–36 month gap described above is the historical norm, but several forces are compressing it for companies positioned to take advantage: a historically large defense budget, sustained demand for a wider base of non-traditional suppliers, and the prospect of conflict in the Middle East are all pushing program offices to move faster and fund more players at once. None of this changes the underlying mechanics of PPBE — the calendar still governs when money becomes available — but it does mean the gap is, if anything, narrower today for a company that shows up with a validated capability than it was in previous budget cycles.

2.2 Capitol Hill — The Congressional Capture Strategy

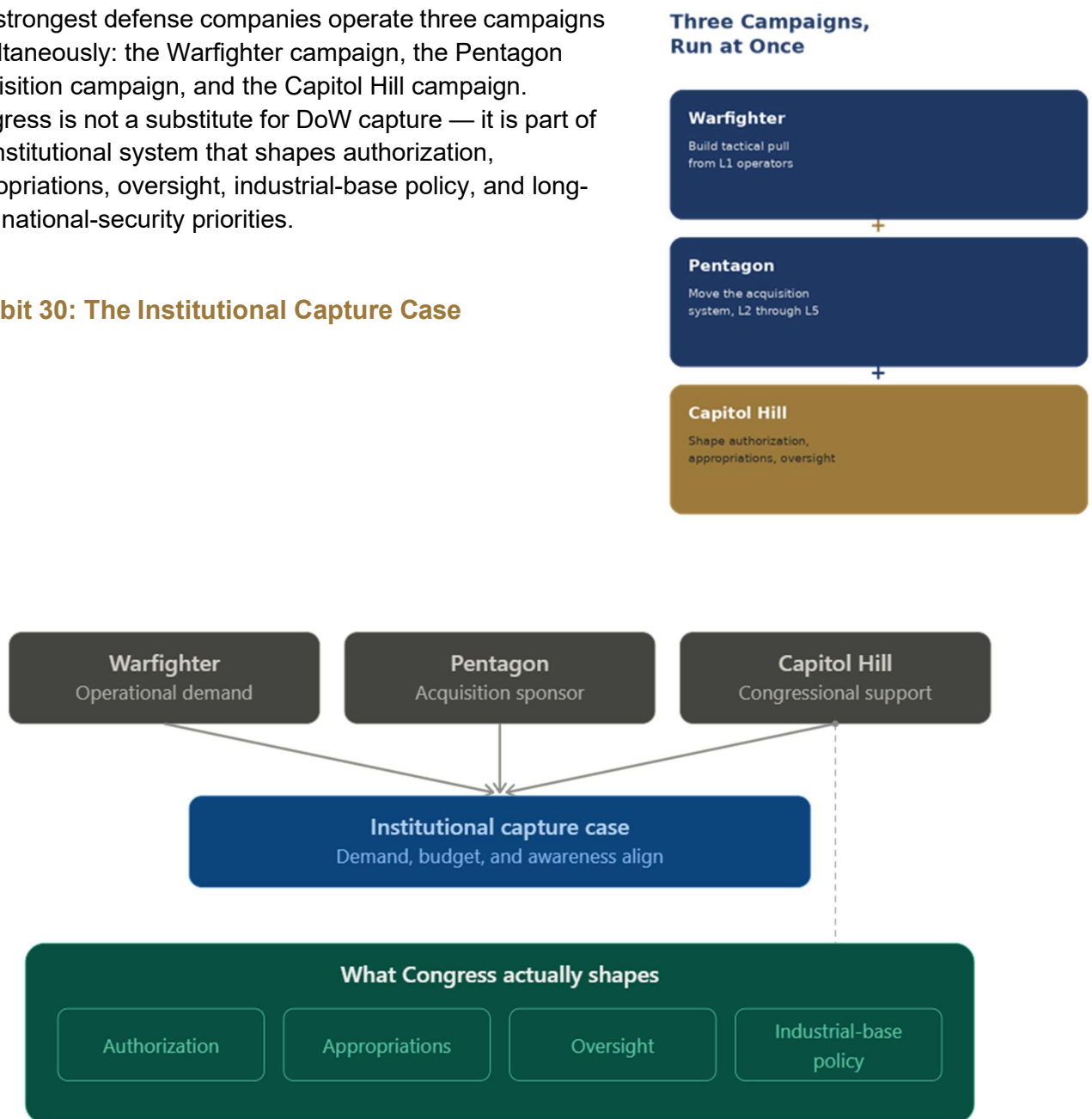
How to shape authorization, appropriations, oversight, industrial-base policy, and congressional support before the funding decision.

Winning inside the Pentagon is necessary but not sufficient. A program can clear every acquisition milestone and still stall if it lacks authorization, appropriated funding, or institutional visibility on Capitol Hill — the same PPBE cycle that governs when money becomes available to a Program Executive Officer is itself shaped, upstream, by Congress. This section turns to that layer: how to engage the legislative side of defense capture without mistaking it for a shortcut around DoW.

2.2.1 Congress Is Not a Customer

The strongest defense companies operate three campaigns simultaneously: the Warfighter campaign, the Pentagon acquisition campaign, and the Capitol Hill campaign. Congress is not a substitute for DoW capture — it is part of the institutional system that shapes authorization, appropriations, oversight, industrial-base policy, and long-term national-security priorities.

Exhibit 30: The Institutional Capture Case



Source: GMA Defense & Technology Practice

A congressional office is not a procurement office. A Member cannot simply decide to purchase a system or direct a Program Executive Officer to award a contract to a particular company. Congress operates through authorization legislation, appropriations, oversight, reporting requirements, policy direction, industrial-base measures, acquisition authorities, and other lawful mechanisms.

The objective, therefore, is not to "go around" DoW by going to Congress — it's to build a coherent institutional case in which operational demand, acquisition sponsorship, budget alignment, and congressional awareness converge.

THE RIGHT QUESTION: What congressional action, if any, would remove an institutional barrier to adoption or reinforce an existing national-security priority?

2.2.2 Authorization and Appropriations

The annual National Defense Authorization Act (NDAA) is the principal authorization vehicle for defense policy and authorities. Appropriations legislation provides the budget authority used to fund federal activities. A company must understand which problem it is trying to solve before selecting a congressional pathway.

Exhibit 31: Authorization and Appropriations

Question	Authorization	Appropriation
Primary purpose	Policy and authority	Budget authority
Principal committees	House/Senate Armed Services Committees	House/Senate Appropriations Committees
Major vehicle	NDAA	Defense Appropriations Act
Policy / authority changes	Primary mechanism	Generally not primary mechanism
Actual funding	Generally no	Yes
Company relevance	Very high	Very high
Scaled Production	Can we deliver at scale and become a lasting defense industrial company?	

Source: GMA Defense & Technology Practice analysis

PRACTICAL RULE Know whether the barrier is policy, authority, requirement, or funding before approaching Congress.

2.2.3 The Congressional Defense Ecosystem

A defense startup should map Congress with the same discipline that it maps the Pentagon.

Exhibit 32: The Congressional Defense Ecosystem

Layer	What it includes	Why it matters
Home delegation	Members representing headquarters, facilities, employees, suppliers, installations	District/state economic and industrial-base relevance
Armed Services Committees	HASC / SASC and relevant subcommittees	Authorization, policy, oversight
Appropriations	House/Senate Appropriations and Defense Appropriations Subcommittees	Funding and budget authority
Professional staff	Committee and subcommittee staff	Deep legislative and program knowledge
District/state staff	Local constituent and economic-development teams	Jobs, facilities, local investment
Issue-focused Members	Members with demonstrated interest in relevant capabilities	Policy validation and coalition building
Scaled Production	Can we deliver at scale and become a lasting defense industrial company?	

Source: GMA Defense & Technology Practice analysis

Service and Veteran Associations as Force Multipliers

Independent of any single company's lobbying, service and veteran associations — the Air & Space Forces Association, the Association of the United States Army, and the National Guard Association of the United States among them — actively push acquisition and modernization issues on Capitol Hill on behalf of their services. These organizations already have standing relationships with the relevant committees and staff; a company whose capability supports a modernization priority they're championing can often get a faster, more credible hearing by aligning with that existing advocacy rather than building a coalition from zero.

2.2.4 Congressional Staff Architecture

The common startup instinct is: “We need to get the Senator.” A more useful objective is: “We need to identify the congressional staff architecture capable of moving the specific issue.”

Exhibit 33: Congressional Staff Architecture

Audience	What they need to understand
Member	Strategic importance, national-security relevance, constituency relevance
Chief of Staff	Overall importance, risk, constituency implications
Legislative Director	Legislative opportunity and timing
Defense Legislative Assistant	Defense substance and policy
Armed Services staff	Authorization and policy implications

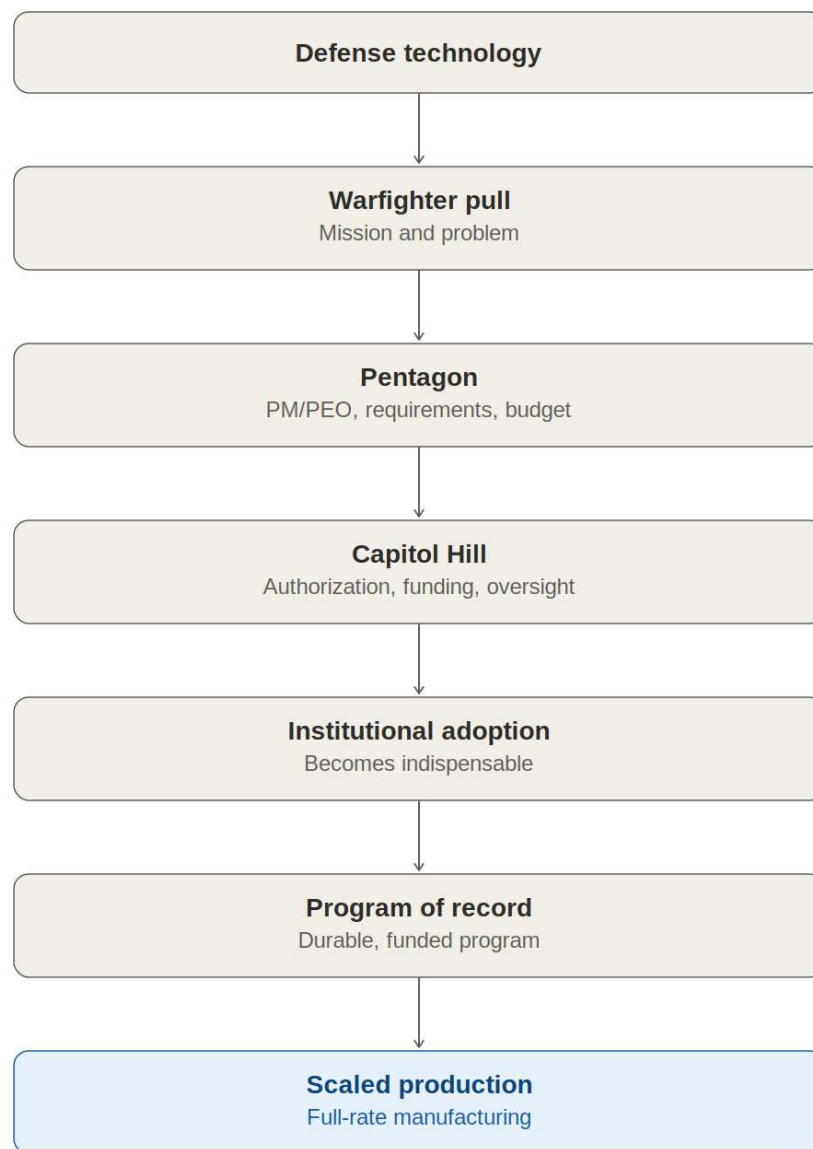
Audience	What they need to understand
Appropriations staff	Funding pathway, account and execution
District staff	Jobs, facilities, constituents
Committee professional staff	Evidence, program relevance, statutory implications
Scaled Production	Can we deliver at scale and become a lasting defense industrial company?

Source: GMA Defense & Technology Practice analysis

2.2.5 Congressional Engagement Before the RFP

The same principle that governs effective defense capture governs congressional engagement: the RFP is late in the process.

Exhibit 34: The Three-Layer Defense Capture Model



Source: GMA Defense & Technology Practice

Congressional engagement belongs inside the capture cycle, not after it.

2.2.6 What Congress Can Actually Do

Exhibit 35: Congress Pathways and Action

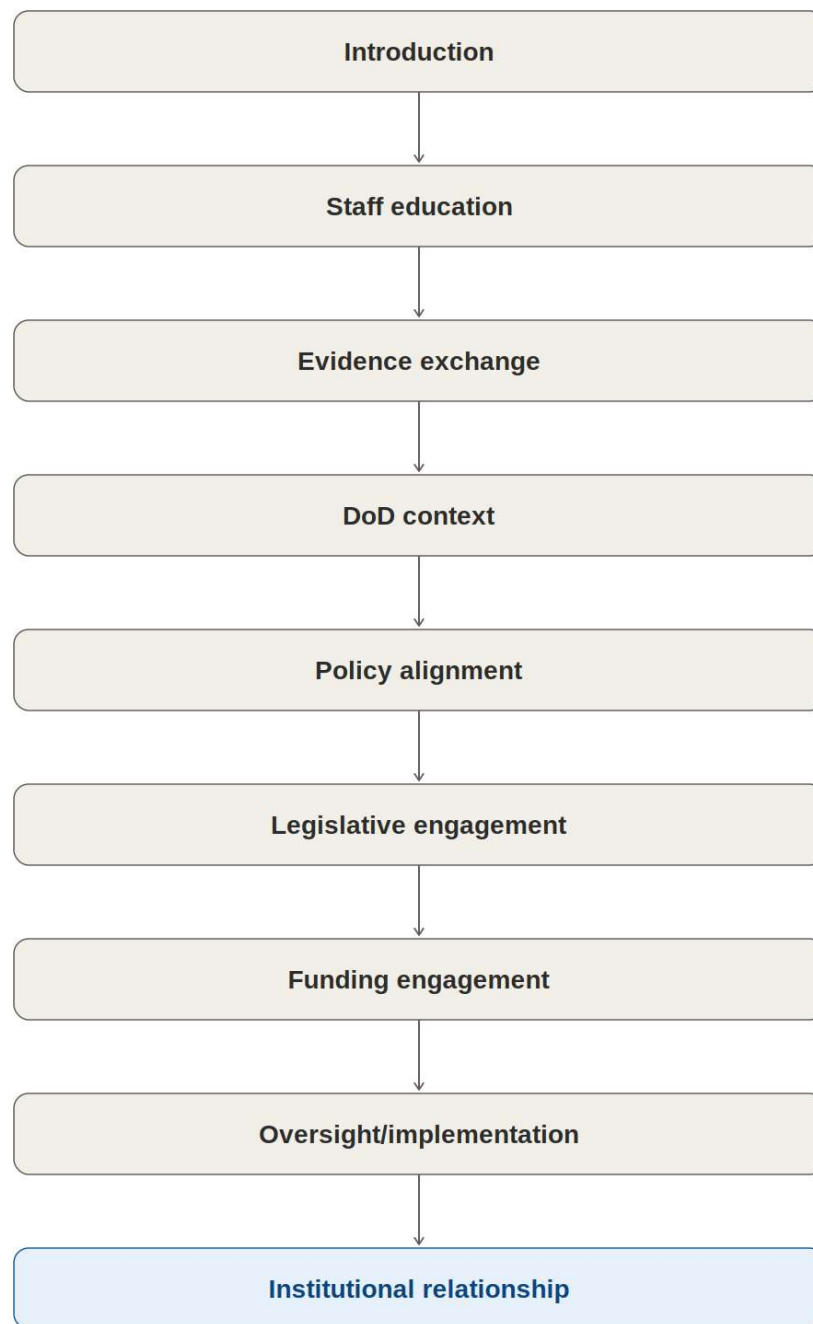
Objective	Pathway	Audience
Educate an office	Brief Member/staff on threat, capability gap, evidence, government context	Member / office staff
Request a DoW briefing	Member or committee engagement with the relevant government organization	Relevant staff
Support report language	Seek a legitimate request for a study, briefing, assessment, or report	Relevant subcommittee
Support authorization policy / new or modified acquisition authority	NDAA, through the appropriate authorizing committee	HASC / SASC or relevant authorizing committee
Support funding (RDT&E or procurement)	Defense appropriations process	Defense appropriations subcommittees
Encourage or establish a pilot	NDAA or report language, aligned with a recognized need	Relevant subcommittee
Request oversight	Raise a legitimate question via committee or Member	Relevant committee
Highlight industrial-base capacity / priority	NDAA + appropriations	Armed Services + Appropriations
Local economic impact	District/state engagement	Home delegation
Supply-chain resilience	Policy + funding	Relevant committees
Facilitate introductions	Where appropriate, connect government stakeholders	Varies
Scaled Production	Can we deliver at scale and become a lasting defense industrial company?	

Source: GMA Defense & Technology Practice analysis

RULE: ONE MEETING, ONE PRIMARY ASK A company should never walk into a congressional office with a vague request for “support.” It should arrive with a specific, procedurally appropriate objective from the table above.

Engagement with an office also matures over time, independent of the specific ask:

Exhibit 36: Building the Congressional Relationship



Source: GMA Defense & Technology Practice

2.2.7 Report Language

Many founders understand bills and appropriations but overlook committee report language. Report language can be an important mechanism for congressional direction and oversight.

A legitimate objective may involve asking DoW to provide a briefing, evaluate an emerging technology, assess an operational capability, establish or evaluate a pilot, report on barriers, assess acquisition pathways, examine industrial-base capacity, or report on transition progress.

The objective is not to insert the company's name into congressional documents. The objective is to establish government attention around a capability or problem that the company is positioned to solve.

2.2.8 Building the Congressional Narrative

The congressional narrative should be bigger than the company. A Member's office isn't in the business of funding startups — it's in the business of solving capability gaps, protecting the industrial base, and serving constituents and national-security priorities. Every conversation with Capitol Hill should be framed by that institutional mission, not the company's balance sheet.

This is a discipline, not a talking point. The same underlying fact — a startup with promising technology and a funding need — can be framed two ways: one version centers the company and its needs and reads as a pitch; the other centers the mission, the operator, and the policy landscape and reads as an institutional case. Congressional staff hear the first version dozens of times a week and are trained to discount it. They act on the second because it gives them something they can actually use: a defined problem, evidence that it's real, and a specific institutional lever to pull.

Exhibit 37: Building the Congressional Narrative

Weak framing	Stronger framing
"Our company needs funding."	"The United States faces a capability gap in X, and our validated capability addresses it."
"Our technology is revolutionary."	"Operators have evaluated the capability and the remaining barrier is transition."
"We need a contract."	"The government has a legitimate acquisition pathway; congressional action could address a defined policy or funding barrier."
"We are creating jobs."	"This capability strengthens U.S. defense production and creates measurable domestic industrial capacity."

Source: GMA Defense & Technology Practice analysis

Exhibit 38: Five Questions Every Company Must Be Able to Answer



Source: GMA Defense & Technology Practice

2.2.9 The District and State Economic Story

Congressional engagement becomes stronger when the national-security case is accompanied by a credible economic-development and industrial-base story.

Exhibit 39: The District and State Economic Story

Measure	Examples
Employment	Current employees; projected hiring
Capital investment	Facilities, tooling, laboratories

Measure	Examples
Domestic supply chain	U.S. suppliers and strategic dependencies
Workforce	Technical training, apprenticeships, university partnerships
Manufacturing	Current and scalable production capacity
Strategic impact	Supply-chain resilience; domestic capability
Constituency relevance	District/state facilities, jobs, installations

Source: GMA Defense & Technology Practice analysis

2.2.10 Defense Industrial Base Strategy

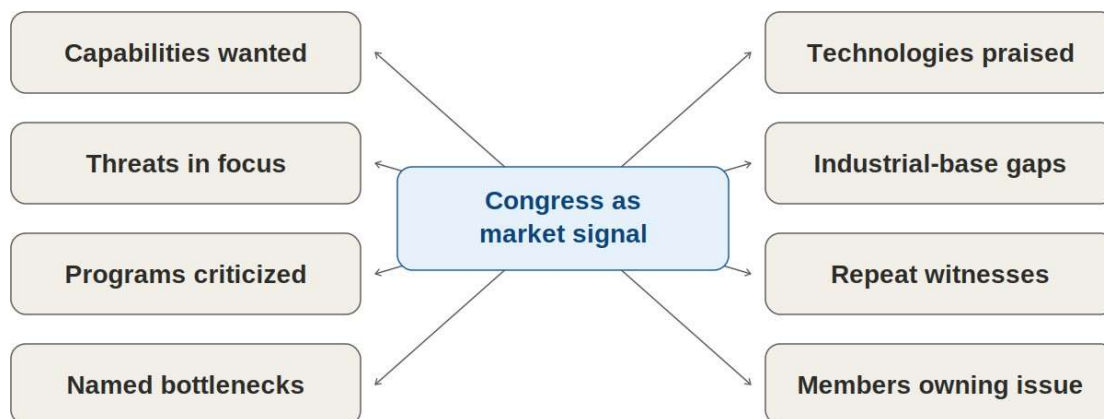
Some of the strongest congressional cases are not primarily about a platform. They concern strategic capacity: domestic production, supply-chain resilience, energetics, semiconductors, shipbuilding, munitions, batteries, sensors, AI infrastructure, cyber resilience, space systems, additive manufacturing, logistics, and workforce capacity.

KEY QUESTION What strategic dependency does this company reduce?

2.2.11 Congressional Hearings as Market Intelligence

Congress should not only be viewed as an institution to influence. It should also be viewed as a source of strategic market intelligence.

Exhibit 40: Congress as a Market Signal



Source: GMA Defense & Technology Practice analysis

The most sophisticated companies ask: “What is Congress telling us about where the defense market is going?”

2.2.12 The Congressional Calendar

Exhibit 41: The Congressional Calendar

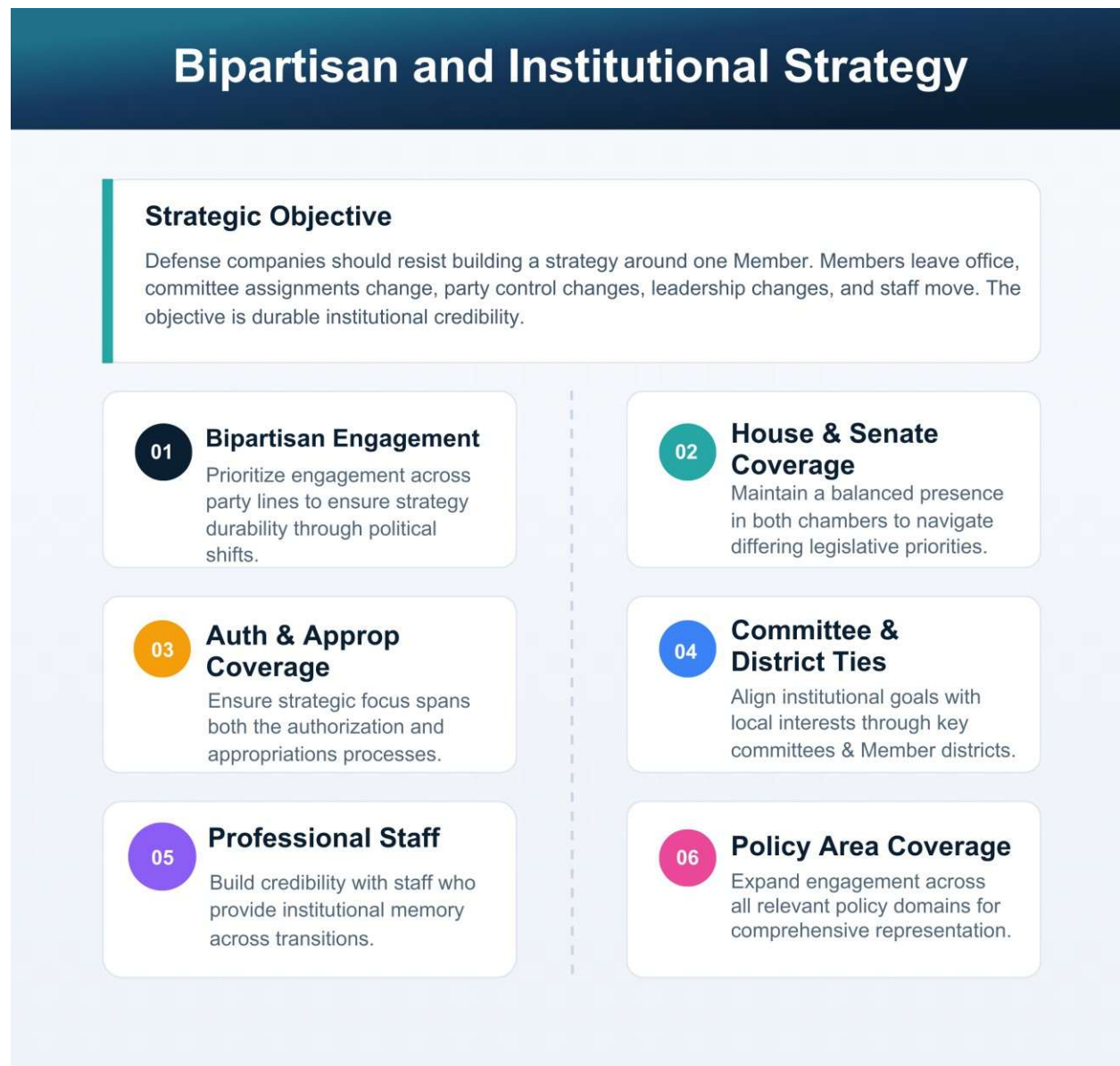
Period	Typical activity	Company objective
Fall / winter	Staff education and issue development	Establish institutional understanding
Budget submission	President's budget and agency requests	Compare requested funding with priorities
Early spring	Hearings and legislative development	Provide evidence and education
Spring	NDAA development	Advance legitimate policy concepts
Late spring / early summer	NDAA markups	Protect authorization objectives
Summer	Appropriations development	Advance legitimate funding priorities
Fall	Negotiations and final legislation	Protect provisions and funding
Year-round	Oversight and staff engagement	Maintain institutional credibility

Source: GMA Defense & Technology Practice analysis

The exact calendar changes from year to year. Companies should work from current committee schedules rather than assume a fixed annual timetable.

2.2.13 Bipartisan and Institutional Strategy

Exhibit 42: Bipartisan and Institutional Strategy



Source: GMA Defense & Technology Practice

2.2.14 The Congressional Briefing Package

Exhibit 43: The Congressional Briefing Package

Section	Required content
The Problem	Operational or national-security problem

Section	Required content
Why Now	Why the issue has become urgent
The Capability	What the company provides
Validation	Who has tested or evaluated it
Government Path	Relevant Service, agency, program office, or acquisition pathway
Industrial-Base Impact	Where and how the capability is produced
District/State Impact	Jobs, investment, suppliers, facilities
Congressional Ask	One or two specific actions

Source: GMA Defense & Technology Practice analysis

The briefing should be short enough for staff to absorb and rigorous enough to survive follow-up questions.

2.2.15 What Not to Do on Capitol Hill

Exhibit 44: What Not to Do on Capitol Hill

Failure pattern	Why it fails
"Please fund my company."	Congress is not venture capital.
Wrong committee	The office may have no relevant jurisdiction.
Going too late	Key legislative and budget decisions may already be shaped.
Asking for a contract	Congressional engagement does not replace acquisition.
Unsupported claims	Credibility can be damaged by overstating DoW or congressional support.
Making the company the story	The national-security problem and evidence should lead.
Treating an introduction as strategy	Access is not institutional adoption.

Source: GMA Defense & Technology Practice analysis

2.2.16 Synchronizing Congress and the Pentagon

The strongest congressional strategy is synchronized with the Pentagon strategy. These two tracks are not independent workstreams that happen to run in parallel — they're two audiences hearing about the same program, at the same time, and any daylight between what they're told becomes a liability. A program office that hears "we need transition funding" while a congressional office hears "the technology still needs validation" isn't looking at two framings of the same effort; it looks, to both sides, like the company doesn't actually know where its own program stands.

This matters because staff and program offices talk to each other. Committee staff routinely check a company's congressional narrative against what the acquisition side is actually saying in hearings, briefings, and budget justifications. A mismatch doesn't just cost credibility with one audience — it raises the question of which version is true, and that doubt travels back to the Pentagon side just as fast as it spreads on the Hill.

The discipline, then, is to treat the Pentagon's current framing as the anchor and translate it — not duplicate it — into the congressional register. Exhibit 39 below shows this in practice: as the program office's stated blocker shifts from validation, to transition funding, to industrial capacity, to integration, to durability, the congressional message shifts with it, always reinforcing the same underlying fact rather than introducing a separate one.

Exhibit 45: Synchronizing Congress and the Pentagon

If the program office says...	Congressional message should...
"We need to validate the technology."	Reinforce the capability gap and evidence-development need.
"The technology is validated; transition funding is the problem."	Focus on legitimate transition and budget barriers.
"We need domestic manufacturing capacity."	Emphasize industrial-base investment and production resilience.
"Integration is the primary obstacle."	Focus on interoperability, standards, testing and implementation.
"We need a long-term program."	Connect the capability to sustainable acquisition, funding and sustainment.

Source: GMA Defense & Technology Practice analysis

ALIGNMENT PRINCIPLE Congress should not hear one story while the acquisition organization hears another.

2.2.17 The Ultimate Objective: Institutional Alignment

The strongest defense companies eventually reach a point where multiple parts of government — the warfighter, the technical community, the Program Manager, acquisition leadership, budget officials, Congress, and the industrial base — independently recognize the importance of the capability.

Congressional engagement contributes to this by functioning as a form of risk reduction. Where the Pentagon side manages technical, integration, schedule, and sustainment risk through testing, program-office engagement, and lifecycle planning, the congressional side manages the same risks through policy awareness, oversight, authorization, appropriations, and long-term visibility.

This is the same institutional alignment that runs through the full capture pathway — from technology and warfighter pull, through validation, sponsorship, and budget alignment, to congressional education, authorization, and appropriations, and on to transition, production, and sustainment.

The company has crossed an important threshold when it is no longer merely trying to sell technology to the government, but has established an institutional pathway for the capability to become part of the defense enterprise. That pathway doesn't end at Program of Record: at scale, the objective shifts from gaining recognition to maintaining it. Sustaining procurement, expanding production, recompeting, pursuing international expansion, and justifying technology refresh all depend on the same durable policy, budget, industrial-base, and oversight alignment built earlier.

2.2.18 GMA Strategic Positioning

This framework broadens the role of GMA beyond proposal development. The larger proposition is that GMA can help defense and dual-use companies identify, shape, fund, and execute the institutional pathway from emerging technology to government adoption.

Exhibit 46: How GMA Supports the Institutional Pathway

Capability	Role
Defense funding intelligence	Identify relevant federal and defense funding opportunities.
Capture strategy	Map operators, labs, program offices, acquisition pathways and budget cycles.
Congressional landscape analysis	Map committees, Members, staff, jurisdictions and legislative timing.
Congressional briefing preparation	Develop evidence-based briefings and specific asks.
Funding-pathway analysis	Connect legitimate program needs to relevant mechanisms.
Industrial-base opportunity identification	Identify domestic production, supply-chain and workforce opportunities.
Proposal development	Translate validated requirements into competitive submissions.
Transition strategy	Support movement from prototype or pilot to enduring adoption.

Source: GMA Defense & Technology Practice analysis

Congressional engagement should be conducted in accordance with applicable federal law, congressional ethics rules, and lobbying-disclosure requirements, with qualified government-affairs or legal professionals involved where appropriate.

2.2.19 Congress as Part of the Institutional Adoption System

Congress should not be treated as the place a defense startup goes when the Pentagon says no. It should be treated as part of the long-cycle institutional strategy required to turn emerging technology into enduring defense capability.

The most effective companies understand the distinction between selling technology, winning a contract, and building institutional adoption. The first requires a good product. The second requires a credible acquisition pathway. The third requires alignment across operators, technical authorities, program offices, acquisition leadership, budget officials, Congress, the industrial base, and the broader national-security establishment.

CORE PRINCIPLE Shape the requirement before the RFP. Shape the budget before the appropriation. Shape congressional understanding before the markup. Build the institutional coalition before the competition begins.

Hill Capture Dashboard

A fill-in worksheet for tracking a single company's Hill capture status.

Exhibit 47: Worksheet for Hill Capture Dashboard

Field	Company answer / owner
Operational problem	
Capability	
Warfighter sponsor	
Program office / PM	
Requirement status	
Acquisition pathway	
Budget account / Program Element	
Relevant HASC jurisdiction	
Relevant SASC jurisdiction	
Relevant appropriations jurisdiction	
Home-state / district Members	
Issue-focused Members	
Key professional staff	
Legislative opportunity	
Congressional ask	
Evidence package	
Next legislative decision point	
Next staff engagement	
Owner	
Status	

Source: GMA Defense & Technology Practice analysis

2.3 Acquisition Pathways & Contract Vehicles

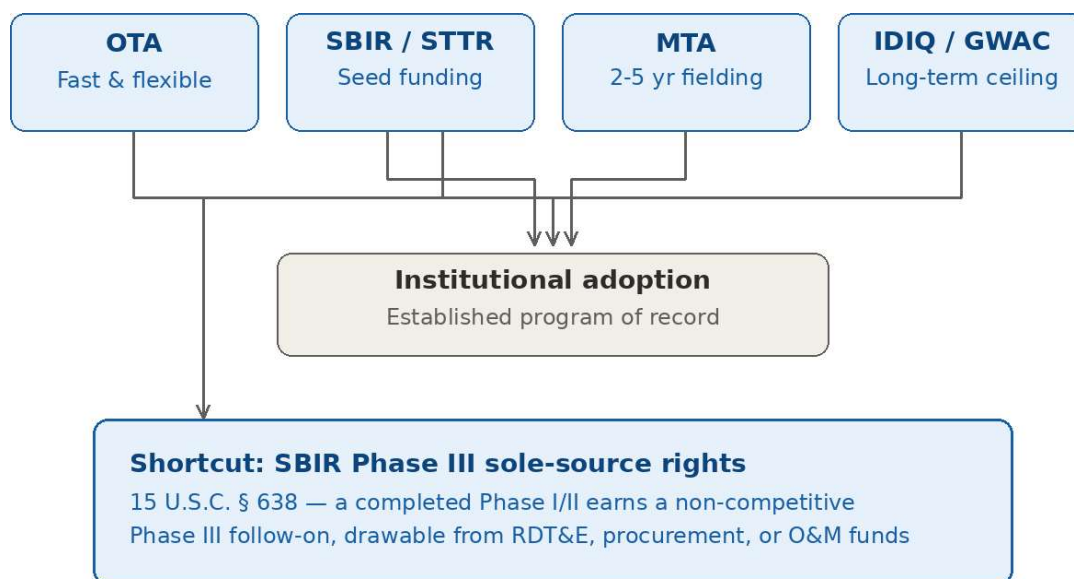
Exhibit 48: Acquisition Pathways & Contract Vehicles

Vehicle / Pathway	Advantages	Strategic Purpose
Other Transaction Authority (OTA)	Speed; bypasses traditional FAR friction; flexible IP	Rapid prototyping, transitioning directly into follow-on non-competitive production OTAs
SBIR / STTR (Phase I → III)	Fast initial seed funding for niche capability development	Use Phase I/II as foot-in-door; pursue Phase III for sole-source production
Middle Tier of Acquisition (MTA)	Rapid Prototyping and Rapid Fielding pathways under Section 804	Fields capability in 2–5 years, bypassing standard JCIDS timelines
IDIQ / GWAC	Broad procurement vehicle with multi-year ceiling capacity	Long-term purchasing mechanism once institutional adoption is established

Source: GMA Defense & Technology Practice analysis

Strategic leverage SBIR Phase III sole-source rights. Under 15 U.S.C. § 638, a company that has completed a Phase I or Phase II SBIR retains data rights and is eligible for non-competitive Phase III follow-on contracts, drawable from any pool of government funds (RDT&E, Procurement, or O&M) without a new open competition. This is one of the few statutory shortcuts around full and open competition available to a startup — but it only exists if you ran an SBIR in the first place; a company that skipped SBIR has to use MTA or a production OTA instead.

Exhibit 49: SBIR Phase III Sole-Source Pathway



Only available if SBIR was run first — skip it, and MTA or a production OTA is the fallback

Source: GMA Defense & Technology Practice analysis

Who Deconflicts Cross-Service Programs: OSD vs. Joint Staff

Two Pentagon-level bodies referee conflicts between services, and founders selling into more than one service should know which is which. The Joint Staff carries more operational say — it represents warfighter requirements and joint operational needs across the services. The Office of the Secretary of Defense, particularly OSD Policy, holds the authority to stop an effort outright on policy or standardization grounds. A classic trigger: if the Marine Corps and the Army each move to buy a different type of howitzer, OSD Policy can and will intervene to force standardization rather than let two incompatible acquisition programs proceed in parallel. A company whose capability could plausibly be adopted by more than one service should expect this kind of cross-service deconfliction question to surface, and should have an answer for it before a program office asks.

Who Can Say No

Joint Staff

More operational say.
Represents warfighter
requirements across
the services.

OSD Policy

Can stop an effort outright
on policy or standardization
grounds — even a funded,
operationally-backed one.

Building SBIR Coalitions

Beyond running your own SBIR/STTR track, building coalitions with other SBIR awardees and non-traditional entities can accelerate adoption. DoW's own posture under the current Secretary favors spreading award dollars across a wider base of new entrants rather than concentrating them in a small set of incumbents — a company that positions itself as part of a broader coalition of capable non-traditional players, rather than a lone vendor, aligns with that posture and can move faster through it.

Winning the Phase III funding solves the money problem, not the credibility problem — the government still has to see the capability perform under conditions it recognizes. The comparison below is what separates a demo that impresses from one that gets funded.

2.4 Designing High-Impact Field Demonstrations

Exhibit 50: High-Impact Field Demonstrations

Failing Demo Design	Risk-Reduction Demo Design
Controlled conditions	Stressed operational environment
Vendor operates the platform	Military operators execute
Isolated environment	Integrated into the C2 network
Focus: "look what it does"	Focus: "clear, repeatable"

Source: GMA Defense & Technology Practice analysis

An effective demonstration addresses risk directly:

Exhibit 51: Designing Demonstrations That Address Risk

Operator control

Active-duty personnel operate the system with minimal instruction.

Adverse conditions

GPS jamming, electronic attack, degraded comms, extreme weather.

System integration

Real-time telemetry, target tracks, or sensor data fed directly into legacy C2 (TAK, DCGS, GCCS-J).

Rapid post-demo reporting

A comprehensive data package (reliability, power consumption, MTBF, network bandwidth) within 48 hours of the demo, not weeks later.

Source: GMA Defense & Technology Practice

Exhibit 52: Field Demo Under Operational Conditions



Source: Stock Image

Shield AI — "Subsystem Leverage."** Where Anduril in Part 1 demonstrates before there's a program to win, Shield AI's case shows what happens after: decoupling a proven autonomy stack from any single airframe so it can be adopted onto hardware the company doesn't own, without waiting on a new platform buy.

2.5 Case Studies

Exhibit 53: Shield AI: Subsystem Leverage



Source: Stock Image

Shield AI subsystem leverage in practice. The same "platform-agnostic" strategy from Part 1 plays out concretely here: Hivemind's integration onto Kratos's Firejet target drone went from project start to first live flight in roughly 120 days by reusing the same technical foundation across platforms, and its integration onto General Atomics' MQ-20 Avenger ran on a twelve-week timeline. Neither of these is Shield AI hardware — they're proof that decoupling the autonomy stack from any single airframe multiplies the number of PMs who can adopt it without waiting for a new hardware program.

Lesson: the statutory commercial-item preference is a real, litigable protection — but it's a slower and more adversarial path than the bottom-up adoption route in Part 1, and it works best as a last resort when an agency is demonstrably ignoring an available commercial alternative, not as a primary go-to-market strategy.

Illustrative pattern the exclusivity trap. A recurring pattern among comms and RF-adjacent startups: a company signs an exclusive distribution agreement with a Tier-1 prime to accelerate market access. The prime integrates the startup's software into its own hardware and markets it as a proprietary upgrade — then deprioritizes the effort once it threatens the prime's existing product line. Bound by the exclusivity clause, the startup can't sell directly to government or partner elsewhere, misses PPBE windows waiting on the prime, and is eventually acquired at a distress valuation.

Lesson: *exclusive distribution agreements with a legacy prime should be treated as a red flag by default. Retain direct L3 PM relationships and negotiate IP/data rights independent of any single prime.*

Program Manager for Ground Based Air Defense awarded Overland a \$20M production OTA — funded through APFIT — to build autonomous ground vehicles supporting the Marine Air Defense Integrated System (MADIS). Overland was explicit about what the award represented: not just revenue, but "the first step toward integrating fully autonomous ground vehicles at scale in USMC programs of record," and the first time any ground-autonomy company had held a production contract as prime rather than as a subcontractor.

The company raised a \$100M Series B eight months prior to the award, and used the funding runway to double its manufacturing capacity ahead of the win — capacity a company without a validated demo and an LOI-stage relationship with PM GBAD would not have known to build.

Lesson for founders: a production OT can be an important transition mechanism, but it does not by itself guarantee a Program of Record, enduring funding, or follow-on demand — and prototype-to-production timing only works if manufacturing capacity is already being built before the award, not after it.



Darkhive, a San Antonio-based tactical autonomy and small-UAS company founded in 2021, spent its first two years accumulating a portfolio of small, unglamorous AFWERX and Air Force Research Laboratory contracts — a \$1.25M voice-enabled C2 award, four small AFWERX drone-technology contracts, a \$1M OSD award — the kind of L1/L2-level activity Part 1 of this playbook describes as foundation-building rather than validation.

That changed in December 2025, when Darkhive won a \$49.7M APFIT contract — the largest single award in the program's history — bringing its total DoW contract value to roughly \$64M. The company didn't wait for the contract revenue to fund delivery: it opened a \$30M Series B round, led by RTX Ventures, the venture arm of the defense prime RTX. RTX Ventures had also led Darkhive's Series A, and stepping up to lead the B round again was, in the company's own words, evidence of "belief in our team, our products, and how far we are positioned to take our products in support of the warfighter."

Lesson for founders: an APFIT or OTA production award is itself a fundraising asset — it is the validated-institutional-demand signal investors are actually pricing, and a returning prime-affiliated investor doubling down on the strength of a contract award is a stronger signal to the next round than almost any commercial traction metric.

Exhibit 54: Year 2 Do's and Don'ts

DO'S AND DON'TS FOR YEAR TWO	
DO	DON'T
● Track the two-year PPBE cycle and ensure your capability is included in the Service's POM submission during the Programming phase ● Use non-competitive SBIR Phase III contracts to transition prototype wins into multi-year production awards. ● Conduct field demonstrations using active-duty military operators rather than company engineers ● Insist on explicit workshare percentages and IP protections in all Teaming Agreements with Prime contractors	■ Rely on prototype OTAs without securing a documented commitment for a follow-on non-competitive production OTA. ■ Run isolated demonstrations that fail to connect directly to legacy systems of record. ■ Sign exclusive distribution agreements that cut off direct access to government PMs or prevent partnerships with other vendors. ■ Treat a verbal expression of interest from a General Officer as a substitute for a funded contract line item.

Source: GMA Defense & Technology Practice

How GMA Helps — Part 2: Year Two

The PPBE calendar doesn't wait for a founder to notice it. GMA tracks DoW funding priorities and emerging policy shifts (data rights language, cost-share changes) in real time and incorporates them into your strategy as they happen, so a demonstration gets timed to an actual POM cycle instead of a Year 0 conversation about Year 0 money.

When a company has run an SBIR and is eligible for sole-source Phase III rights under 15 U.S.C. § 638, or is weighing OTA versus MTA versus IDIQ, GMA's team helps navigate the pathway and the paperwork that has to be right the first time including the specific commercialization narrative Phase III reviewers look for, which our proposal team has refined across a deep bench of SBIR/STTR submissions.



Part 3: Year Three — Becoming a Major Defense Supplier

“In defense procurement, manufacturing capability is part of the core product, not a downstream operational detail.”

3.1 Scaling for Industrial Advantage

By Year Three, the evaluation shifts to industrial scale, production throughput, supply chain security, and long-term sustainment:

- Advanced manufacturing high-rate, resilient production facilities capable of scaling to surge demand during crisis scenarios.
- Secure supply chain architecture — map statutory, regulatory, program-specific, security, and supply-chain requirements at the component level; sourcing restrictions vary by system, component, funding, country of origin, and contract clause, and should be confirmed against the specific program rather than assumed.

Exhibit 55: Surge-Capable Production Line



Source: Stock Image

- Enterprise compliance & accreditation — Facility Security Clearances, CMMC Level 2/3, ATOs across multiple classification levels (Unclassified through TS/SCI).
- Global sustainment & logistics — Field Service Representatives, depot-level repair pipelines, decade-long spares/maintenance support models.

Exhibit 56: Scaling for Industrial Advantage



Source: GMA Defense & Technology Practice

3.2 Manufacturing as a Competitive Advantage

Exhibit 57: U.S. Aerospace Manufacturing Line



Source: The Boeing Company, defense aerospace manufacturing facility imagery

In defense procurement, manufacturing capability is part of the core product, not a downstream operational detail.

Exhibit 58: Evolution of Manufacturing

Traditional Model	New Defense Model
Secure contract award first	Build industrial base first
Design manufacturing line	Demonstrate scaled capacity
Risk: delivery delays	Outcome: low risk to PM

Source: GMA Defense & Technology Practice

Manufacturing scaling framework:

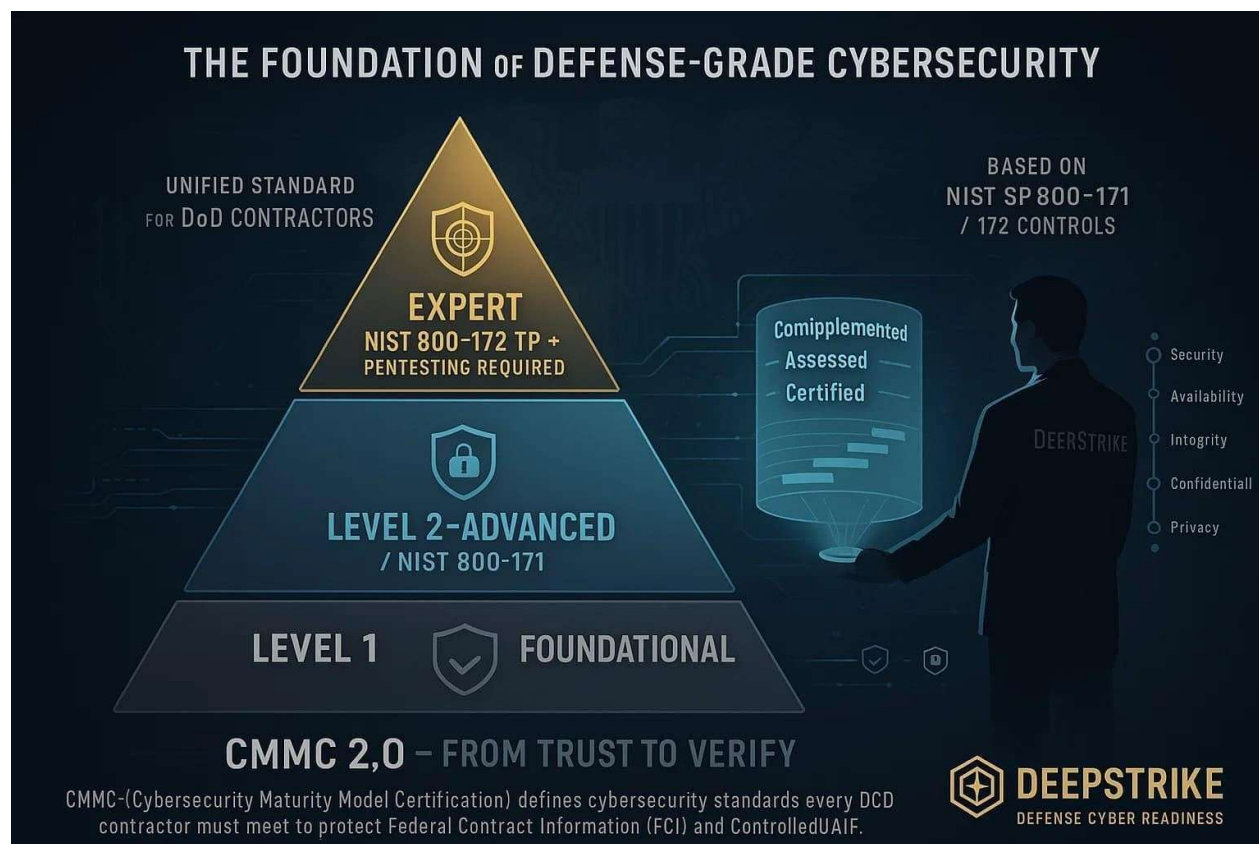
- **Design for Manufacturability (DFM)** — standardize sub-assemblies, minimize custom fasteners, use COTS components wherever possible.
- **Capital investment ahead of the contract** — automated tooling, environmental test chambers, and production infrastructure funded before the major award, not after.
- **Surge capacity testing** — demonstrate to PEOs an actual, executed 3–5x production expansion within 90 days, not a modeled one. A PM sizing a wartime reserve requirement is going to ask for evidence.

3.3 The Back-Office Compliance Regimes That Blindside Scaling Companies

FCL, CMMC, and ATOs get the attention because they gate whether you can touch classified or sensitive systems at all. Two other regimes get far less attention but reliably surprise Year 2–3

companies once contract size crosses certain thresholds — and both changed materially as of mid-2026:

Exhibit 59: CMMC 2.0 Certification Levels



Source: Deepstrike (defense cybersecurity readiness firm), CMMC 2.0 certification level reference chart

Note: CMMC implementation status has continued to evolve; as of this writing, Phase 1 implementation is paused while core safeguarding obligations under NIST SP 800-171 and DFARS remain in force. Confirm current CMMC policy status and the specific clause set in the target solicitation before treating any level as a settled gate.

Certified cost or pricing data (formerly "TINA," now the Truthful Cost or Pricing Data Statute). The threshold jumped from \$2.5M to \$10M for contracts entered into after June 30, 2026 (10 U.S.C. § 3702, as amended by NDAA Section 1804).

Above the threshold: you must certify your cost/pricing data is accurate, complete, and current as of the agreement date — and DCAA can audit it afterward for "defective pricing."

Legacy contracts: awards signed on or before June 30, 2026 stay under the old \$2.5M threshold, and modifications to those contracts can still trigger certification at the old number.

The practical effect: a company with a mix of pre- and post-June 2026 awards has to track which threshold applies to which contract.

Cost Accounting Standards (CAS). The same NDAA raised the CAS applicability threshold from \$2.5M to \$35M, and the full-coverage threshold (all 19 — soon to be 15 — standards) from \$50M to \$100M.

Exhibit 60: CAS Applicability Thresholds

Contract value	CAS coverage
Below \$35M	Generally exempt
\$35M–\$100M	"Modified" coverage — a lighter-weight subset of standards
Above \$100M	Full coverage — accounting system must be built to the standard, not retrofitted after the fact

Source: GMA Defense & Technology Practice analysis

Nontraditional defense contractor (NDC) exemption. If you haven't held a full CAS-covered DoW contract or subcontract in the prior year, you likely qualify as an NDC — which, under the same NDAA, gets you additional relief from certified cost/pricing data and FAR Part 31 cost-principle requirements, even above the standard thresholds. Confirm NDC status with your contracts team before assuming you need full compliance infrastructure; overbuilding compliance you don't yet need is a common and avoidable Year 2 cost.

Exhibit 61: CAS and Certified Cost/Pricing Data Thresholds

FEDERAL CONTRACTING THRESHOLD CHANGES	
	TRUTHFUL COST OR PRICING DATA Formerly known as TINA. The FY2026 NDAA raised the threshold from \$2.5M to \$10M for contracts entered into after June 30, 2026. Key Note: Contracts signed on or before June 30, 2026, remain subject to the old \$2.5M threshold. Modifications to legacy awards can still trigger certification at the \$2.5M level.
	COST ACCOUNTING STANDARDS (CAS) The applicability threshold was raised from \$2.5M to \$35M. • Exempt: Below \$35M. • Modified Coverage: \$35M – \$100M. • Full Coverage: Above \$100M. Full coverage (all 19 standards) requires accounting systems to be built to standard from the ground up, not retrofitted.
	NONTRADITIONAL CONTRACTOR (NDC) If you haven't held a full CAS-covered DoD contract in the prior year, you likely qualify as an NDC. The Benefit: Additional relief from certified cost/pricing data and FAR Part 31 requirements even above standard thresholds. Confirm NDC status before overbuilding compliance infrastructure.

Source: GMA analysis; 10 U.S.C. § 3702 (Truthful Cost or Pricing Data Statute), as amended by NDAA FY2026 Section 1804

3.4 Congressional Engagement & Policy Alignment

Exhibit 62: Engagement & Policy Alignment

Authorization Process	Appropriations Process
HASC / SASC	HAC-D / SAC-D
Sets policy framework and procurement ceilings via the annual NDAA	Allocates actual dollars to specific budget line items and Program Element numbers

Source: GMA Defense & Technology Practice analysis

- **Protect Program Elements** — monitor annual subcommittee markups to ensure your target PE line items survive.
 - **NDAA legislative provisions** — coordinate with staffers to draft language directing the Services to evaluate or field capabilities in your domain.
 - **Local economic impact** — highlight your facility's regional footprint, supply chain partners, and engineering headcount during legislative briefings; a jobs-and-supply-chain narrative resonates with appropriators in a way a pure capability narrative doesn't.
-

3.5 Case Studies

Exhibit 63: Palantir: The DCGS-A2 Protest



Source: *Palantir USG, Inc. v. United States*, 904 F.3d 980 (Fed. Cir. 2018), *aff'g* 129 Fed. Cl. 218 (2016)

Palantir Technologies: The same case, viewed as a supply-side lesson. The DCGS-A2 protest (detailed in Part 2) also demonstrates a Year Three point: existing procurement statutes offer real, enforceable protections for commercial technology companies once you're large enough to litigate for them. That's a scaled-company move, not a Year One move — but it's worth knowing the tool exists.

[Palantir Defense Solutions](#)

Illustrative pattern the single-source supply chain failure. A recurring pattern among hardware-heavy startups: a company secures a major production contract while its engineering team still relies on specialized, non-cleared foreign vendors for components like optical lenses or PCBs. An audit during performance discovers the components breach NDAA supply chain restrictions. The result is a government-issued stop-work order, frozen payments, and months of schedule slip while the company redesigns the hardware and requalifies domestic suppliers.

Lesson: NDAA-compliant, domestic/allied supply chains have to be built during the design phase, not qualified after a production contract is already signed — replacing non-compliant

components late in the process is far more expensive than designing around them from the start.

Exhibit 64: Year 3 Do's and Don'ts

DO'S AND DON'TS FOR YEAR THREE	
DO	DON'T
● Invest in domestic, automated manufacturing capacity ahead of major procurement contracts ● Enforce federal statutory preferences for commercial items under FAR Part 12 and 10 U.S.C. § 3453. ● Track Congressional markups across HASC, SASC, HAC-D, and SAC-D to protect your line-item funding. ● Build a complete sustainment framework — including Field Service Representatives (FSRs), depot repair networks, and spares logistics — to support long-term deployments.	■ Rely on non-compliant, single-source foreign component suppliers for critical hardware assemblies. ■ Treat manufacturing design as an afterthought to be solved after winning a contract. ■ Neglect Facility Security Clearances (FCL) or Cybersecurity Maturity Model Certification (CMMC Level 2/3) compliance ■ Assume that a contract award guarantees funding execution without ongoing Congressional engagement.

Source: GMA Defense & Technology Practice

How GMA Helps — Part 3: Year Three

Scaling into a major supplier means the compliance regimes in 3.3 stop being theoretical CMMC readiness, CUI protocols, and NDAA-driven thresholds all become live risk. GMA supports opportunity intelligence, capture planning, funding-pathway analysis, proposal development, and strategic messaging around this compliance picture, and coordinates with qualified cybersecurity assessors, accounting/cost specialists, and export-control counsel where a client's needs require specialized, licensed, or credentialed advice — reducing the risk of disqualification on non-technical grounds at the exact stage where a mistake is most expensive.

At this stage, some companies also start encountering classified and sensitive-program requirements — GMA helps clients understand eligibility and onboarding pathways for initiatives like the DoW Special Access Program Corporate Portfolio and DARPA's BRIDGES program, and acts as a secure liaison between commercial teams and government sponsors so engagements stay within approved export-control and information-security boundaries from the outset.



Part 4: Domain-Specific Execution Matrix & Tactical Reference Manual

“The right conference and lab contact should feed directly into a named PEO/PM relationship and a specific contract pathway — not three parallel, disconnected activities.”

4.1 Cross-Domain Operational Matrix

This maps directly onto the L1–L5 advocate model from Part 1: the "Primary Technical Labs" column is where your L2 evaluators sit, and the "Target PEOs & PMs" column is where your L3–L4 advocates sit.

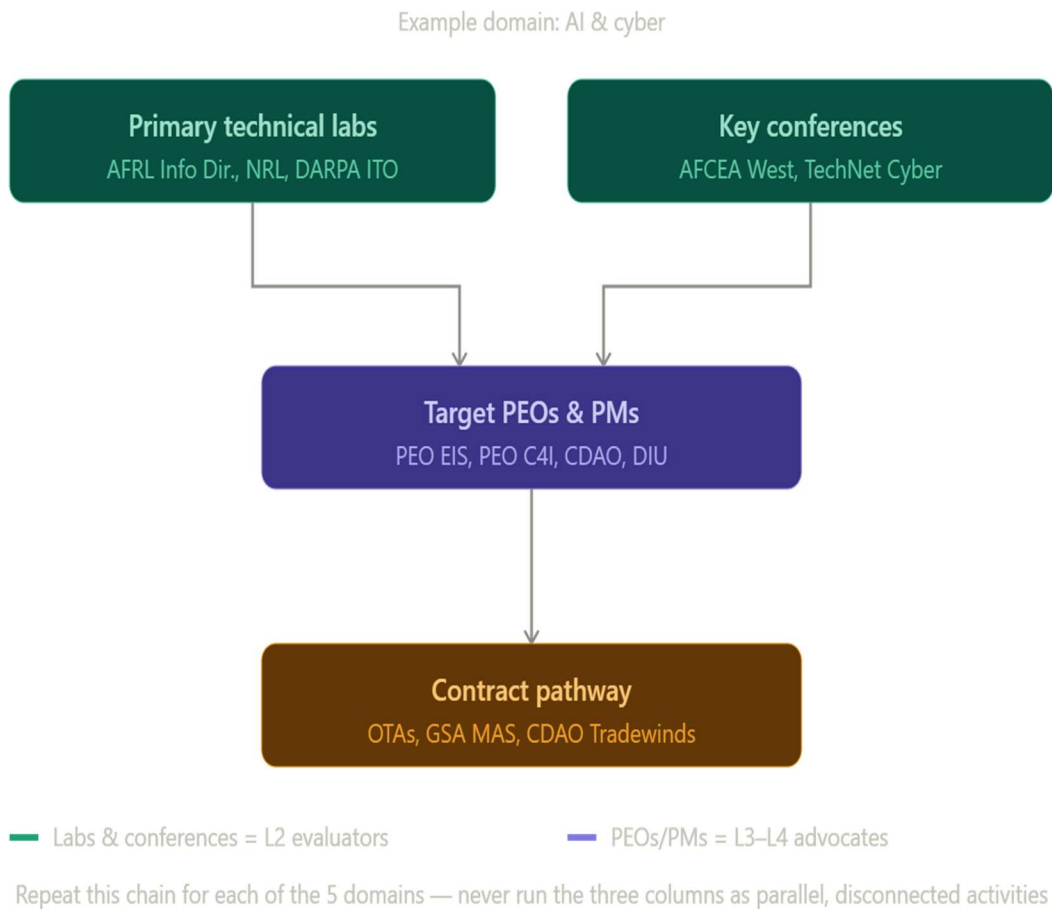
Exhibit 65: Cross-Domain Execution Matrix

Domain	Target PEOs & PMs	Primary Technical Labs	Key Conferences	Contract Pathways
AI & Cyber	PEO EIS, PEO C4I, CDAO, DIU	AFRL Info. Directorate, NRL, DARPA ITO	AFCEA West, Cyber Edge, TechNet Cyber	OTAs, GSA MAS, CDAO Tradewinds
Domain	Target PEOs & PMs	Primary Technical Labs	Key Conferences	Contract Pathways
Drones & Autonomy	PEO Soldier, PEO Unmanned Aviation, DIU	DEVCOM ARL, NSWC Crane, AFRL Munitions	SOF Week, AUSA, AUVSI XPONENTIAL	DIU Blue UAS, MTA
Maritime Systems	PEO USC, PEO Ships, NAVSEA	NSWC Carderock, NSWC Panama City	Sea-Air-Space, SNA Symposium	Rapid Prototyping OTAs, Phase III SBIR
Space Systems	SSC, SDA	AFRL Space Vehicles Directorate	Space Symposium, GEOINT	SDA Tranche Contracts, Space OTAs
Hypersonics	RCCTO, PEO Missiles & Space, DARPA	AEDC, NAVAIR China Lake	NDIA Hypersonics, AIAA	Section 804 MTA, Production OTAs

Source: GMA Defense & Technology Practice analysis

Use this matrix to route BD effort: for any domain, the right conference and lab contact should feed directly into a named PEO/PM relationship and a specific contract pathway — not run as three parallel, disconnected activities.

Exhibit 66: Domain BD Routing Chain



Source: GMA Defense & Technology Practice analysis

4.1.1 Domain Notes — What "Working the Matrix" Actually Looks Like



AI & Cyber. CDAO's Tradewinds Solutions Marketplace is the fastest on-ramp for AI/ML tools because it lets program offices browse pre-vetted solutions and issue task orders without running a full new competition — the leverage point for a startup is getting listed and demonstrated there before a PEO has a stated requirement, not after. DIU's AI portfolio moves on roughly 60–90 day commercial solutions opening (CSO) cycles; missing a cycle costs a full quarter, so track the CSO calendar rather than waiting for it to surface at a conference.

<https://www.tradewindai.com/>



Drones & Autonomy. The clearest recent case study is the Army's Company-Level Small UAS Directed Requirement: Anduril's Ghost-X and Performance Drone Works' C-100 were selected in September 2024, and both were already on DIU's Blue UAS Cleared List — the NDAA-compliance vetting list — before the Army requirement was even competed. Ghost-X was also pulled into the Replicator initiative's second tranche through the Army's Medium Range Reconnaissance program.

The lesson: Blue UAS listing is a prerequisite, not a milestone — companies that wait to pursue it until a PEO shows interest are structurally too late. Replicator itself is also instructive as a

cautionary example: as of late 2025, CRS and multiple outlets reported the initiative had fielded “hundreds,” not the “multiple thousands” originally targeted by August 2025, citing production capacity, unfinished designs at time of selection, and C2 integration as the main failure points. It has since been folded into the Defense Autonomous Working Group rather than shut down — a reminder that even a high-profile rapid-acquisition vehicle can slip badly, and founders should treat Replicator-style programs as accelerants, not guaranteed revenue.

<https://www.congress.gov/crs-product/IF12611>

Exhibit 67: Blue UAS



Source: GMA Defense & Technology Practice analysis of public company disclosures

Maritime Systems. NAVSEA's unmanned surface/subsurface portfolio (PEO USC) leans heavily on rapid prototyping OTAs because traditional shipbuilding acquisition timelines are incompatible with commercial iteration speed — lead with an OTA-shaped proposal, not a FAR Part 15 proposal, even when the same office also runs traditional procurements for crewed ships.

<https://www.navsea.navy.mil/>

Exhibit 68: U.S. Navy Medium USV Selection



Source: U.S. Navy Medium Unmanned Surface Vessel (MUSV) program; Naval News, "U.S. Navy Selects 7 Contenders for the MUSV Program" (May 2026)



Space Systems. SDA's Proliferated Warfighter Space Architecture is structured in "Tranches" — Tranche 0, 1, 2, and so on — each a distinct competitive award cycle rather than one long-running contract. A vendor that misses a given Tranche competition typically waits 18–24 months for the next one, which makes SDA's own published tranche timeline more decision-relevant than any conference date.

[Space Development Agency – Increasing Warfighters' Lethality, Maneuverability, and Survivability](#)

Exhibit 69: SDA Proliferated Warfighter Space Architecture



Source: Stock Image

Hypersonics. This is the domain where the Section 804 Middle Tier Acquisition pathway matters most, because traditional major-capability acquisition timelines are widely viewed as incompatible with the pace of adversary hypersonics development. RCCTO exists specifically to bypass that timeline — a hypersonics-adjacent company not tracking RCCTO's rapid prototyping/rapid fielding project list is targeting the wrong office.

<https://www.aiaa.org/>

Exhibit 70: RCCTO Hypersonics Rapid Prototyping



Source: Stock Image

4.2 The Strategic Founder Checklist

Tool: The Pursuit Decision Tree

Before committing capture resources to a specific pursuit, run it through these gates. A “no” means fix that gap or drop the pursuit — not proceed and hope it resolves itself. A pursuit failing 2+ gates should be logged as “watch,” not “active.”

1. Is there a named operational need? (A specific mission thread or capability gap — not a general priority.)
2. Is there a sponsor? (An individual or office with a stake in the outcome.)
3. Is there a technical validation route? (A lab, test org, or demo venue that can generate credible evidence.)
4. Is a budget/program element identifiable? (A PE number or POM line — not “they seemed interested.”)
5. Is an appropriate contracting pathway available? (SBIR/STTR, OTA, CSO, BAA, direct contract.)
6. Is the company contract-ready? (SAM.gov, CAGE code, reps & certs, accounting adequacy where required.)

7. Is the company production-ready, if production is being asked for? (Not just a working prototype.)
8. Does the expected value justify the pursuit cost and opportunity cost?

Year One: Building the Engine

Team structure — Hire a dedicated Capture Director and Acquisition Leads with prior experience inside your target PEOs.

- Prioritize candidates who held budget or requirements authority inside the target PEO (former PM, deputy PM, or requirements officer), not just former operators — operators validate the technology, acquisition insiders navigate the money and paperwork.
- Treat this as a Q1 hire, not a Year Two hire.

User discovery — Structured discovery interviews with L1 operators to document specific, named capability gaps.

- Target company- and battalion-level operators, not just senior officers at conferences.
- Output a written capability gap memo per interview — this becomes the evidence base for both the technical data package and, later, POM justification.

Technical data package — Empirical data on stress-test results, power draw, and environmental limits, not marketing decks.

- Include failure-mode data, not just best-case performance.
- Build it in a format an L2 evaluator can hand to their own technical review board without translation.

Lab validation — Complete technical evaluation with an accredited Service Laboratory or Warfare Center.

- Choose the lab that owns the relevant test environment (e.g., NSWC Crane for trusted electronics/EW, AEDC for hypersonic ground test).
- Sequence this before your first big trade show push, not after.

Event strategy — Targeted, pre-scheduled technical exchanges, not passive booth exhibitions.

- Request one-on-one technical interchange meetings (TIMs) with named PEO/PM technical staff in advance.
- Bring the technical data package to the TIM — the goal is a follow-up evaluation request, not booth traffic.

Exhibit 71: Year One Checklist: Building the Engine



Source: GMA Defense & Technology Practice

Year Two: Structuring the Program

POM alignment — Get your capability mapped into the Service's POM submission during Programming.

- The Programming phase runs 18–24 months ahead of the fiscal year of execution — pitching in the same fiscal year you want funding means you've already missed the window.
- Ask the PM directly which Program Element number your capability would sit under, and whether that PE has room, is fully subscribed, or needs a new line.

Operational demos — Field trials with active-duty operators under contested conditions.

- Jammed GPS, degraded comms, adverse weather — this is the data a PM actually uses to defend the program internally.
- Capture results in the same technical-data-package format so evidence accumulates instead of resetting each time.

Transition vehicle — Non-competitive Phase III SBIR, MTA, or production OTA.

- Phase III sole-source authority requires a Phase I/II SBIR lineage; if you never ran one, MTA or a production OTA are your realistic paths.
- Negotiate the production-OTA follow-on language into the prototype agreement itself, not after delivery.

Open architecture compliance — Native compliance with MOSA, SOSA, OMS, CMOSS.

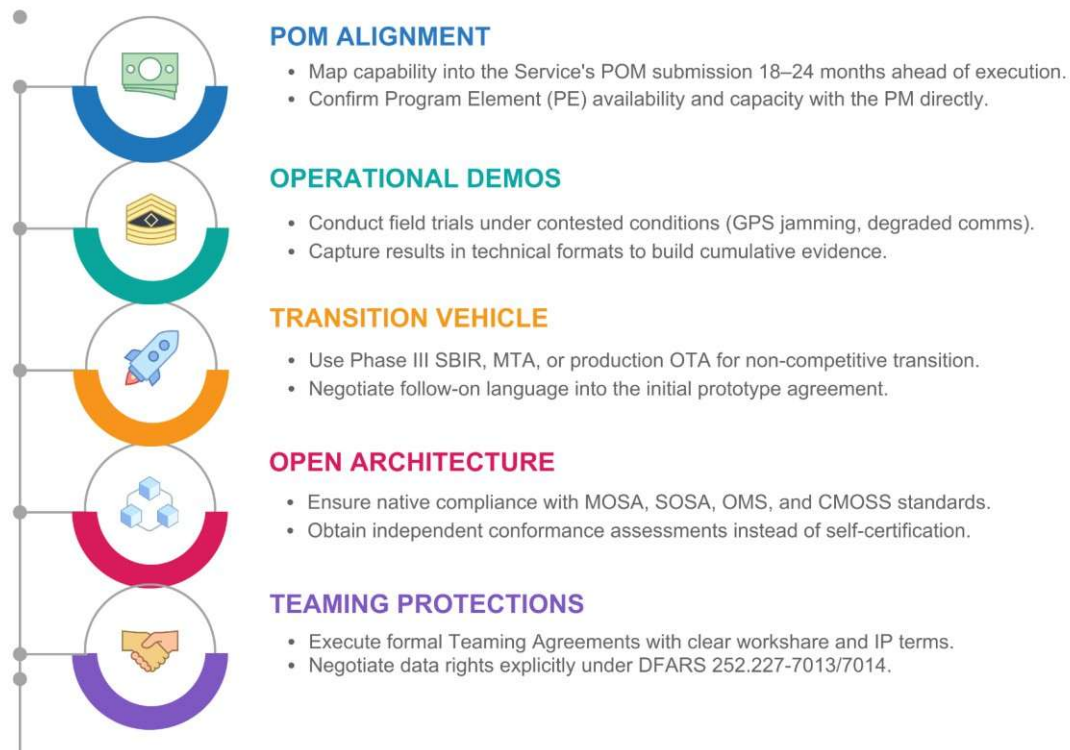
- Get an independent conformance assessment, not just self-certification.
- Treat this as a Year One design constraint, not a retrofit.

Teaming protections — Formal Teaming Agreements with clear workshare and IP terms.

- Negotiate data rights (technical data and computer software) explicitly under DFARS 252.227-7013/7014.
- Put minimum workshare percentages and any exclusivity window in writing.

Exhibit 72: Year Two Checklist: Structuring the Program

Year Two: Structuring the Program



Source: GMA Defense & Technology Practice

Year Three: Scaling the Supplier

- **Manufacturing infrastructure** — Demonstrated 3–5x surge capacity.
 - Run an actual surge production test and document throughput — projections don't satisfy a PM sizing a wartime reserve requirement.
 - Dual-source critical subcomponents before scaling, not after.
- **Compliance & security** — CMMC Level 2/3, ATOs, Facility Security Clearance.
 - Start FCL sponsorship at the start of Year Two — clearance timelines vary materially based on sponsorship, ownership, foreign-interest mitigation, facility readiness, and government workload, so initiate early and verify current requirements for the target work.
 - Budget C3PAO assessment as a recurring line item, not a one-time cost.
- **Supply chain hardening** — Sub-tier component sourcing mapped against NDAA and program-specific restrictions at the component level; requirements vary by system, funding source, and contract clause.
 - Map the bill of materials at the component level — a compliant assembly can still contain a non-compliant part.
 - Build redundancy into the compliant base itself; a single compliant sole-source supplier just relocates the risk.
- **Congressional engagement** — Brief Appropriations staff to protect your target Program Element.
 - Engage Members and staff whose districts contain your manufacturing and supply chain footprint.
 - Time engagement to the Appropriations markup calendar, not your own PR calendar.
- **Sustainment framework** — FSR model, spare parts pipeline, depot repair capability.
 - Price sustainment into the original contract structure, not as a follow-on negotiation.
 - Stand up forward-deployed FSR presence before full fielding, not after the first failure report.

Exhibit 73: Year Three Checklist: Scaling the Supplier

Year Three: Scaling the Supplier



Source: GMA Defense & Technology Practice

4.3 Common Failure Patterns

- **Booth-only BD strategy** — visibility isn't pipeline; conferences without pre-scheduled technical exchanges generate leads that never convert.
- **Selling to the wrong PEO** — pitching the office that owns the platform when budget and requirement authority sits with a different PEO or cross-cutting office (CDAO, DIU).
- **Premature scaling** — building production capacity ahead of a signed transition vehicle. Replicator's own fielding shortfall shows even a well-funded DoW initiative can under-deliver against its targets — a startup's balance sheet has far less runway to absorb that gap.
- **Unprotected IP in teaming arrangements** — entering a prime relationship without negotiated data rights, then discovering the prime can recompute using your own technical data.
- **Missing the PPBE window** — pitching in the fiscal year you want funding rather than 18–24 months earlier.
- **Sustainment as an afterthought** — winning the initial contract without a priced sustainment plan, then losing leverage once the government controls the recompute.

How GMA Helps — Part 4: Domain-Specific Execution Matrix

The Founder Self-Assessment Scorecard in Appendix A is close to what GMA's Key Considerations Analysis does before a client commits resources to a pursuit — a structured go/no-go read on whether an opportunity is actually winnable.

Our AI-powered Grant Matching Engine screens across 10,000+ federal, state, and foundation programs to surface the domain-specific opportunities (SBIR, STTR, BAA, OTA, and beyond) that fit a company's actual position on this matrix, and our execution team works with clients to carry the highest-probability ones through to a submitted, fundable proposal.

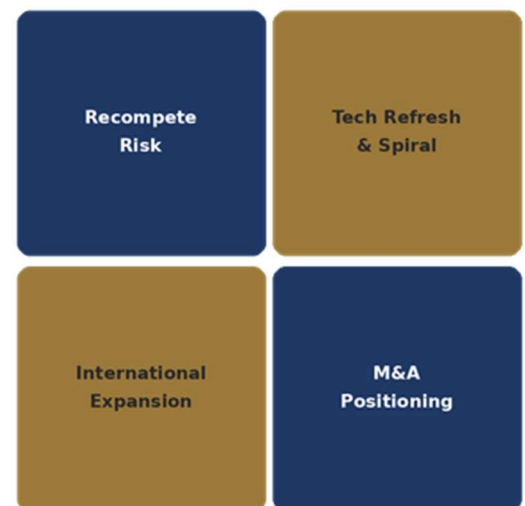


Part 5: Year Four and Beyond — Becoming an Enduring Powerhouse

“Your program is never permanently won.”

Everything through Year Three builds toward a single Program of Record. But a Program of Record is not a powerhouse — it's one revenue stream, one customer, one recompute clock, one set of political sponsors. What happens next determines whether the company outlives its first program. Four fronts have to be managed simultaneously from this point forward: recompute risk, tech refresh, international expansion, and M&A positioning.

**Four Fronts,
Held at Once**



5.1 Recompute Risk — Your Program Is Never Permanently Won

Every Program of Record carries an expiration date — written into the contract as a base-plus-options structure, or implied by the government's obligation to periodically test the market. Founders who treat their first award as a finish line make a predictable mistake: they staff down capture, let teaming relationships atrophy, and stop tracking the customer's evolving requirements — only to discover, eighteen months before recompute, that a better-funded competitor has quietly built a bid for the same work. The government owes incumbency no loyalty; it owes taxpayers competition. A company that can't articulate, in writing, why it will win its next recompute two years out has already put the program at risk.

The discipline mirrors Year One capture, but runs continuously instead of episodically: a living relationship with the program office between cycles, not just during proposal windows; tracking every modification, funding plug, and requirements shift as an early signal of the next solicitation's emphasis; and actively managing past performance — CPARS ratings, on-time delivery, cost control — as a strategic asset rather than a byproduct of doing the work..

Exhibit 74: MQ-1 Predator



Source: General Atomics Aeronautical Systems, Inc. (GA-ASI), MQ-1 Predator program materials

Winning a Program of Record starts a new clock, not a finish line. Production OTAs and MTA pathways typically run 2–5 years before facing either a recompetete or a transition to a standard FAR-based procurement, at which point the "nontraditional" advantages that got you the initial award (speed, flexible IP, OTA structure) may no longer apply. Treat the day you win the program as the day you start building the evidence base — cost performance, schedule performance, sustainment metrics — for the recompetete, rather than waiting until the recompetete RFP drops.

5.2 Tech Refresh and Spiral Development

Sustaining a Program of Record means resisting a trap that killed many legacy primes: freezing the design at first delivery and treating every year after as pure production. The Department has shifted toward spiral development — fielding a capable system now, then upgrading in disciplined increments as the threat and technology evolve, rather than waiting for a "perfect" next-gen program that arrives late and over budget. A company not already scoping its Block II improvements while delivering Block I is ceding that ground to a competitor who is.

This takes two things founders often underinvest in: a standing internal R&D line — funded from margin, IRAD, or follow-on SBIR/STTR Phase III work — that keeps the core technology moving even when near-term funding is production-focused; and a formal channel for routing operator feedback into the engineering roadmap, so spirals are driven by field data, not guesses. Programs rarely lose their edge all at once — they lose it one skipped refresh cycle at a time.

Exhibit 75: Spiral Development in Practice



Source: Stock Image

Where 5.1 is about whether the government keeps paying you at all, this is about whether you're the one delivering what they pay for next. Programs of Record are built around a spiral/increment model — the government expects the capability to evolve over the life of the program rather than freeze at initial fielding.

A company that treats its Year Three win as a finished product rather than the first increment of an ongoing relationship cedes the tech-refresh opportunity to whoever shows up with the next capability upgrade. Budget and staff for spiral development from day one of production, not as a reactive response to a competitor's newer offering.

5.3 International Expansion — FMS and Direct Commercial Sales

Once a capability is established with a U.S. Program of Record, Foreign Military Sales (FMS, government-to-government, State Department-administered) and Direct Commercial Sales (DCS, company-to-foreign-government, subject to ITAR/EAR export licensing) become real revenue paths — and a validated U.S. program is usually the credibility prerequisite allied governments look for.

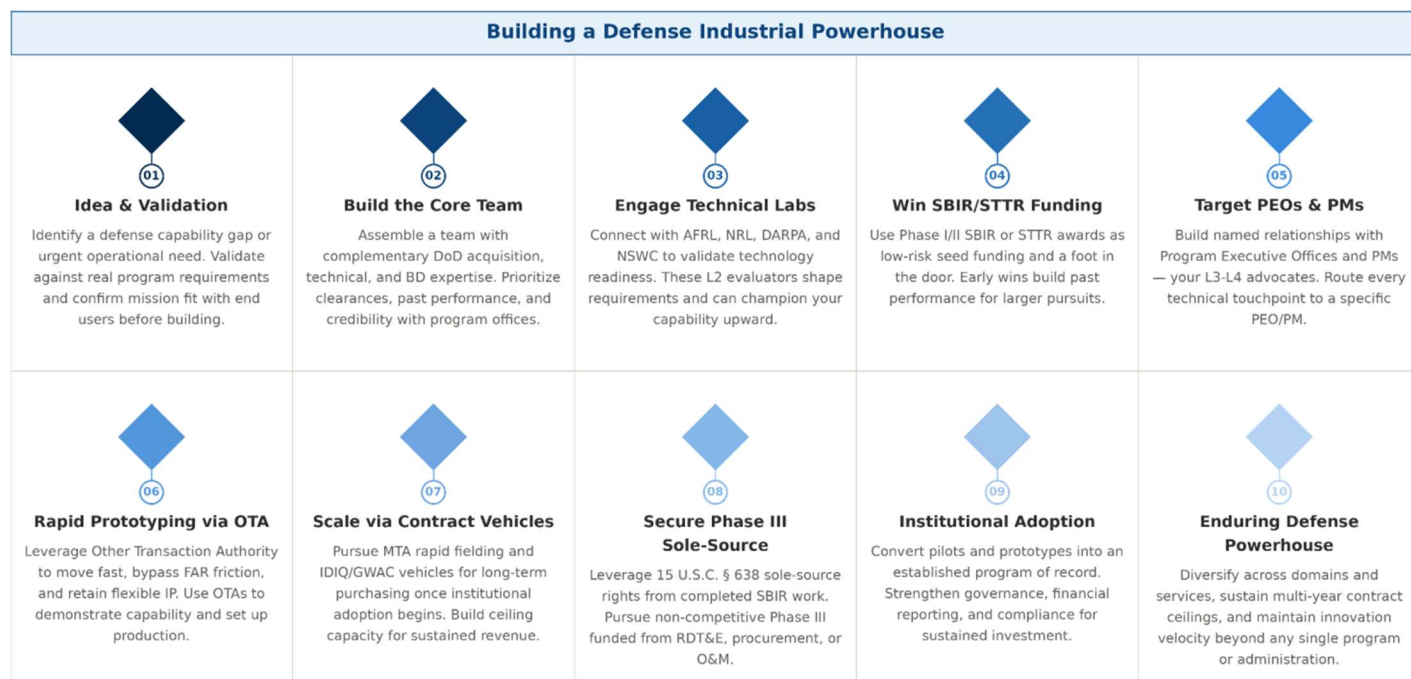
This is not theoretical — it's the standard sequencing for platforms that reach international markets today. Shield AI's V-BAT was first established with U.S. Navy, Army, and Coast Guard programs; only afterward did it move into allied sales, including an eight-aircraft order from the Netherlands Ministry of Defence in July 2025 to equip the Royal Netherlands Navy and Marine Corps, and selection by the Japan Maritime Self-Defense Force to operate from its warships.

The lesson for a founder: international sales conversations are worth having early, but they close faster and easier once you have a U.S. Program of Record to point to — sequence accordingly.

5.4 M&A — Buyer or Target

By Year Four, most companies find themselves on one side or the other of consolidation. Being acquired isn't automatically a failure outcome (see the Palantir/DCGS-A2 case, where the underlying commercial-item leverage that won the protest also made the company an attractive scaled acquirer of adjacent capability rather than an acquisition target) — but a company that hasn't deliberately decided which side it intends to be on typically ends up on the wrong one, acquired opportunistically by a prime that wants the technical data package more than the team.

Exhibit 76: Building a Defense Industrial Powerhouse



Source: GMA Defense & Technology Practice

The teaming-agreement discipline from Part 2 (protect your data rights, avoid unprotected exclusivity) is what preserves the option to be a buyer rather than only ever being available as a target.

Strategic Company Program Roadmap



Technical Validation

- Have you completed a technical evaluation with an accredited Service Laboratory or Warfare Center (not just internal testing)?
- Do you have documented failure-mode data, not just best-case performance metrics?
- Have you run structured discovery interviews with company- or battalion-level operators (L1), with written capability-gap memos?



Institutional Adoption

- Can you name the specific PM who owns the requirement you're pursuing?
- Do you know the Program Element (PE) number your capability would sit under?
- Has a demonstration been run by active-duty operators under contested conditions — not by your own engineers under controlled conditions?
- If you're relying on a prime relationship, do you have negotiated data rights (DFARS 252.227-7013/7014) rather than a verbal or informal understanding?



Program of Record

- Is your capability included in the Service's current POM submission?
- Do you have a documented (not verbal) commitment to a follow-on production OTA, or Phase III SBIR eligibility confirmed?
- Have you validated open-architecture compliance (MOSA/SOSA/OMS/CMOSS) via independent conformance assessment?



Scaled Production

- Have you demonstrated (not modeled) 3–5x surge production capacity?
- Are your sub-tier components mapped and verified NDAA-compliant at the individual part level?
- Do you know whether your next major proposal will cross the \$35M CAS or \$10M certified cost/pricing data thresholds, and is your accounting system ready if so?
- Is sustainment (FSR model, spares, depot repair) priced into your contract structure rather than planned as a future negotiation?



Enduring Powerhouse

- Do you have a recompute evidence base being built continuously, rather than assembled reactively when the recompute RFP drops?
- Is spiral/tech-refresh development budgeted as an ongoing line item, not a response to competitive pressure?
- Have you had at least one substantive FMS/DCS conversation with an allied government?

How GMA Helps — Part 5: Year Four and Beyond

Recompete risk and spiral development mean the evidence base for "why fund us again" has to be built continuously — not assembled after an RFP drops. GMA offers ongoing, retainer-based support, including funding-pathway analysis and preparation support for post-award contract negotiation, compliance, and execution, coordinated with legal and contracting specialists where the negotiation itself requires licensed advice. Funding intelligence, compliance tracking, and performance documentation stay active between pursuits rather than restarting cold each cycle. The recompete evidence base described in 5.1 becomes a byproduct of how the program is already being run, not a scramble assembled after the fact.

For founders moving into FMS/DCS conversations or evaluating M&A, GMA's team supports allied-market opportunity intelligence and growth-stage strategic advisory, and coordinates deal structuring and negotiation preparation with qualified legal, accounting, and investment-banking specialists. This extends past pure grant-writing into the broader funding and exit strategy an enduring defense supplier needs, without GMA substituting for licensed legal or banking advice.

On the M&A side specifically, GMA works closely with mid-market investment banks that specialize in defense and government-services transactions and can make introductions and help prepare a company for that process. A company deciding whether it intends to be a buyer or a target — the question posed in 5.4 — has that conversation with bankers who already understand the technical data rights and teaming-agreement discipline built up since Part 2, rather than starting the education process from zero once a term sheet is already on the table.

Conclusion: The Winning Framework

How This Maps to GMA's Work

Each stage of this playbook pairs with a specific GMA deliverable and the decision it's meant to support — summarized below before the closing framework that follows.

Playbook stage	GMA deliverable	Client decision supported
Market entry	Defense Readiness Diagnostic and target-market map	Enter / pause / redirect
Customer discovery	Mission-gap interview plan and stakeholder map	Confirm target problem
Technical validation	Evidence package and demonstration strategy	Fund demonstration or defer
Transition	Acquisition-pathway and budget-alignment plan	Select funding/contract route

Playbook stage	GMA deliverable	Client decision supported
Scale-up	Industrial-readiness gap assessment	Invest in production/compliance
Congressional alignment	Issue brief and lawful stakeholder engagement plan	Advance a defined institutional objective
Sustainment/recompete	Program-health and recompile readiness review	Protect/expand position

Deliverable names above should be confirmed against GMA's current service catalog before this table is used externally.

Exhibit 78: Human-Machine Teaming in the Field



Source: Stock Image

Long-term success in defense technology is rarely a function of technical capability alone — it's a function of how low-friction the adoption path is for the people who have to buy, integrate, and sustain the system. The Ghost-X case shows what it looks like when this works: Blue UAS listing, lab-grade credibility, and alignment with an active-duty requirement converged to produce a directed-requirement award and a Replicator tranche slot.

Shield AI's Hivemind shows the software-side version: prove it on your own hardware, then decouple it so every prime's airframe becomes a potential adoption path instead of a competitor. Palantir's DCGS-A2 case shows what's available if adoption stalls anyway — a real, litigable statutory protection, best used as a last resort once you're large enough to invoke it.

And Replicator's own fielding shortfall is the department-level version of every “premature scaling” failure pattern in this playbook: speed and visibility outrunning production reality and system integration.

The organizations that win align to the warfighter's operational need, retire integration and schedule risk for the Program Manager, work the PPBE budget cycle deliberately rather than reactively, and invest early in scalable domestic manufacturing — while pricing sustainment and protecting IP from day one rather than after the first production contract. Do those things consistently, and the platform stops being an experimental prototype and becomes a program of record.

Private Company Profiles

The companies below aren't a comprehensive market map they're the pattern-matches GMA's Defense & Tech practice tracks against the framework in Parts 1–5: funded, post-seed defense-tech and dual-use companies executing the same capture playbook this document describes, at various points on the same maturity curve.

Space / Orbital

Defense Hardware, Space Domain Awareness (SDA) & Infrastructure

Rhea Space Activity

HQ / Founded	Washington, DC Founded 2018
Founders	Shawn Usman (Founder & CEO, astrophysicist and former U.S. Intelligence Community official) and Cameo Lance (Co-Founder & COO, physicist).
Funding	~\$7M total raised (including a ~\$6M Series A). Clean SBIR-to-Phase III transition story, expanding early Phase I/II work into a \$1.8M AFRL contract (FA9451-25-C-X014), a \$1.9M SpaceWERX Direct-to-Phase II award, and a \$1.9M SpaceWERX "Vanguard" award leveraging NASA-developed AutoNav.

What They Do: Builds autonomous optical navigation algorithms, infrared sensors, and space domain awareness analytics platforms (such as the LUNINT Dashboard) for cislunar operations, intelligence/surveillance/reconnaissance (ISR), and directed energy applications.

Capture-Curve Position: Institutional Adoption — building the government relationships and non-dilutive award trail that precede a funded program.

Why It's Interesting: Illustrates an effective strategy for bootstrapping deep-tech R&D via non-dilutive defense awards—scaling intelligence-focused algorithms into core tactical maneuver tools for the U.S. Space Force and Intelligence Community.

Watch Item: Funding is thin (~\$7M total) relative to the technical ambition — the SBIR-to-Phase-III pathway has to keep converting, or the company risks running out of runway before a Program of Record materializes.

Exhibit 79: Rhea Space Activity



Source: Rhea Space Activity public disclosures, press releases, and funding announcements

ThinkOrbital

HQ / Founded	Colorado Founded 2021
Founders	Lee Rosen (Co-Founder & CEO, former SpaceX VP of Mission & Launch Operations and retired U.S. Air Force Colonel).
Funding	\$8M raised; multiple Phase II SBIR awards, plus a \$4M TACFI and a \$40M STRATFI.

What They Do: Builds autonomous in-space manufacturing, high-precision electron-beam metal welding, and orbital inspection platforms ("ThinkX," "ThinkPlatform") utilizing penetrating X-rays for Space Domain Awareness.

Capture-Curve Position: Technical Validation / Demonstrated Capability — three space flights and multiple Phase II government awards demonstrate continued technical and programmatic progress.

Why It's Interesting: ThinkOrbital has now flown in space three times and continues to advance autonomous in-space manufacturing and orbital inspection capabilities. Its technology combines electron-beam welding with non-destructive X-ray inspection and is also being applied to terrestrial use cases. The company reports a patented drone-to-drone X-ray imaging and material characterization application for military, law enforcement, first responders, and energy infrastructure inspection.

Recent Momentum: ThinkOrbital reports **\$8M** raised, three space flights, multiple Phase II SBIR awards, a **\$4M TACFI**, and a **\$40M STRATFI**. The company is also expanding its technology into terrestrial X-ray imaging and material characterization applications.

Exhibit 80: ThinkOrbital



Source: ThinkOrbital public disclosures, press releases, funding announcements, and company-provided updates

Array Labs

HQ / Founded	Palo Alto, CA Founded 2021
Founders	Andrew Peterson (Co-Founder & CEO) and Isaac Robledo (Co-Founder).

What They Do: Operates formation-flying satellite clusters using multistatic Synthetic Aperture Radar (SAR) interferometry to build real-time, 3D volumetric maps of Earth for mission planning, ISR, and Airborne Moving Target Indication (AMTI).

Capture-Curve Position: Technical Validation — still proving the core technology works before institutional adoption begins.

Why It's Interesting: Directly challenges legacy 2D optical/SAR imagery by providing all-weather 3D wide-area radar tracking—positioning itself as a critical payload and constellation supplier for the military's "Golden Dome" architecture.

Watch Item: No funding or contract figures are disclosed in public sources — the Golden Dome positioning is aspirational so far, not yet backed by a named award.

Exhibit 81: Array Labs



Source: Array Labs public disclosures, press releases, and funding announcements

Agile Space Industries

HQ / Founded	Durango, CO Founded 2009 (originally Advanced Mobile Propulsion Test)
Founders	Daudi Barnes (Founder & CTO, veteran propulsion engineer) & Chris Pearson (CEO, ex-Airbus/Redwire aerospace executive).
Funding	~\$41.5M raised (Series A led by Caruso Ventures and Howdy Partners). Backed by Lockheed Martin Ventures and Veteran Ventures Capital. Supplied hypergolic propulsion hardware to True Anomaly for USSF's VICTUS HAZE Tactically Responsive Space mission.

What They Do: Designs, 3D-prints, and hot-fire tests chemical propulsion thrusters, bipropellant rocket engines, and fuel tanks under one roof.

Capture-Curve Position: Institutional Adoption — building the government relationships and non-dilutive award trail that precede a funded program.

Why It's Interesting: Eliminates long aerospace supply-chain lead times by combining additive manufacturing with co-located engine testing, becoming a primary tier-1 propulsion supplier for responsive space defense and space-based intercept missions.

Watch Item: Growth runs through other primes' programs (it supplied True Anomaly's VICTUS HAZE mission) rather than owning a program of its own — its trajectory is tied to its customers' award cycles, not its own.

Exhibit 82: Agile Space Industries



Source: Agile Space Industries public disclosures, press releases, and funding announcements

Antaris

HQ / Founded	Los Altos, CA Founded 2021
Founders	Tom Barton (Co-Founder & CEO, ex-Planet and enterprise software executive) and Shankar Chandran (Co-Founder).
Funding	~\$42M raised (\$28M Series A led by WestWave Capital). Backed by Lockheed Martin Ventures. Awarded a \$1.7M SpaceWERX Orbital Prime Direct-to-Phase II contract, validated self-healing spacecraft for USSF, and secured an MDA SHIELD contract.

What They Do: Provides a cloud-native SaaS suite (TrueTwin digital twins, SatOS) that automates satellite design, hardware-in-the-loop simulation, and fleet operations.

Capture-Curve Position: Institutional Adoption — building the government relationships and non-dilutive award trail that precede a funded program.

Why It's Interesting: Replaces custom, slow-to-build ground infrastructure with a standardized, software-defined operating system that reduces satellite fleet integration timelines from years to months.

Watch Item: As a SaaS layer for satellite operations, its moat depends on becoming the default tool before a government-native or prime-built alternative locks in first.

Exhibit 83: Antaris



Source: Antaris public disclosures, press releases, and funding announcements

Cislunar Industries

HQ / Founded	Denver/Loveland, CO Founded 2017
Founders	Gary Calnan (Founder & CEO, finance/aerospace entrepreneur) and Joe Bush (Co-Founder).
Funding	~\$5.7M raised (including a \$1M Colorado ONE Fund seed). Holds NASA SBIRs, a \$1.7M SpaceWERX Direct-to-Phase II contract, a DARPA LunA-10 architecture study award, and a \$1.9M Sustained Space Maneuver Challenge award.

What They Do: Engineers micro-foundries for in-space metal recycling and manufactures Power Processing Units (PPUs) for electric propulsion spacecraft engines.

Capture-Curve Position: Technical Validation — still proving the core technology works before institutional adoption begins.

Why It's Interesting: Solves the logistics bottleneck of deep-space missions by enabling on-orbit metal processing from orbital debris or spent rocket bodies, converting space junk into structural materials and thruster fuel.

Watch Item: Still in the study-and-SBIR stage (DARPA LunA-10 is an architecture study, not a build contract) — in-space manufacturing at any real scale remains years from an on-orbit test.

Exhibit 84: Cislunar Industries



Source: Cislunar Industries public disclosures, press releases, and funding announcements

Commercial / Civil Focus

Starcloud (formerly Lumen Orbit)

HQ / Founded	Redmond, WA Founded 2024
Founders	Philip Johnston (Co-Founder & CEO) and Adi Oltean (Co-Founder & CTO, ex-Microsoft).
Funding	~\$200M raised (\$170M Series A at a ~\$1.1B valuation led by Benchmark and EQT Ventures).

What They Do: Operates high-density orbital data centers equipped with GPU clusters to process AI payloads in space.

Market Position: Well-capitalized commercial play (\$200M raised, ~\$1.1B valuation) with early DoW interest but no disclosed defense contract

Why It's Interesting: Bypasses terrestrial power constraints for large-scale AI compute while providing low-latency edge processing for orbital sensors, drawing early DoW interest for hosted payloads.

Watch Item: The defense narrative is “early interest,” not a signed award — worth confirming whether any hosted-payload discussion has actually converted before treating this as a defense comparable.

Exhibit 85: Starcloud



Source: Starcloud public disclosures, press releases, and funding announcements

Arinna

HQ / Founded	South San Francisco, CA Founded 2024
Founders	Stanford University researchers in the Breakthrough Energy Fellows program.
Funding	~\$4M seed led by Spacecadet Ventures, Anorak Capital, and Breakthrough Energy Foundation.

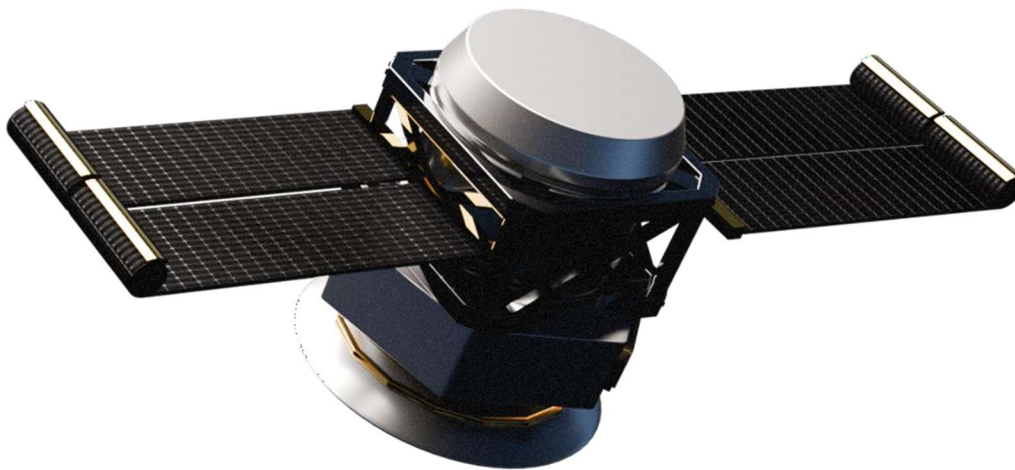
What They Do: Manufactures ultra-thin flexible solar sheets using transition-metal-dichalcogenide (TMD) photovoltaics.

Market Position: Pre-seed materials-science play (\$4M seed) with no defense contract or pilot yet disclosed

Why It's Interesting: A pure material-science play that vastly improves watts-per-kilogram power density ratios for deep-space and long-duration orbital platforms.

Watch Item: *The profile cites no government engagement at all; this is a materials bet whose defense relevance is still entirely prospective.*

Exhibit 86: Arinna



Source: Arinna public disclosures, press releases, and funding announcements

Aviation

Dual-Use & Next-Gen Aviation Platforms

JetZero

HQ / Founded	Long Beach, CA Founded 2021
Founders	Tom O'Leary (Co-Founder & CEO) and Mark Page (Co-Founder & CTO, aerodynamicist who pioneered the modern BWB at McDonnell Douglas).
Funding	>\$1.0B in funding and commitments, including a \$235M USAF/DIU demonstrator contract and a \$175M Series B led by B Capital with Northrop Grumman and RTX.

What They Do: Builds full-scale Blended Wing Body (BWB) aircraft (Z4) that cut fuel burn and emissions by 50% for commercial passenger and military tanker/cargo missions.

Capture-Curve Position: Program of Record (transitioning) — moving from prototype/demonstrator funding toward a formal Program of Record.

Why It's Interesting: Represents the first serious effort in decades to displace traditional "tube-and-wing" aircraft, offering massive range extensions critical for Pacific theater air logistics.

Watch Item: A full-scale blended-wing-body aircraft is a genuinely novel airframe — first-of-kind demonstrator execution risk is real no matter how strong the funding and prime backing look on paper.

Exhibit 87: JetZero



Source: JetZero public disclosures, press releases, and funding announcements

Boom Supersonic

HQ / Founded	Denver, CO Founded 2014
Founders	Blake Scholl (Founder & CEO, technology entrepreneur and aviation founder).
Funding	Multi-hundred million capital raised. Holds a \$60M AFWERX STRATFI award, USAF SBIR Phase II contracts, and a strategic partnership with Northrop Grumman for executive transport and fast-response ISR variants.

What They Do: Builds commercial supersonic passenger airliners (Overture), custom engines (Symphony), and supersonic flight testbeds (XB-1).

Capture-Curve Position: Institutional Adoption — building the government relationships and non-dilutive award trail that precede a funded program.

Why It's Interesting: Revives civil supersonic flight while providing military planners with a commercial, off-the-shelf platform for time-critical executive transport and high-speed multi-mission operations.

Watch Item: *The core business is commercial supersonic passenger flight; the military variant is a secondary bet riding on the commercial program's success, not an independent line of funding.*

Exhibit 88: Boom Supersonic



Source: Boom Supersonic public disclosures, press releases, and funding announcements

MightyFly

HQ / Founded	San Leandro, CA Founded 2019
Founders	Manal Habib (Founder & CEO, MIT aerospace engineer and flight control software expert).
Funding	Backed by Draper Associates. Winner of a \$1.25M AFWERX SBIR Phase II contract; conducted autonomous point-to-point military payload demonstrations for the USAF.

What They Do: Manufactures autonomous, long-range hybrid-eVTOL cargo aircraft ("Cento") featuring an automated internal cargo loading and unloading system.

Capture-Curve Position: Technical Validation — still proving the core technology works before institutional adoption begins.

Why It's Interesting: Fully automates mission logistics—loading, flying up to 600 miles, unloading, and returning without human ground crew intervention.

Watch Item: Funding is SBIR-scale (\$1.25M award, Draper-backed) relative to the certification and manufacturing challenge of a certified autonomous cargo aircraft — no Series A has been disclosed.

Exhibit 89: MightyFly



Source: MightyFly public disclosures, press releases, and funding announcements

Natilus

HQ / Founded	San Diego, CA Founded 2016
Founders	Aleksey Matyushev (Co-Founder & CEO) and Anatoly Kvapil (Co-Founder).
Funding	\$28M Series A (led by Draper Associates & The Veterans Fund). Bidding alongside defense partners on the \$40B USAF Next-Generation Air-refueling System (NGAS) program.

What They Do: Designs blended-wing-body autonomous cargo drones (KONA) and commercial passenger airliners (Horizon) engineered to maximize volumetric payload efficiency.

Capture-Curve Position: Institutional Adoption — building the government relationships and non-dilutive award trail that precede a funded program.

Why It's Interesting: Increases internal volume by 60% compared to traditional aircraft of equal weight, delivering an affordable alternative for contested military airlift operations.

Watch Item: NGAS is a bid, not an award — the entire near-term thesis rides on winning a competitive process against much larger, better-capitalized primes.

Exhibit 90: Natilus



Source: Natilus public disclosures, press releases, and funding announcements

Defense Hardware, Autonomy & Sensors

Anello Photonics

HQ / Founded	Santa Clara, CA Founded 2018
Founders	Dr. Mario Paniccia (CEO, former Intel Fellow and silicon photonics pioneer) and Hugh Thomas (Co-Founder).
Funding	Series B-2 raised (\$25M led by MESH). Backed by In-Q-Tel and Lockheed Martin Ventures. Secured a DIU contract, Navy SBIR Phase II, and a \$20M APFIT production award.

What They Do: Manufactures chip-scale Silicon Photonics Optical Gyroscopes (SiPhOG) providing precision inertial navigation in GPS-denied environments.

Capture-Curve Position: Program of Record — has converted institutional interest into a funded, budgeted contract vehicle.

Why It's Interesting: Replaces heavy, expensive Fiber-Optic Gyros (FOGs) with microchip-scale optical tech, allowing small uncrewed aerial vehicles and munitions to maintain pinpoint navigation under heavy electronic warfare.

Watch Item: The strongest-positioned of the space/sensor group (APFIT production award plus a DIU contract) — the open question is whether gyroscope demand scales fast enough to justify the Series B-2 valuation.

Exhibit 91: Anello Photonics



Source: Anello Photonics public disclosures, press releases, and funding announcements

Seasats

HQ / Founded	San Diego, CA Founded 2020
Founders	Mike Flanigan (Co-Founder & CEO) and Dylan Hicks (Co-Founder).
Funding	>\$40M raised (\$20M Series A led by Konvoy & Shield Capital; strategic investment from L3Harris). Holds an \$89M SBIR Phase III IDIQ and a \$24M APFIT award.

What They Do: Manufactures low-cost, solar-powered, highly survivable Autonomous Surface Vessels (Lightfish, Heavyfish) for maritime domain awareness and electronic warfare.

Capture-Curve Position: Program of Record — has converted institutional interest into a funded, budgeted contract vehicle.

Why It's Interesting: Delivers months-long ocean endurance in small, affordable surface craft that are difficult to target, giving naval commanders cheap, distributed sensor grids across vast oceans.

Watch Item: *The autonomous surface vessel category is increasingly crowded; the SBIR Phase III IDIQ and APFIT award are real, but survivability claims need to keep holding up as more USV competitors enter maritime domain awareness.*

Exhibit 92: Seasats



Source: Seasats public disclosures, press releases, and funding announcements

Defense Unicorns

HQ / Founded	Colorado Springs, CO / San Antonio, TX Founded 2021
Founders	Rob Slaughter (Co-Founder & CEO, former USAF cyber operations officer) and Arthur Ferrell (Co-Founder).
Funding	\$136M Series B led by Bain Capital Tech Opportunities at a >\$1B valuation. Awarded a \$300M DoW software contract and a \$15M SpaceWERX STRATFI.

What They Do: Develops open-source software delivery platforms (Zarf, Leapfrog) that enable air-gapped deployment of modern AI, mission software, and zero-trust applications into classified environments.

Capture-Curve Position: Scaled Production — operating as an established production or infrastructure supplier to the defense industrial base.

Why It's Interesting: Solves the "last mile" software delivery problem for defense tech—allowing edge systems, subsea craft, and classified satellites to receive continuous, cloud-native code updates without touching the internet.

Watch Item: The open-source delivery-platform model means the moat depends on becoming the default air-gapped deployment layer before a competing standard locks in across DoW.

Exhibit 93: Defense Unicorns



Source: Defense Unicorns public disclosures, press releases, and funding announcements

Firestorm Labs

HQ / Founded	San Diego, CA Founded 2022
Founders	Dan Magy (Co-Founder & CEO) and Ian Munsell (Co-Founder & CTO).
Funding	~\$153M raised (\$82M Series B). Backed by Lockheed Martin Ventures, In-Q-Tel, and Booz Allen Ventures. Holds a \$100M USAF IDIQ and a ~\$30M Indo-Pacific drone contract.

What They Do: Manufactures modular 3D-printed UAVs (Tempest) and mobile, containerized automated factories ("xCell") for forward-deployed drone production.

Capture-Curve Position: Program of Record — has converted institutional interest into a funded, budgeted contract vehicle.

Why It's Interesting: Shifts drone production directly to combat theater bases—letting commanders print custom, mission-specific airframes in shipping containers within hours of an operational need.

Watch Item: Forward-deployed containerized drone factories are a novel manufacturing model — scaling xCell production capacity across multiple theaters simultaneously is unproven at the volumes the contracts imply.

Exhibit 94: Firestorm Labs



Source: Firestorm Labs public disclosures, press releases, and funding announcements

Swarm Aero

HQ / Founded	Oxnard, CA Founded 2022
Founders	Founded by Danny Goodman (CEO); team draws heavily from Scaled Composites, SpaceX, Northrop Grumman, Boeing, Airbus, and Archer, with billions of dollars in prior DoW contract experience among them.
Funding	~\$59M raised to date (Two Sigma Ventures, Founders Fund, Khosla Ventures, Coatue, Construct Capital, a16z, MaC Venture Capital, Silent Ventures). \$10M+ in Air Force Research Laboratory (AFRL) contracts for high-rate composite manufacturing R&D and aircraft component production.

What They Do: Builds large, multimission uncrewed aerial vehicles (Group 5-class UAS) paired with swarm command-and-control software, enabling small operator teams to coordinate large numbers of aircraft in communications-degraded, contested environments.

Capture-Curve Position: Institutional Adoption — building the government relationships and non-dilutive award trail that precede a funded program.

Why It's Interesting: Pairs a proven aerospace manufacturing bench (Scaled Composites/SpaceX veterans) with a composite production process built for volume from day one — targeting the "affordable mass" gap in large UAS, backing it with a new 80,000 sq. ft. Fayetteville, AR manufacturing center built specifically to scale output.

Watch Item: The AFRL funding to date is for manufacturing R&D and production readiness, not a unit order — a real acquisition commitment for the aircraft itself hasn't been disclosed yet.

Exhibit 95: Swarm Aero



Source: Swarm Aero public disclosures, press releases, and funding announcements

Swarm Defense

HQ / Founded	Detroit, MI Founded 2024
Founders	Spun out of Firefly Drone Shows by drone show pioneers and robotics hardware engineers.
Funding	~\$1.44M in grant/accelerator funding. Selected for the DoW "Drone Dominance" Phase I gauntlet.

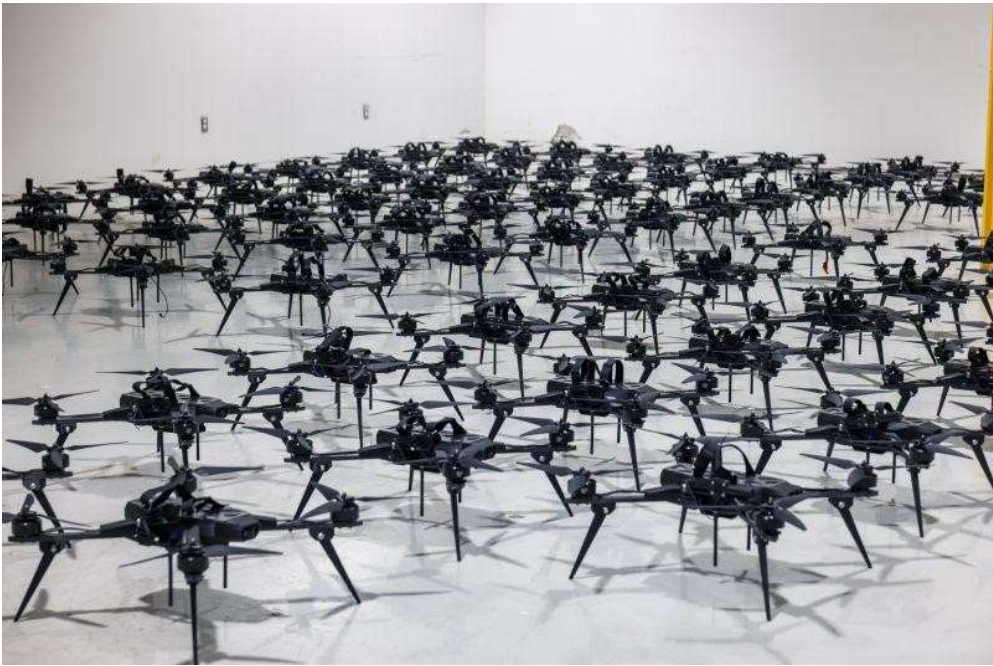
What They Do: Converts commercial light-show drone formation algorithms into tactical attack swarms and threat emulators.

Capture-Curve Position: Technical Validation — still proving the core technology works before institutional adoption begins.

Why It's Interesting: Repurposes proven civilian swarm coordination software to deploy high-density, NDAA-compliant attack swarms at low unit costs.

Watch Item: *Extremely early (under \$2M, still in a Phase I gauntlet with no guarantee of advancing) — the light-show-to-attack-swarm concept still needs to prove out under real electronic warfare conditions.*

Exhibit 96: Swarm Defense



Source: Swarm Defense public disclosures, press releases, and funding announcements

Grid Aero

HQ / Founded	San Leandro, CA Founded 2024
Founders	Founded by experienced aerospace autonomous flight engineers.
Funding	~\$26M total (\$20M Series A co-led by Bison Ventures & Geodesic Capital). Awarded an AFWERX Direct-to-Phase II SBIR.

What They Do: Builds low-cost, autonomous long-range cargo aircraft ("Lifter-Lite") tailored for contested island logistics.

Capture-Curve Position: Technical Validation — still proving the core technology works before institutional adoption begins.

Why It's Interesting: Addresses the critical Indo-Pacific logistical bottleneck by providing autonomous, uncrewed payload transport over extended maritime distances without requiring runways.

Watch Item: Contested-logistics cargo UAS is a crowded category (competing directly with Natilus's KONA, among others) and no production contract has been disclosed yet.

Exhibit 97: Grid Aero



Source: Grid Aero public disclosures, press releases, and funding announcements

Vanguard Defense

HQ / Founded	West Covina & San Francisco, CA Founded 2024
Founders	Tech founders partnering directly with U.S. military veterans.
Funding	~\$5M seed led by First In; partnered with MIT Lincoln Laboratory and DoW SkillBridge.

What They Do: Cleans, labels, and structures complex military sensor telemetry into AI-ready training datasets using a veteran workforce.

Capture-Curve Position: Technical Validation — still proving the core technology works before institutional adoption begins.

Why It's Interesting: Solves the primary bottleneck in defense AI—converting messy, classified operational telemetry into structured training data while creating jobs for transitioning veterans.

Watch Item: *Data-labeling is a services business more than a defensible product — the competitive barrier to entry here is lower than a hardware or software platform play.*

Exhibit 98: Vanguard Defense



Source: Vanguard Defense public disclosures, press releases, and funding announcements

Certus Core

HQ / Founded	Tampa, FL Founded 2021
Founders	Veteran-founded SDVOSB team specializing in tactical software engineering.
Funding	~\$1.6–2.3M raised; holds a GSA IT Schedule contract (47QTCA24D009F) with active AFWERX ties.

Knox Systems

HQ / Founded	NYC & Washington, DC Founded 2024
Founders	Enterprise cloud security software leaders.
Funding	~\$31.5M raised, including a \$25M Series A (Mar 2026) led by B Capital with M12 (Microsoft's Venture Fund), Okta Ventures, MongoDB Ventures, Hearst Ventures, and Bench strength.
Federal Footprint	Holds Authorizations to Operate across 16 federal civilian and defense agencies (per Knox, April 2026), including FEMA, HHS, DHS, Department of Commerce, FDA, and the U.S. Marine Corps; achieved FedRAMP High in March 2026.

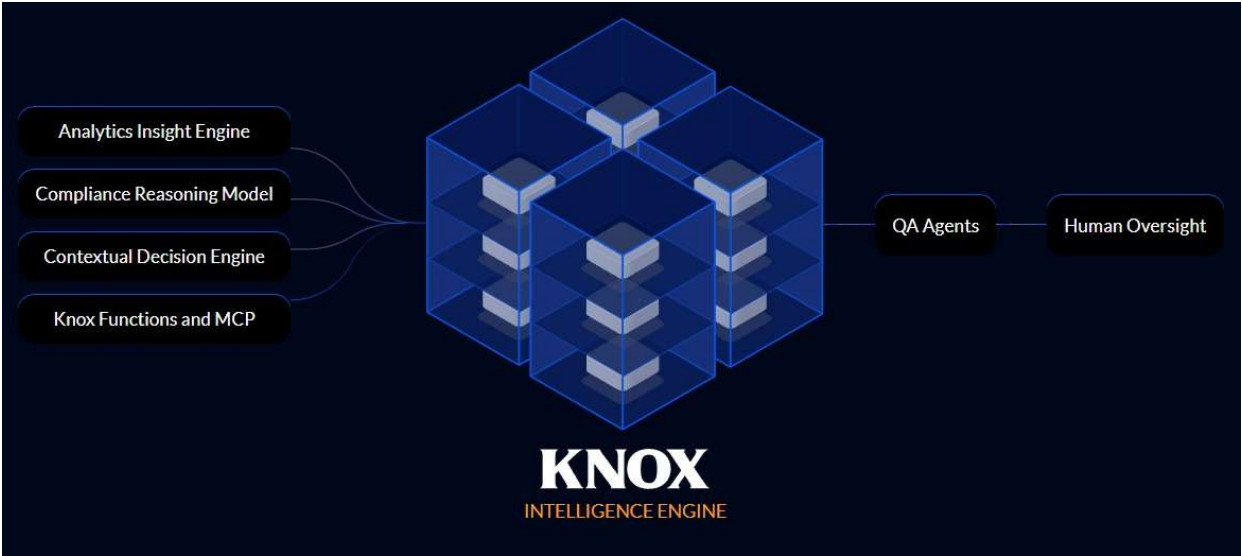
What They Do: Operates a pre-authorized, multi-cloud federal managed boundary spanning AWS, Microsoft Azure, and Google Cloud, enabling SaaS providers to achieve FedRAMP / DoD ATOs in ~90 days at roughly 90% lower cost.

Capture-Curve Position: Institutional Adoption — building the government relationships and non-dilutive award trail that precede a funded program; now hosts 70+ commercial software vendors inside its boundary.

Why It's Interesting: Removes the single largest administrative barrier for commercial tech startups entering the defense space — radically accelerating the cloud authorization timeline and turning FedRAMP from a two-year barrier into a 90-day on-ramp.

Watch Item: Knox is positioned as a partner to, not a competitor of, the hyperscalers — with formal collaborations across AWS (Marketplace listing and AWS Partner), Microsoft (Azure Government Cloud partnership and M12 investment), and Google Cloud (multi-cloud boundary).

Exhibit 100: Knox Systems



Sources: Company data; Knox Systems press releases (Mar-Jul 2026); Washington Technology (Jul 2026); PRNewswire (Apr 2026).

PAD Group (Precision Aerospace & Defense Group)

HQ / Founded	Overland Park, KS Public SPAC Roll-up
Founders	Defense industrial manufacturing roll-up executives.
Funding	Announced SPAC merger with FACT II Acquisition Corp at an implied ~\$320M enterprise value (\$80M BC Partners financing).

What They Do: Consolidates precision CNC machining, non-destructive testing, and legacy aircraft sustainment suppliers into a unified industrial platform.

Capture-Curve Position: Scaled Production — operating as an established production or infrastructure supplier to the defense industrial base.

Why It's Interesting: Strengthens the sub-tier defense industrial base by acquiring and modernizing critical, fragile middle-market machining and component suppliers.

Watch Item: *The roll-up strategy depends on successfully integrating acquired suppliers post-merger — SPAC structures carry their own execution and market-perception risk independent of the underlying industrial thesis.*

Exhibit 101: PAD Group



Source: PAD Group public disclosures, press releases, and funding announcements

Smack Technologies

HQ / Founded	Austin, TX Founded 2024
Founders	Founded by former MARSOC special operations operators.
Funding	\$93M total raised (\$61M Series B led by Costanoa & First In). Awarded seven-figure prototype contracts by the Joint Fires Network and Marine Corps Warfighting Lab.

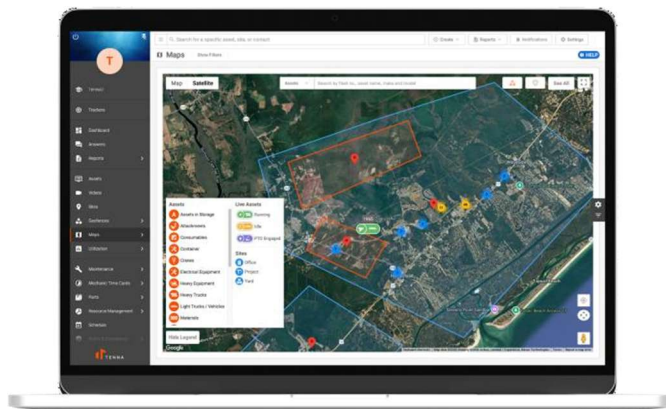
What They Do: Builds reinforcement-learning decision models (Omega/Alpha) for tactical-edge command and control.

Capture-Curve Position: Institutional Adoption — building the government relationships and non-dilutive award trail that precede a funded program.

Why It's Interesting: Fuses combat-proven doctrine with cutting-edge reinforcement learning to give commanders real-time tactical decision recommendations at machine speed.

Watch Item: Prototype contracts with the Joint Fires Network and Marine Corps Warfighting Lab are a strong signal but are still pre-Program-of-Record; the \$93M raised is large relative to the contract value disclosed so far.

Exhibit 102: Smack Technologies



Source: Smack Technologies public disclosures, press releases, and funding announcements

Tenna Systems

HQ / Founded	New York, NY Founded 2023
Founders	Electronic warfare and spectrum intelligence researchers.
Funding	\$13.5M seed led by Costanoa Ventures; active U.S. Army, USAF, DHS, and Golden Dome engagements.

What They Do: Turns existing radio systems into passive RF sensors to geolocate tactical jamming and spoofing (Arena, Tracer, Halo).

Capture-Curve Position: Technical Validation — still proving the core technology works before institutional adoption begins.

Why It's Interesting: Transforms standard tactical comms equipment into an electronic warfare detection net without requiring expensive new hardware retrofits.

Watch Item: “Active engagements” across four agencies is breadth without disclosed depth — worth confirming whether any one relationship has converted into a funded contract yet.

Exhibit 103: Tenna Systems: Arena/Tracer/Halo



Source: Tenna Systems public disclosures, press releases, and funding announcements

Robotics / Humanoids (Commercial & Industrial)

Apptronik

HQ / Founded	Austin, TX Founded 2016
Founders	Jeff Cardenas (Co-Founder & CEO) alongside Dr. Luis Sentis and researchers from UT Austin's Human-Centered Robotics Lab.
Funding	~\$1B raised across strategic and venture rounds.

What They Do: Builds "Apollo," a general-purpose bipedal humanoid robot designed for industrial logistics, assembly, and warehouse material handling.

Market Position: Very well-capitalized industrial robotics play (~\$1B raised) with no disclosed defense contract

Why It's Interesting: Deployed commercially on live factory floors (Mercedes-Benz, GXO) to automate heavy logistics alongside human workers.

Watch Item: Capital raised is enormous, but the profile cites no defense-specific contract — defense applicability here is inferred from the logistics use case, not yet evidenced by an award.

Exhibit 104: Apptronik



Source: Apptronik public disclosures, press releases, and funding announcements

Bedrock Robotics

HQ / Founded	San Francisco, CA Founded 2024
Founders	Boris Sofman (Co-Founder & CEO, former Waymo autonomous trucking lead and Anki founder) alongside Kevin Peterson, Tom Eliaz, and Ajay Gummalla.
Funding	Raised \$350M+ (\$270M Series B led by CapitalG and Valor Equity Partners).

What They Do: Builds autonomous retrofit kits ("Bedrock Operator") that convert heavy construction machinery into self-driving equipment.

Market Position: Well-funded construction-automation play (\$350M+ raised) with no disclosed defense contract

Why It's Interesting: Solves critical construction labor shortages by automating earthmoving and heavy site work on major infrastructure projects.

Watch Item: As with several others in this cluster, defense relevance is plausible (autonomous heavy equipment) but not yet supported by any cited government engagement.

Exhibit 105: Bedrock Robotics



Source: Bedrock Robotics public disclosures, press releases, and funding announcements

Anvil

HQ / Founded	Palo Alto, CA Founded 2025
Founders	Founded by Palo Alto hardware engineers and modular robotics experts.
Funding	~\$5.5–7M seed led by Matter Venture Partners and Humba Ventures.

What They Do: Develops modular, plug-and-play hardware development kits ("Legos for robots") that simplify custom robot assembly.

Market Position: Seed-stage robotics-components play (~\$6M) with no disclosed defense contract

Why It's Interesting: Dramatically accelerates physical prototyping timelines for robotics startups by standardizing bus connectors and motor modules.

Watch Item: "Legos for robots" is a platform-enabling bet — its value depends entirely on ecosystem adoption by other robot builders, a chicken-and-egg problem before any defense relevance can be assessed.

Exhibit 106: Anvil



Source: Anvil public disclosures, press releases, and funding announcements

Anyware Robotics

HQ / Founded	Fremont, CA Founded 2022
Founders	Thomas Tang (CEO), Bruce Fan (CTO), Sam Zhou, and Torsten Schreiber—UC Berkeley robotics PhDs and FANUC alumni.
Funding	~\$17.7M raised in early venture funding.

What They Do: Builds "Pixmo," an autonomous mobile robot featuring a 6-DOF arm designed to unload cargo shipping containers.

Market Position: Early-stage logistics-robotics play (~\$18M raised) with no disclosed defense contract

Why It's Interesting: Automates one of the most physically punishing jobs in logistics—manually picking heavy boxes out of hot shipping containers.

Watch Item: No government engagement is cited; the defense angle (contested logistics, warehousing) is inferred from the commercial use case rather than evidenced.

Exhibit 107: Anyware Robotics



Source: Anyware Robotics public disclosures, press releases, and funding announcements

Konnex

HQ / Founded	British Virgin Islands / CA Founded 2025
Founders	Web3 builders and robotics advisors.
Funding	\$15M raised from crypto VCs.

What They Do: Operates a decentralized physical infrastructure (DePIN) network that incentivizes localized robot hardware deployments via token rewards.

Market Position: Very early crypto-VC-backed DePIN robotics network (\$15M raised)

Why It's Interesting: Applies crypto-economic incentives to crowdsource funding and compute resources for industrial robot fleets.

Watch Item: A crypto-VC funding structure is unusual for anything defense-adjacent and would likely draw additional scrutiny (foreign ownership/control considerations) in any government vetting process.

Exhibit 108: Konnex



Source: Konnex public disclosures, press releases, and funding announcements

Lightberry

HQ / Founded	San Francisco, CA Founded 2025
Founders	Y Combinator-backed artificial intelligence researchers.
Funding	~\$500K YC seed round.

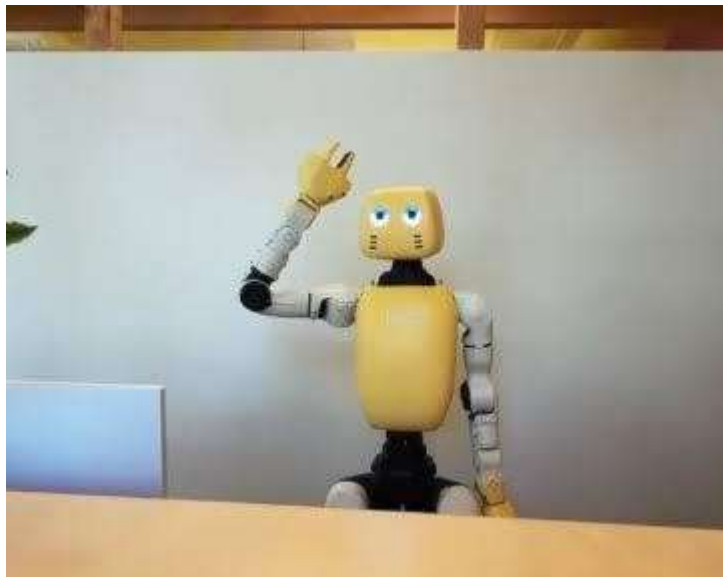
What They Do: Builds conversational AI software acting as the social and emotional "brain" for interactive companion robots.

Market Position: Pre-seed consumer-robotics AI play (~\$500K YC round)

Why It's Interesting: Leverages multimodal LLMs to give physical robots natural voice interfaces and emotional context awareness.

Watch Item: *The furthest from an actual defense program of any company in this set — consumer companion-robot AI has no stated path to a government customer.*

Exhibit 109: Lightberry



Source: Lightberry public disclosures, press releases, and funding announcements

Trener.ai

HQ / Founded	San Francisco, CA & Norway Founded 2024
Founders	Industrial automation engineers.

Funding	\$32M Series A.
----------------	-----------------

What They Do: Builds "Acteris," a software platform that allows factory robot arms to perform automated CNC machine tending without custom code.

Market Position: Series A manufacturing-automation software play (\$32M raised) with no disclosed defense contract

Why It's Interesting: Eliminates the expensive integration engineering costs that prevent small machine shops from adopting automation.

Watch Item: Relevant here mainly as a manufacturing-automation comparable rather than a direct defense play — no government engagement is cited.

Exhibit 110: Trener.ai



Source: Trener.ai public disclosures, press releases, and funding announcements

Antioch.ai

HQ / Founded	New York, NY Founded 2025
Founders	Simulation engineers backed by A* and Category Ventures, with Palantir CTO Shyam Sankar as an angel.
Funding	~\$44M raised.

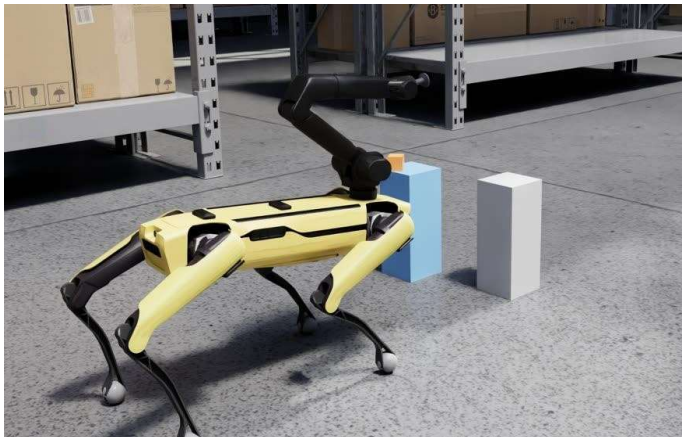
What They Do: Operates GPU-accelerated cloud simulation environments for testing complex autonomous robotic movements.

Market Position: Well-funded robotics-simulation infrastructure play (~\$44M raised)

Why It's Interesting: Provides scalable cloud infrastructure that allows robotics developers to run millions of simulated stress tests before deploying physical hardware.

Watch Item: *Its customers are other robotics companies, not government end users directly — a step removed from the capture playbook this document otherwise describes.*

Exhibit 111: Antioch.ai



Source: Antioch.ai public disclosures, press releases, and funding announcements

RoboForce

HQ / Founded	Milpitas, CA Founded 2023
Founders	Heavy machinery engineers backed by YZi Labs.
Funding	\$67M raised.

What They Do: Builds "Titan," a heavy-duty outdoor industrial robot designed for solar field assembly and mining operations.

Market Position: Growth-stage industrial-robotics play (\$67M raised) with no disclosed defense contract

Why It's Interesting: Solves severe labor bottlenecks in outdoor green-energy infrastructure construction.

Watch Item: No defense contract or engagement is cited in available sources; relevant here as an industrial-robotics comparable, not a capture-pathway example.

Exhibit 112: RoboForce



Source: RoboForce public disclosures, press releases, and funding announcements

Chaos Industries

HQ / Founded	Los Angeles, CA Founded 2022
Founders	Co-founded by John Tenet and Dr. Bo Marr — builders from Silicon Valley alongside defense and government leaders.
Funding	\$1B+ raised (Series C: \$275M at \$2B valuation; \$510M follow-on led by Valor Equity Partners at \$4.5B valuation).

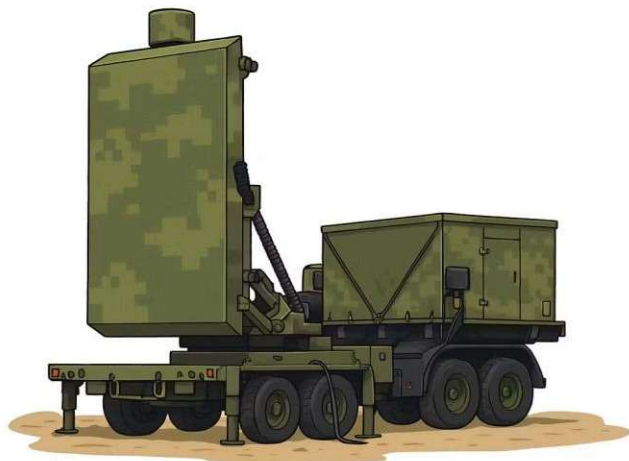
What They Do: Builds Coherent Distributed Networks (CDN) — advanced detection, monitoring, and communication systems, including the Vanquish radar for early warning against drones, missiles, and aircraft.

Capture-Curve Position: Program of Record — has converted institutional interest into a funded, budgeted contract vehicle.

Why It's Interesting: Directly answers the exploding UAS/drone threat environment (e.g., Russia's escalating drone barrages against Ukraine); already embedded with Eglin Air Force Base and scaling fast toward a defensible sensing-layer moat.

Watch Item: Valuation (\$4.5B) is running well ahead of any disclosed program-of-record revenue figure — the gap between valuation and funded, named contracts is the thing worth tracking.

Exhibit 113: Chaos Industries



Source: Chaos Industries public disclosures, press releases, and funding announcements

Castelion

HQ / Founded	El Segundo, CA Founded 2022 (emerged from stealth late 2023)
Founders	Trio of ex-SpaceX engineers — Bryon Hargis (CEO), Sean Pitt, and Andrew Kreitz (CFO).
Funding	100M Series A in Jan 2025; \$350M Series B led by Lightspeed and Altimeter Capital, valuing the company in the billions).

What They Do: Mass-produces low-cost, long-range hypersonic strike weapons (flagship system "Blackbeard") using a SpaceX-style build-fast/test-often/vertically-integrated approach.

Capture-Curve Position: Program of Record — has converted institutional interest into a funded, budgeted contract vehicle.

Why It's Interesting: Positioned as a non-nuclear deterrent play directly against China and Russia's hypersonics lead; already has Air Force and Navy contracts and is building a 1,000-acre production facility ("Project Ranger") to hit volumes legacy primes can't match.

Watch Item: Air Force and Navy contracts are real, but a 1,000-acre production facility is a massive capital commitment ahead of confirmed production-volume orders — the bet is that the contracts scale to match the facility, not the other way around.

Exhibit 114: Castelion



Source: Castelion public disclosures, press releases, and funding announcements

Hadrian

HQ / Founded	Torrance, CA Founded 2020
Founders	Founder and CEO Chris Power, an Australian entrepreneur who built the company after observing rising U.S.–China geopolitical tension.
Funding	260M Series C led by Founders Fund; backers include Andreessen Horowitz, Lux Capital, Construct Capital).

What They Do: Operates AI-driven, software-defined automated factories that mass-produce precision-machined parts (CNC, welding, casting, additive) for aerospace and defense customers 10x faster and at roughly half the cost of traditional shops.

Capture-Curve Position: Scaled Production — operating as an established production or infrastructure supplier to the defense industrial base.

Why It's Interesting: Tackles the aging, fragmented small-machine-shop supplier base underpinning the entire defense industrial base; expanding into munitions and maritime parts and building new capacity (Arizona "Factory 3") to scale as a "factories-as-a-service" backbone for primes and startups alike.

Watch Item: Positioned as manufacturing infrastructure for the broader defense industrial base rather than a single program — success depends on primes and startups actually adopting it as shared capacity rather than building in-house.

Exhibit 115: Hadrian



Source: Hadrian public disclosures, press releases, and funding announcements

Vannevar Labs

HQ / Founded	Palo Alto, CA (also Arlington, VA) Founded 2019
Founders	Co-founded by Brett Granberg (CEO) and Nini Moorhead (COO), Stanford GSB classmates with national security and intelligence backgrounds.
Funding	\$91M+ raised (Series B: \$75M led by investors including Felicis, DFJ Growth, Point72, Costanoa, and General Catalyst).

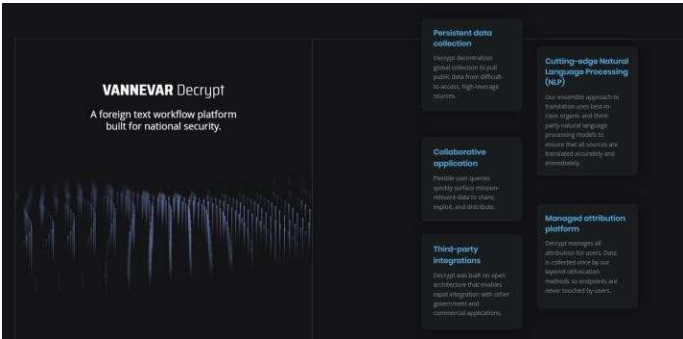
What They Do: Builds AI-powered intelligence software — flagship product "Decrypt" — that uses LLMs and NLP to process foreign-language and open-source intelligence data for military and counter-disinformation missions.

Capture-Curve Position: Program of Record — has converted institutional interest into a funded, budgeted contract vehicle.

Why It's Interesting: Supports 125+ active deployments across DoW and other agencies; moving in 2026 to unify its tools into an "agentic warfare" platform purpose-built to compete with China across gray-zone and cross-Strait scenarios — a bet on AI-native intelligence displacing legacy analyst workflows.

Watch Item: 125+ active deployments is a strong adoption signal, but LLM-based intelligence analysis on foreign-language and open-source data raises its own data-security and classification-handling questions that a pure hardware company wouldn't face.

Exhibit 116: Vannevar Labs: Decrypt



Source: Vannevar Labs public disclosures, press releases, and funding announcements

Appendix A: Founder Self-Assessment Scorecard

Score each stage honestly — most founders overestimate where they actually are. A "no" on any item is a specific, fixable gap, not a reason to feel behind.

Self-Assessment Scorecard — Template
(plot the share of checked items per stage)



Plot the share of “yes” answers per stage to see your actual profile — the dashed line is an illustrative example, not a benchmark to match.

Stage 1 — Technical Validation

- ☐ Have you completed a technical evaluation with an accredited Service Laboratory or Warfare Center (not just internal testing)?
- ☐ Do you have documented failure-mode data, not just best-case performance metrics?
- ☐ Have you run structured discovery interviews with company- or battalion-level operators (L1), with written capability-gap memos?

Stage 2 — Institutional Adoption

- ☐ Can you name the specific PM who owns the requirement you're pursuing?
- ☐ Do you know the Program Element (PE) number your capability would sit under?

- ☐ Has a demonstration been run by active-duty operators under contested conditions — not by your own engineers under controlled conditions?
- ☐ If you're relying on a prime relationship, do you have negotiated data rights (DFARS 252.227-7013/7014) rather than a verbal or informal understanding?

Stage 3 — Program of Record

- ☐ Is your capability included in the Service's current POM submission?
- ☐ Do you have a documented (not verbal) commitment to a follow-on production OTA, or Phase III SBIR eligibility confirmed?
- ☐ Have you validated open-architecture compliance (MOSA/SOSA/OMS/CMOSS) via independent conformance assessment?

Stage 4 — Scaled Production

- ☐ Have you demonstrated (not modeled) 3–5x surge production capacity?
- ☐ Are your sub-tier components mapped and verified NDAA-compliant at the individual part level?
- ☐ Do you know whether your next major proposal will cross the \$35M CAS or \$10M certified cost/pricing data thresholds, and is your accounting system ready if so?
- ☐ Is sustainment (FSR model, spares, depot repair) priced into your contract structure rather than planned as a future negotiation?

Stage 5 — Enduring Powerhouse

- ☐ Do you have a recompute evidence base being built continuously, rather than assembled reactively when the recompute RFP drops?
- ☐ Is spiral/tech-refresh development budgeted as an ongoing line item, not a response to competitive pressure?
- ☐ Have you had at least one substantive FMS/DCS conversation with an allied government?

Appendix B: Glossary

Term	Definition
ATO	Authority to Operate — cybersecurity accreditation required before a system can connect to government networks
CAS	Cost Accounting Standards — federal accounting rules for how contract costs are measured and allocated
CMMC	Cybersecurity Maturity Model Certification — DoW's cybersecurity compliance framework for contractors
CSO	Commercial Solutions Opening — a competitive acquisition mechanism (used heavily by DIU) for commercial technology
DFARS	Defense Federal Acquisition Regulation Supplement — DoW-specific procurement regulations layered on top of the FAR
FAR	Federal Acquisition Regulation — the government-wide rules governing federal procurement
FCL	Facility Security Clearance — clearance for a company (not an individual) to access classified information
JCIDS	Joint Capabilities Integration and Development System — DoW's formal process for validating and prioritizing capability requirements
MOSA / SOSA / OMS / CMOSS	Modular Open Systems Approach / Sensor Open Systems Architecture / Open Mission Systems / C4ISR-EW Modular Open Suite of Standards — open-architecture standards that let systems from different vendors interoperate
MTA	Middle Tier of Acquisition — a Section 804 rapid acquisition pathway (rapid prototyping and rapid fielding)
NDC	Nontraditional Defense Contractor — a company that hasn't held a full CAS-covered DoW contract in the prior year, qualifying it for reduced compliance burdens
OTA	Other Transaction Authority — a flexible, non-FAR contracting mechanism used for rapid prototyping and, on follow-on, production
PE	Program Element — the specific budget line item a capability is funded under
PEO	Program Executive Officer — senior official overseeing a portfolio of related Programs of Record

Term	Definition
PM	Program Manager — the official who owns a specific program's schedule, budget, and contract execution
PoR	Program of Record — a formally established, budgeted, multi-year acquisition program
POM	Program Objective Memorandum — the document through which a Service requests budget for future years during PPBE's Programming phase
PPBE	Planning, Programming, Budgeting, and Execution — DoW's internal budget development and allocation cycle
SBIR / STTR	Small Business Innovation Research / Small Business Technology Transfer — federal programs providing phased seed funding to small businesses, with statutory sole-source Phase III rights
TINA	Truth in Negotiations Act, now formally the Truthful Cost or Pricing Data Statute — requires certified cost/pricing data above a dollar threshold on negotiated contracts

Military Terms & Organizational Structure

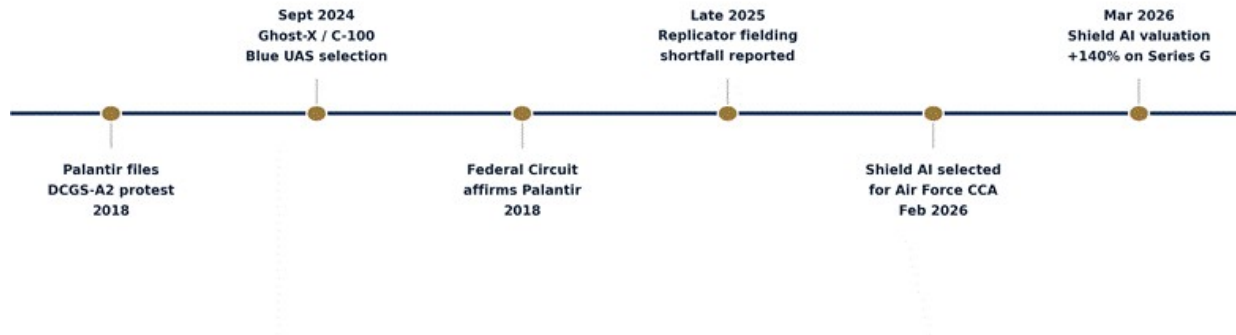
The acronym glossary above covers acquisition and contracting terms. The list below adds common military organizational and rank terminology a founder is likely to encounter in the field or in a program office, but that doesn't show up on a solicitation.

Term	Definition
COCOM	Combatant Command — a unified command with broad, ongoing mission responsibility across a geographic region (e.g., INDOPACOM, CENTCOM) or function (e.g., SOCOM, TRANSCOM).
O-3 / O-5 / O-6	Officer pay grades — roughly Captain/Lieutenant (O-3), Lieutenant Colonel/Commander (O-5), and Colonel/Captain (O-6) depending on service. Useful shorthand for gauging seniority in a program office.
GS-15	The senior-most General Schedule civilian pay grade below the Senior Executive Service — often the civilian counterpart to an O-6 in seniority and program authority.
SES	Senior Executive Service — the senior civilian leadership corps of the federal government, roughly equivalent in seniority to a general officer.
NCO	Non-Commissioned Officer — an enlisted leader (e.g., sergeants, petty officers) who often controls day-to-day operational reality on the ground and is a key L1 stakeholder.
OCONUS / CONUS	Outside the Continental United States / Continental United States — shorthand used constantly in deployment, logistics, and testing contexts.

Term	Definition
TDY	Temporary Duty — official travel away from a service member's home station, often how government evaluators get to a company's demonstration site.
Echelon	A level of command or organization (e.g., squad, platoon, company, battalion, brigade) — useful for understanding how far up or down a chain of command a point of contact sits.
Service Secretary	The civilian head of a military department — Secretary of the Army, Navy, or Air Force — an L5-level decision-maker in the advocate model in Part 1.
NDS / NSS	National Defense Strategy / National Security Strategy — the top-level strategic documents that telegraph future service acquisition priorities, discussed in Part 1.

Appendix C: Sources

Selected Proof Points Referenced in This Playbook



- CRS, "DoW Replicator Initiative: Background and Issues for Congress," updated 2026.
- DefenseScoop, "DOD touts "successful transition" for Replicator initiative — but questions linger" (Sept. 2025).
- Breaking Defense, "'It's alive': Biden-era Replicator drone initiative lives on as DAWG" (Dec. 2025).
- Breaking Defense / Army.mil, "Army selects 2 'attritable' UASs for company-level recon" (Sept. 2024).
- Anduril, "Ghost Approved for the Blue UAS Cleared List" (June 2024).
- Shield AI, "Shield AI Looks To Unleash Its Hivemind Autonomy Software On Multiple Platforms" (Oct. 2024).
- Shield AI, "Autonomy for the World: MQ-20 Avenger and X-62 VISTA."
- Shield AI, "Shield AI awarded U.S. Air Force production contract for Collaborative Combat Aircraft mission autonomy" (2026).
- Palantir USG, Inc. v. United States, No. 2017-1465 (Fed. Cir. Sept. 13, 2018).
- GAO, Palantir USG, Inc., B-412746 (2016).
- American Bar Association, "Palantir USG, Inc. v. United States Adds Teeth to Government Preference for Buying Commercial."
- Crowell & Moring, "The FY 2026 National Defense Authorization Act."
- Inside Government Contracts, "The \$10 Million TINA Threshold Is Now in Effect" (2026).
- Forvis Mazars, "2026 NDAA – Potentially Historic Changes for NDCs" (2026).
- TechCrunch, "Defense startup Shield AI lands \$12.7B valuation, up 140%, after US Air Force deal" (March 2026).
- Prism News, "Shield AI Raises \$1.5 Billion, Valuation Jumps to \$12.7 Billion After Air Force Selection" (March 2026).
- Sacra, "Anduril revenue, valuation & funding."
- Contrary Research, "Anduril Industries Business Breakdown & Founding Story."

Appendix D: Defense Agencies in Detail

This appendix profiles the 19 defense agencies and 10 unified combatant commands shown in the DoW organizational structure chart, with mission summaries and current leadership as of September 2026.

Defense Agencies



Defense Advanced Research Projects Agency (DARPA)

Founded in 1958 in response to Sputnik, DARPA's mission is to prevent — and create — strategic technological surprise for U.S. national security. It's headquartered in Arlington, Virginia, and runs on a lean, program-manager-driven model: roughly 100–200 program managers, each on a limited-term rotation, own a portfolio of high-risk, high-payoff research bets. Past bets include precision weapons and stealth technology; current ones include AI, autonomy, hypersonics, and biotechnology.

DARPA funds work through broad agency announcements, SBIR/STTR, and Other Transaction Agreements (OTAs). These tools let it move fast and engage non-traditional performers — including early-stage startups — more easily than most of DoW's acquisition system. Notably, DARPA doesn't build or field systems itself; it hands off mature technology to the military services or other agencies.

Current director: Stephen Winchell (since May 2025)



Defense Commissary Agency (DeCA)

Established in 1990–1991 and headquartered at Fort Lee (Fort Gregg-Adams), Virginia, DeCA operates nearly 240 military commissaries worldwide. It sells groceries and household goods at

cost plus a small surcharge — a tax-free discount that functions as a non-cash part of military compensation.

DeCA serves active-duty, Guard, and Reserve members, retirees, and their families, both in the U.S. and overseas. It works closely with troop-support vendors and distributors, making it a recurring — if non-technical — point of contact for consumer goods and logistics suppliers.

Current director: John E. Hall, Director and Chief Executive Officer



Defense Contract Audit Agency (DCAA)

DCAA is a roughly 4,000-person audit organization operating under the Under Secretary of Defense (Comptroller), with an annual operating budget near \$650 million. It performs contract audits for DoW and other federal agencies, examining contractor cost and pricing data — including compliance with the Truth in Negotiations Act on sole-source awards.

DCAA doesn't award contracts itself, but its findings directly shape what defense contractors, including smaller subcontractors, can recover and how they must document costs.

Current director: Jennifer L. Desautel (since October 2024)



Defense Contract Management Agency (DCMA)

Headquartered at Fort Lee, Virginia, with roughly 10,000 civilian and military personnel, DCMA administers contracts once they're awarded. It monitors contractor performance, cost, schedule, and quality, and administers Foreign Military Sales contracts.

DCMA is the agency most likely to have a resident presence inside a major contractor's own facility. For growing defense-tech companies, that involvement typically begins once a contract moves from award into execution.

Current director: VADM Stephen R. Tedford



Defense Finance and Accounting Service (DFAS)

Established in 1991 and headquartered in Indianapolis, Indiana, DFAS employs more than 12,000 people and handles essentially all DoW payments — military and civilian pay, retiree and annuitant payments, and payments to contractors and vendors.

It processes well over 100 million pay transactions and tens of millions of commercial invoices each year, manages trust funds for foreign military sales and military retirement, and prepares DoW's consolidated financial statements. In short: it's the back-office system through which contract payments to defense-tech vendors ultimately flow.

Current director: Jonathan Witter (since November 2025)



Defense Health Agency (DHA)

Stood up in 2013 and headquartered in Falls Church, Virginia, DHA is a combat support agency that manages the Military Health System. It has administered military hospitals and clinics directly since 2021, and enables the Army, Navy, and Air Force medical services to keep forces medically ready.

DHA serves more than 9 million beneficiaries and oversees the MHS Genesis electronic health records rollout. It's a growing customer for medtech and health-IT R&D through its own research and development arm.

Current director: Dr. David J. Smith, Acting Director (since February 2025)



Defense Information Systems Agency (DISA)

Headquartered at Fort Meade, Maryland, DISA provides, operates, and defends the DoW Information Network (DODIN) — the command, control, and communications backbone that combatant commands and the wider department depend on. Its director is dual-hatted as Commander of the DoW Cyber Defense Command, tying network operations directly to cyber defense.

For IT and cybersecurity vendors, DISA is one of the largest single buyers in the federal government, and it manages several major enterprise-wide contract vehicles.

Current director: LTG Paul T. Stanton (since October 2024)



Defense Intelligence Agency (DIA)

Formed in 1961 and headquartered at Joint Base Anacostia-Bolling in Washington, D.C., DIA is DoW's primary all-source military intelligence agency and a member of the U.S. Intelligence Community. It produces foreign military intelligence for the Secretary of Defense, the Joint Chiefs, and combatant commanders, and chairs the Military Intelligence Board that coordinates the broader defense intelligence enterprise.

Directorship rotates among three-star Army, Navy, Air Force, and Marine Corps officers roughly every three years.

Current director: LTG James H. Adams III (since February 2026)



Defense Legal Services Agency (DLSA)

Established in 1981 under DoW Directive 5145.1, DLSA provides legal services and support to the defense agencies and DoW field activities that don't have their own general counsel.

Its director is dual-hatted as the General Counsel of the Department of War — the department's chief legal officer and principal legal adviser to the Secretary of Defense. That gives DLSA visibility into legal policy across the entire department, including contracting and acquisition law.

Current director: Earl G. Matthews, DoW General Counsel / DLSA Director (since July 2025)



Defense Logistics Agency (DLA)

DLA is DoW's largest combat logistics support agency. It manages nine global supply chains — food, fuel, medical materiel, uniforms and clothing, construction and equipment, and industrial/weapon-system spare parts — supplying roughly 98% of the department's fuel and the majority of spare parts for military weapon systems.

DLA also oversees the National Defense Stockpile of critical materials and operates a worldwide network of distribution centers, making it a primary counterparty for hardware, materials, and sustainment-focused defense-tech companies.

Current director: LTG Mark T. Simerly (since 2024)



Defense POW/MIA Accounting Agency (DPAA)

Formed in January 2015 from the merger of the Joint POW/MIA Accounting Command, the Defense Prisoner of War/Missing Personnel Office, and part of the Air Force's Life Sciences Lab, DPAA is headquartered at the Pentagon, with scientific laboratories at Offutt Air Force Base, Nebraska, and Joint Base Pearl Harbor–Hickam, Hawaii.

Its roughly 800-person workforce researches, investigates, recovers, and identifies the remains of U.S. personnel unaccounted for from WWII through the Persian Gulf War. It reports to the Under Secretary of Defense for Policy.

Current director: Kelly McKeague (since September 2017)



Defense Security Cooperation Agency (DSCA)

Headquartered at the Pentagon, DSCA administers DoW's security cooperation programs — most visibly Foreign Military Sales (FMS) and International Military Education and Training (IMET) — transferring defense articles, training, and services to allied and partner nations.

Through its Defense Security Cooperation University, DSCA also trains and certifies the DoW workforce that manages these relationships. That makes it the key gateway for U.S. defense-tech companies pursuing allied and coalition sales.

Current director: Michael F. Miller



Defense Security Service — now Defense Counterintelligence and Security Agency (DCSA)

Renamed from Defense Security Service in 2019 (the org chart in this playbook uses the legacy name), DCSA is headquartered at Marine Corps Base Quantico, Virginia, and is the federal government's largest security and vetting organization.

It conducts roughly 2 million background investigations a year for more than 100 federal entities, oversees security for about 10,000 cleared contractor facilities under the National Industrial Security Program, and leads DoW counterintelligence and critical-technology-protection efforts. That makes it directly relevant to any defense-tech company pursuing a facility clearance or cleared personnel.

Current director: Justin Overbaugh, Acting Director (since November 2025)



Defense Threat Reduction Agency (DTRA)

Stood up in 1998 from the merger of several predecessor organizations (tracing back to the Manhattan Project's Armed Forces Special Weapons Project) and headquartered at Fort Belvoir, Virginia, DTRA is both a defense agency and a combat support agency. Its mission: countering weapons of mass destruction — chemical, biological, radiological, nuclear, and high-yield explosive threats — and supporting the U.S. nuclear deterrent.

With roughly 2,100 civilian and military personnel and a budget near \$2 billion, DTRA funds applied research and fields WMD-defeat and detection technology directly to combatant commands — a natural fit for CBRN-detection and countermeasure startups.

Current director: Maj. Gen. Lyle K. Drew, Acting Director



Missile Defense Agency (MDA)

Headquartered in Huntsville, Alabama, with major test and integration sites elsewhere, MDA develops, tests, and fields the nation's layered ballistic missile defense system (BMDS) — radars, interceptors, command-and-control, and sensor networks designed to defend the homeland, deployed forces, and allies against missile attack, including hypersonic threats.

MDA runs a significant R&D and prototyping portfolio and works closely with the military services' own missile defense commands, making it one of the more accessible agencies for sensor, tracking, and interceptor technology companies.

Current director: LTG Heath A. Collins



National Geospatial-Intelligence Agency (NGA)

Formed in 1996 (originally as the National Imagery and Mapping Agency, renamed in 2003) and headquartered at Fort Belvoir/Springfield, Virginia, NGA is a combat support agency and Intelligence Community member with roughly 14,500 employees.

It collects, analyzes, and distributes geospatial intelligence (GEOINT) — imagery, mapping, and location-based data — supporting both national decision-makers and deployed forces. NGA has increasingly leaned on commercial satellite imagery and analytics providers, making it one of the more commercially engaged intelligence agencies for space and data-analytics startups.

Current director: LTG Michele Bredenkamp



National Reconnaissance Office (NRO)

Established in 1961 and declassified as an organization in 1992, NRO designs, builds, launches, and operates the nation's fleet of intelligence-gathering satellites — providing signals, imagery, and communications intelligence to the Intelligence Community and military.

Structured as a joint DoW/Intelligence Community organization, NRO has significantly expanded commercial and proliferated-architecture space acquisitions in recent years. That makes it an active buyer for commercial space, launch, and remote-sensing companies alongside its classified satellite programs.

Current director: Dr. Christopher J. Scolese (since 2019)



National Security Agency/Central Security Service (NSA/CSS)

Established in 1952 and headquartered at Fort Meade, Maryland, NSA leads the U.S. government's signals intelligence (SIGINT) mission. Alongside CSS, it also leads the cryptologic and national security systems protection mission — defending classified and sensitive national security networks while producing foreign signals intelligence for policymakers and military commanders.

Its director has, at various points including currently, been dual-hatted as Commander of U.S. Cyber Command, tying NSA's signals and cyber expertise directly to DoW's offensive and defensive cyber operations.

Current director: Gen. Joshua M. Rudd (since March 2026; dual-hatted as Commander, U.S. Cyber Command)



Pentagon Force Protection Agency (PFPA)

Established in May 2002 in response to the September 11 attacks, and growing out of the earlier Defense Protective Service, PFPA is a federal law enforcement agency. It's responsible for force protection, security, and law enforcement at the Pentagon Reservation, the Mark Center, DHA headquarters, and other DoW facilities in the National Capital Region.

Its roughly 1,200 government personnel and 500 contractors include the uniformed Pentagon Police, criminal investigators, threat-management agents, and CBRN defense and explosives specialists. PFPA reports to the DoW Director of Administration and Management.

Current director: Chris Bargery

Unified Combatant Commands (10)



U.S. Africa Command (AFRICOM)

Established in 2007 and headquartered in Stuttgart, Germany — it has never relocated onto the African continent — AFRICOM is the geographic command responsible for U.S. military relations and operations across Africa, except Egypt, which falls under CENTCOM.

Its mission centers on building partner-nation military capacity, countering violent extremist organizations such as al-Shabaab and ISIS affiliates, and coordinating disaster response and security cooperation across a 53-country area of responsibility, with a comparatively light permanent U.S. troop footprint.

Current commander: Gen. Dagvin Anderson, USAF (since August 2025)



U.S. Central Command (CENTCOM)

Established in 1983 out of the earlier Rapid Deployment Joint Task Force and headquartered at MacDill Air Force Base, Florida — with a forward headquarters at Al Udeid Air Base, Qatar — CENTCOM's area of responsibility spans the Middle East, Central Asia, and parts of South Asia, plus Egypt.

It has been the primary American command for major operations of the last three decades, including Desert Storm, the Afghanistan war, and the Iraq war, and continues to lead counterterrorism and regional deterrence operations, including against Iran-aligned groups.

Current commander: Adm. Charles B. Cooper II, USN (since August 2025)



U.S. Cyber Command (CYBERCOM)

Elevated to a full unified combatant command in 2018 — having started in 2010 as a sub-unified command under STRATCOM — and headquartered at Fort Meade, Maryland, CYBERCOM directs, synchronizes, and conducts DoW's cyberspace operations. That includes defending military networks, supporting combatant commanders with offensive and defensive cyber capabilities, and, through its Cyber National Mission Force, defending the nation against significant cyberattacks.

Its commander has historically been, and currently is, dual-hatted as Director of NSA, linking cyber operations directly to the nation's signals intelligence capability.

Current commander: Gen. Joshua Rudd, USA (since March 2026; dual-hatted as Director, NSA)



U.S. European Command (EUCOM)

Tracing its lineage to 1947 and headquartered in Stuttgart, Germany, EUCOM is responsible for U.S. military operations and relations across Europe and parts of the Arctic and Atlantic, coordinating with roughly 50 countries as the U.S. military's principal interface with NATO.

Its commander is dual-hatted as NATO's Supreme Allied Commander Europe (SACEUR), one of only two strategic NATO commanders — giving the role direct authority over both U.S. and allied force posture, a command relationship that has taken on heightened significance since Russia's invasion of Ukraine.

Current commander: Gen. Alexis G. Grynkewich, USAF (since summer 2025; also NATO Supreme Allied Commander Europe)



U.S. Indo-Pacific Command (INDOPACOM)

The oldest and largest of the geographic combatant commands, tracing back to 1947 and headquartered at Camp H. M. Smith, Hawaii, INDOPACOM covers roughly half the Earth's surface and population, from the west coast of the U.S. to the western border of India.

Its priorities center on deterring China in the Indo-Pacific theater and countering North Korea, while sustaining a dense network of treaty allies and partners including Japan, South Korea, Australia, and the Philippines — making it the command most frequently cited in discussions of great-power competition and next-generation defense-tech demand.

Current commander: Adm. Samuel Paparo, USN (since May 2024)



U.S. Northern Command (NORTHCOM)

Established in 2002 in the aftermath of September 11 and headquartered at Peterson Space Force Base, Colorado, NORTHCOM is responsible for homeland defense and, in coordination

with civil authorities, defense support during domestic emergencies. Its area of responsibility covers the continental U.S., Alaska, Canada, Mexico, and surrounding approaches.

Its commander is dual-hatted as commander of NORAD, the binational U.S.-Canada aerospace warning and defense command, giving NORTHCOM a unique combined role in homeland air and missile defense.

Current commander: Gen. Gregory M. Guillot, USAF (since February 2024)



U.S. Southern Command (SOUTHCOM)

One of the oldest unified commands, tracing to 1947 and headquartered at Doral, Florida (near Miami), SOUTHCOM covers Central and South America and the Caribbean — 31 countries and 16 territories.

Its focus areas include countering transnational criminal organizations and drug trafficking, humanitarian assistance and disaster response, and building partner-nation military capacity, with counter-narcotics operations taking on renewed prominence in recent policy discussions.

Current commander: Gen. Francis L. Donovan, USMC (since February 2026)



U.S. Special Operations Command (SOCOM)

Established in 1987 in the wake of the failed Iran hostage rescue mission and headquartered at MacDill Air Force Base, Florida, SOCOM is the functional command responsible for organizing, training, and equipping special operations forces across the Army, Navy, Air Force, and Marine Corps. Its mission covers counterterrorism, unconventional warfare, direct action, and other special operations worldwide.

SOCOM holds its own acquisition authority — a rarity among combatant commands — and has historically been an early adopter of small-company and rapid-prototyping technology, making it a frequently cited entry point for defense-tech founders.

Current commander: Adm. Frank M. Bradley, USN (since October 2025)



U.S. Strategic Command (STRATCOM)

Established in 1992 at the end of the Cold War, absorbing the old Strategic Air Command's nuclear mission, and headquartered at Offutt Air Force Base, Nebraska, STRATCOM oversees the U.S. nuclear deterrent (the nuclear triad), global strike planning, missile defense integration, and space and cyber deterrence.

It co-locates with the Defense Threat Reduction Agency's WMD-countering mission at Fort Belvoir facilities, and is the command most directly tied to strategic-level nuclear command, control, and communications (NC3) modernization.

Current commander: Adm. Richard A. Correll, USN (since December 2025)



U.S. Transportation Command (TRANSCOM)

Established in 1987 and headquartered at Scott Air Force Base, Illinois, TRANSCOM is the functional command providing global air, land, and sea transportation for DoW — coordinating the deployment and sustainment of forces worldwide through its component commands (Air Mobility Command, Military Sealift Command, and Surface Deployment and Distribution Command).

TRANSCOM manages a vast logistics network spanning strategic airlift, sealift, and surface transportation, making it a natural customer for logistics-technology, tracking, and supply-chain-visibility companies.

Current commander: Gen. Randall Reed, USAF (since October 2024)

Appendix E: Additional Resource Tools

Congressional Readiness Checklist

National Security

- What threat are we addressing?
- What capability gap exists?
- Why does the gap matter now?
- What happens if the government does nothing?

Operational Validation

- Which operators have evaluated the capability?
- Under what conditions?
- What evidence exists?
- What remains unproven?

Pentagon Alignment

- Which Service or agency owns the problem?
- Which PM/PEO is relevant?
- What acquisition pathway is appropriate?
- What budget account would logically support the capability?

Congressional Alignment

- Which authorizing committee has jurisdiction?
- Which appropriations subcommittee is relevant?
- Which Members represent our facilities and workforce?
- Which Members have demonstrated interest?
- Which professional staff should be educated?

Legislative Objective

- What specifically are we asking Congress to do?
- Is the ask authorization, appropriations, oversight, report language, or education?
- Why is congressional involvement appropriate?
- What evidence supports the request?

District / Industrial Base

- How many jobs are involved?
- What capital investment is involved?
- What are domestic suppliers involved?
- Does the company address strategic supply chain vulnerability?
- Can the company scale production?

Timing

- What is the relevant congressional calendar?
- When do committees develop relevant legislation?
- When are hearing?
- When are markups?
- When are appropriate decisions made?
- What must happen before the next budget cycle?

Appendix F: Where This Leaves You

Everything in this playbook—the L1–L5 advocacy model, the PPBE calendar, the compliance thresholds, the case studies—comes down to one question: *where does your company actually stand on Appendix A's scorecard right now, versus where your pitch deck says you stand?*

Most founders and capture leads know the gap exists but haven't had someone map it against their specific program, PEO, and timeline.

That's the conversation GMA's Defense & Technology Practice has with founders: no cost, no obligation, before you commit resources to a pursuit. We've supported aerospace and defense-related companies including MightyFly, Precision AI, Honeywell, Sovereign Skies Defense, and other drone and dual-use companies across exactly this kind of capture and funding strategy work. Our team works closely with registered lobbyists on Capitol Hill, giving clients a direct channel into the Congressional engagement work described in Part 3.4: protecting Program Element line items, shaping NDAA language, and building the appropriator relationships that a pure capability narrative can't reach on its own.

What that conversation covers:

- A read on your current stage against the Appendix A checklist
- Where your capture engine has real gaps (advocacy level, TRL evidence, PPBE timing)
- Which of the pathways in Part 2.3 (OTA, SBIR Phase III, MTA, IDIQ) actually fits your position
- Where GMA's capture, compliance, and Hill relationships can move your timeline faster than doing it alone

The GMA Defense Team

Kristin Cooper



Role: Founder & CEO

Background: Kristin founded GMA in 2009 and has led the firm to secure over \$2.5 billion in client funding at a 90.2% win rate. Before founding GMA, she was a Full Professor at California State University, Chico's College of Engineering, where she directed grant-funded programs across five colleges and served as founding director of the university's Environmental Resource Program. She also served as a registered federal lobbyist, where she personally secured a \$100 million line item in the Department of Energy budget — the same authorization-and-appropriations process this playbook walks founders through, done rather than studied. She holds a Certified Professional Contracts Manager (CPCM) credential and brings 30+ years of federal funding strategy experience to every client engagement.

Mark Sue



Role: CFO & Defense & Technology Practice Lead

Background: Mark brings a background in engineering and finance, with foundational experience at top Wall Street investment banks, to his role as CFO of Grant Management Associates. He holds a Master's degree in Aeronautical Science, giving him a technical grounding in the aerospace and defense domains uncommon among CFOs. He leads GMA's Defense & Technology practice vertical, applying a CFO's eye for runway and unit economics to the capital, budget-cycle, and funding-pathway questions defense-tech founders bring to the table — CFO first, defense specialist second, which is exactly the combination founders need when a capture strategy also has to survive a term sheet and a board meeting.

Sonia Vohnout



Role: OppsSpot, LLC Founder & President, GMA Partner

Background: Sonia has over 30 years of experience in systems engineering, quality systems, business development, and program management, holding a B.S. and M.S. in Systems Engineering from the University of Arizona. She is a nationally recognized expert in SBIR/STTR programs, commercialization strategy, and non-dilutive funding, and has helped clients secure tens of millions of dollars from DoW, Navy, DARPA, NASA, NIH, and other federal agencies. She's personally guided more than fifty companies through the SBIR/STTR pathway into the commercial and defense marketplace — the Phase I-to-III sole-source pathway in this playbook, walked by someone who's done it dozens of times over.

Elizabeth Kujan



Role: Senior Technical Writer

Background: Beth holds a Ph.D. in Materials Science and Engineering from Cornell University and a B.S. in the same field from MIT. She has 10 years of experience as a technical and grant writer with a strong SBIR/STTR track record, having brought in funding from the Department of Homeland Security, U.S. Air Force, National Science Foundation, and the New York State Energy Research and Development Authority. Her doctoral-level materials science background means she writes directly into hard-tech defense proposals — sensors, coatings, advanced materials — rather than translating secondhand from a founder's explanation.

Matthew Seabaugh



Role: Senior Materials Science Consultant

Background: Matthew is Founder and Chief Science Officer of Material Impact Consulting. He spent years at Nexceris and NexTech Materials working on solid oxide fuel cells, electrochemistry, and ceramic materials, including DOE-funded R&D such as published

pyrochlore-catalyst reforming research conducted with the National Energy Technology Laboratory. That bench-level materials science and energetics background is the kind of subject-matter depth a TRL or CMMC conversation with a program office actually requires — earned in the lab, not the boardroom.

Robert A. Moore



Role: Brigadier General, U.S. Army National Guard (Retired)

Background: Robert is a 30-year Army officer who rose from an Officer Candidate School commission to Brigadier General. His career centered on the budget and force-management side of defense: Director of Global Force Management, Mobilization, and Reserve Component Readiness within the Office of the Secretary of Defense for Personnel & Readiness; Director of Mobilization for the Assistant Secretary of Defense for Reserve Affairs; and Principal for Resources and Contracting, G-8, Army National Guard. He commanded a battalion in Afghanistan and holds the Army Distinguished Service Medal, Defense Superior Service Medal, and Legion of Merit. Robert has lived the PPBE budget cycle and force-readiness process from inside the Pentagon and the National Guard Bureau — this playbook’s Part 2, from someone who ran it.

Brandon Mendez-Lynch



Role: Co-Founder, Black Mountain Strategy

Background: A U.S. Air Force Academy graduate and former Intelligence Officer with a Special Operations background, Brandon managed current intelligence operations at the J2 level and led the Latin America Crisis Action Team through multiple regional crises. He now works as a Senior Defense Innovation Strategist helping dual-use startups build capture strategies, navigate SBIR proposals, and align commercial technology with government acquisition pathways — the founder-facing work this playbook teaches, done daily from inside the intelligence community he once served.

Charles Hatley



Role: Co-Founder, Black Mountain Strategy

Background: Charles is a U.S. Air Force veteran with 1,500+ combat hours in the RC-135 and EC-130 as a Mission Crew Commander, and served as an instructor at the Air Force's premier officer leadership school, training over a hundred leaders in critical thinking, management, and joint warfare. He has since built and exited a consumer venture (Beyond Alpha Nutrition) and co-founded another (Wlvs Den) — pairing operational military leadership with the hands-on realities of building a company from scratch.

